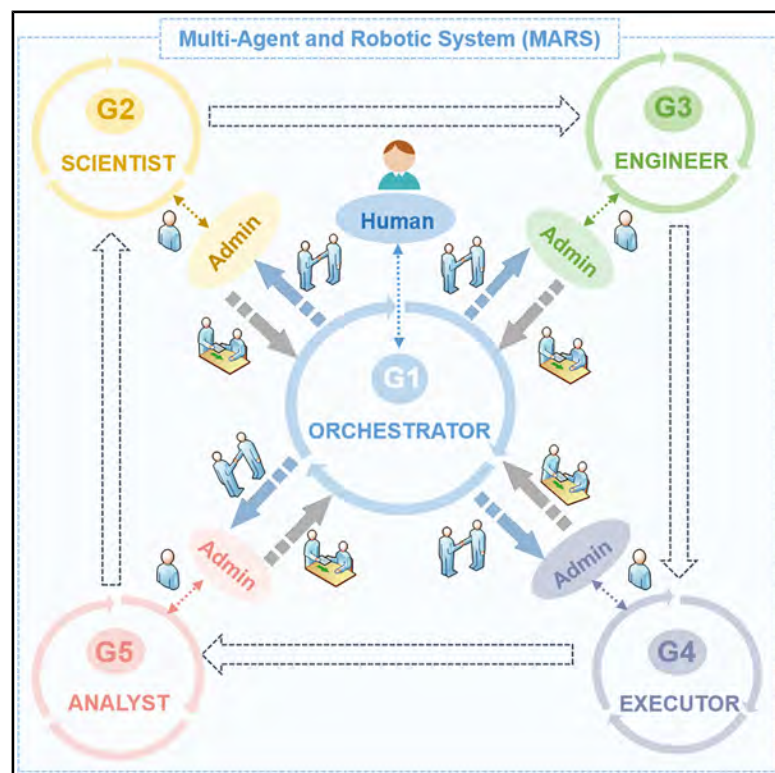


Knowledge-driven autonomous materials research via collaborative multi-agent and robotic system

Graphical abstract



Highlights

- Hierarchical multi-agent system with specialized tools mimics human research teams
- LLM-robot integration drives end-to-end autonomous materials research
- Hybrid retrieval enables cross-domain knowledge integration for professional designs
- Knowledge-based reasoning across domains leads to materials design

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In brief

The collaborative multi-agent and robot system (MARS), a hierarchical knowledge-driven framework coordinating 19 LLM agents with 16 domain-specific tools, is developed to accelerate materials research. By coordinating specialized agents with domain-specific tools, MARS autonomously executes full-cycle materials development, from design to synthesis and optimization. The system demonstrated a 60-fold acceleration in developing a water-stable perovskite nanocomposite through cross-domain knowledge integration. This paradigm shift enables researchers to focus on creative problem-solving while AI handles repetitive tasks, fundamentally changing materials innovation approaches.



Benchmark

First qualification/assessment of material properties and/or performance



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Article

Knowledge-driven autonomous materials research via collaborative multi-agent and robotic system

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PROGRESS AND POTENTIAL Large language models (LLMs) provide new possibilities for accelerating materials research, yet their application in complex materials science remains limited. Here, we developed the collaborative multi-agent and robot system (MARS), a knowledge-driven hierarchical architecture coordinating 19 LLM agents with 16 domain-specific tools organized into functional modules, achieving closed-loop autonomous materials discovery by integrating robotic experimentation. MARS offers accurate and professional guidance for materials development and relieves the hallucination inherent to current LLMs through a hybrid retrieval-augmented generation approach. As experimental validation, the system demonstrates its capabilities in perovskite nanocrystal development, optimizing synthesis protocols within 10 experimental iterations and designing a biomimetic “core-shell-corona” structure for water-stable perovskite composites within 3.5 h compared to conventional methods requiring 4–6 months. Through an interactive natural language interface, researchers can orchestrate experiments while tracking real-time agent coordination. Beyond efficiency gains, MARS represents a paradigm shift toward truly autonomous materials research where knowledge-driven AI can initiate scientific discovery. MARS can combine knowledge from different scientific fields to create solutions that human researchers might overlook, as demonstrated by its biomimetic design inspired by cell membranes. Looking forward, we envision MARS accelerating discoveries across critical areas including renewable energy materials, medical compounds, and sustainable technologies. This could help address urgent global challenges like climate change and healthcare needs by removing research bottlenecks. As the system evolves, it could democratize advanced materials research by reducing dependence on specialized infrastructure and expertise, allowing scientists worldwide to focus on creative problem-solving rather than repetitive tasks. MARS represents not just an improvement in research efficiency but also a fundamental shift in how we approach materials innovation.

SUMMARY

Large language models (LLMs) provide new possibilities for accelerating materials research, yet their application in complex materials science remains limited. Here, we developed the collaborative multi-agent and robot system (MARS), a knowledge-driven hierarchical architecture coordinating 19 LLM agents with 16 domain-specific tools for closed-loop autonomous materials discovery. MARS combines scientific knowledge with decision-making capabilities while mitigating hallucination through retrieval-augmented generation with a customized knowledge base. In experimental validation, the system



optimized perovskite nanocrystal synthesis within 10 iterations and designed a biomimetic “core-shell-corona” structure for water-stable perovskite composites in 3.5 h versus conventional methods requiring 4–6 months. This acceleration automates literature review and experimental planning, allowing researchers to focus on creative thinking while interacting through a natural language interface. This work establishes an integrated AI-driven framework for accelerating materials innovation and presents a paradigm for AI-enabled scientific discovery.

INTRODUCTION

Materials are the foundation of human understanding and transformation of the world, and innovations in this field are critical to societal progress and technological advancement.^{1,2} The discovery and application of new materials is typically a complex, lengthy, and costly process that requires the accumulation of multidisciplinary knowledge, extensive study of prior research, expertise in preparation techniques, time-intensive experimental iterations, and thorough validation in practical scenarios.^{3–6} To enhance the efficiency of materials innovation, scientists have made sustained and long-term efforts, such as the launch of the Materials Genome Initiative in 2011, which aims to accelerate materials discovery using high-throughput computational and experimental platforms.^{7–9} Despite these advances, the development of new materials typically still requires years, highlighting the urgent need for new methodologies to accelerate the pace of materials research.

Since OpenAI released ChatGPT, large language models (LLMs) have attracted widespread research interest, leading to substantial breakthroughs across various fields.^{10,11} LLMs can serve as invaluable research assistants and aid researchers in efficiently mining literature, quickly accessing critical data, and providing potential optimization strategies.^{12–15} Besides, LLMs have recently shown exceptional abilities in code generation, specialized tool integration, and strategic planning.^{16–23} These advances have enabled several pioneering studies that successfully couple LLM-powered agents with robotic experimentation platforms, achieving autonomous chemical synthesis,^{24,25} accelerated reaction optimization,²⁶ and intelligent synthesis route recommendation,²⁷ thereby demonstrating the transformative potential of AI-driven laboratory automation. These capabilities are particularly valuable for materials science, where the inherently multidisciplinary nature demands precise knowledge extraction and integration from an ever-expanding scientific literature.²⁸ This is particularly evident in nanomaterials research, a field characterized by an explosion of scientific literature over the past two decades, yet struggles with challenges of research redundancy and reproducibility.^{29,30} While promising, leveraging the full potential of LLMs for autonomous materials research requires addressing key challenges, including accurate comprehension of domain-specific knowledge, mitigation of hallucinations, robust control of automated synthesis, efficient management of complex tasks, and development of systematic workflows.

Herein, we develop a knowledge-driven multi-agent and robot system (MARS) for end-to-end autonomous materials discovery. MARS employs a collaborative network of 19 specialized LLM agents with access to 16 domain-specific tools to orchestrate complex material research tasks by integrating the robotic plat-

form. Critically, MARS leverages a comprehensive materials science knowledge base that guides its reasoning and mitigates hallucinations, providing professional guidance and designing multi-step synthesis processes compatible with automated robotic platforms. Subsequently, MARS can control multiple robotic platforms to autonomously execute iterative experiments (synthesis, analysis, and optimization) to achieve research objectives, demonstrated by successfully tuning green-emitting perovskite nanocrystals to target blue emission within 10 experimental iterations. These capabilities enable MARS to function as a human research team with superior domain-specific expertise compared to existing LLM-based systems. Furthermore, demonstrating its accelerated discovery potential, MARS autonomously designed and synthesized CsPbBr₃@LP/PEG composite with enhanced water stability and biocompatibility within 3.5 h, significantly accelerating materials development by 60-fold compared with the traditional method. By automating literature review, experimental planning, and protocol optimization, MARS allows human researchers to redirect their focus toward higher-level scientific insights, hypothesis generation, and creative problem-solving. Moreover, a user-friendly web interface facilitates intuitive interaction with MARS, enabling researchers to specify research objectives in natural language and monitor the automated workflow, creating a synergistic human-AI partnership that maximizes scientific productivity and accelerates the pace of materials innovation.

RESULTS

The architecture of MARS

This automated materials research framework employs a multi-agent system architecture that is analogous to a well-coordinated research laboratory. The MARS framework coordinates a network of 19 specialized agents, enabling complex multi-objective tasks in materials design, synthesis, and analysis through collaborative intelligence. These agents constitute a higher-level framework built on LLMs, leveraging their core reasoning and generative capabilities while incorporating task definition, dynamic utilization of 16 professional toolkits, and feedback loops to achieve complex objectives beyond the capacity of standalone LLMs. As shown in Figure 1A, the system is hierarchically structured into five synergistic functional groups with distinct responsibilities to emulate the real-world scientific research team, including the ORCHESTRATOR group (G1), SCIENTIST group (G2), ENGINEER group (G3), EXECUTOR group (G4), and ANALYST group (G5) (see Note S1).

The ORCHESTRATOR Group serves as the central coordinator, plans sub-tasks based on user requests, and selects the appropriate agent group to complete the sub-task, while

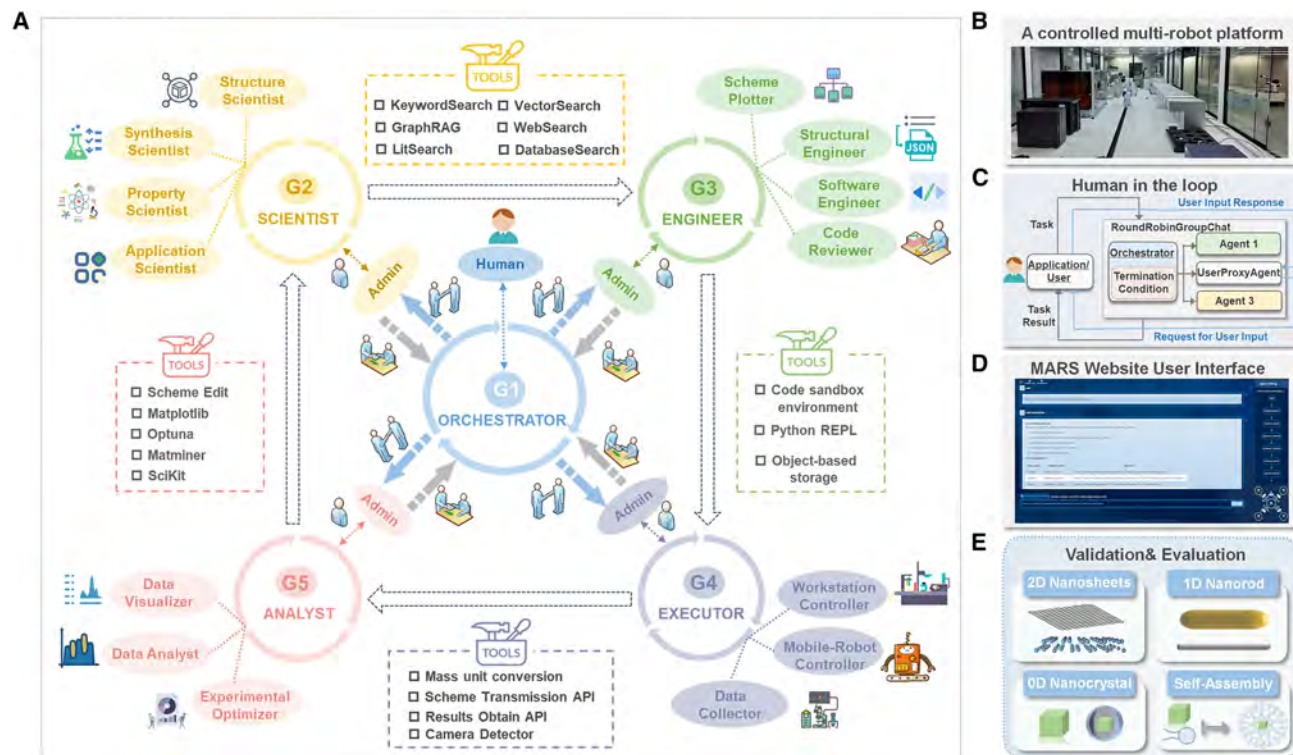


Figure 1. Architecture of MARS

(A) Overview of the MARS system. Circles represent 5 functional groups, and dashed boxes indicate callable toolkits.
(B) Physical setup of the multi-robot control platform.
(C) The mechanism of human-in-the-loop.
(D) Web-based user interface of MARS.
(E) Experimental validation and evaluation.

managing the message queue between agent groups. The SCIENTIST group, equipped with the hybrid retrieval-augmented generation (RAG) strategy and several customized information retrieval tools, focuses on materials knowledge retrieval and solution design, satisfying materials synthesis, structure, property, and application requirements. The ENGINEER group translates scientific proposals into executable protocols, bridging the gap between theoretical design and practical implementation. As presented in Figure 1B, the EXECUTOR group interacts with automated experimental platforms to carry out these protocols, controlling instrumentation, collecting data, and ensuring the reliable execution of designed experiments. Finally, the ANALYST Group analyzes experimental results and generates refinement strategies, driving iterative improvement and optimization based on experimental data. This modular design allows for specialized expertise and efficient task distribution among the agent groups.³¹

An effective, hierarchical message-passing mechanism is crucial to MARS, enabling multimodal (text, code, data, and image) communication between and within functional groups. Similar to information flow in research laboratories, the ORCHESTRATOR (G1) acts as a central hub, receiving user requests, analyzing requirements, planning, and distributing sub-tasks to administrator agents (function as team leaders) within

each group (G2–G5). These administrator agents further hand off sub-tasks to specialized agents within their respective groups. Leveraging domain expertise and dedicated tools (e.g., GraphRAG for SCIENTIST, Python REPL for ENGINEER, and Matplotlib for ANALYST), these agents execute their assigned tasks. Results are aggregated by the administrators and returned to the ORCHESTRATOR, which evaluates them and adjusts the whole sub-task list, assigning tasks or returning final results. This hierarchical structure, with clearly defined roles and communication pathways, ensures efficient collaboration, robust task management, and a streamlined, adaptable workflow for materials design.

As illustrated in Figure 1C, MARS integrates a human-in-the-loop mechanism, enabling dynamic task adjustment and optimization through user interaction and feedback with the ORCHESTRATOR. This allows expert intervention and validation at critical decision points when agents request assistance from experts, effectively combining computational efficiency with human expertise to ensure practical validity while maintaining autonomous operation. Figure 1D presents the user-friendly interface that facilitates user interaction and real-time monitoring of the experimental workflow (see Note S2 and Figures S1 and S2). As shown in Figure 1E, the capabilities of MARS have been validated by designing and synthesizing

diverse nanomaterials, demonstrating the effectiveness in practical research scenarios.

Knowledge-driven automated experimentation

Retrieval-augmented generation (RAG) enhances language model outputs by incorporating relevant information from external knowledge sources before generating responses. This approach grounds model outputs in factual, domain-specific information, thereby reducing hallucinations that commonly occur when models rely exclusively on knowledge embedded during training.³² MARS integrates materials knowledge retrieval with experimental execution, forming complementary cognitive and physical dimensions that enable the system to transition from conceptual understanding to real-world synthesis. Research on materials typically begins with comprehensive literature review and information retrieval. The SCIENTIST group accelerates this process through a hybrid RAG approach that transforms unstructured scientific literature into structured experimental guidance. Our hybrid RAG framework combines traditional vector similarity search with graph-based retrieval methods (GraphRAG), enabling more comprehensive and contextually aware information retrieval for complex materials science queries.

As illustrated in Figure 2A, four specialized LLM-based agents within the SCIENTIST group focus on distinct areas: structure, synthesis, properties, and applications, each contributing distinct expertise while leveraging specialized retrieval tools. The core retrieval system combines three primary approaches: KeywordSearch for word-level query matching, VectorSearch for semantic-level similarity-based matching, and GraphRAG for topic-level view through the knowledge graph. This hybrid approach ensures comprehensive query results by balancing sentence matching, paragraph matching, and topic view.

As shown in Figure 2B, MARS utilizes an extensive knowledge base carefully constructed from academic papers, books, and web resources to support this retrieval system (see Note S3). These diverse sources are converted to markdown format for natural segmentation and then enhanced with vector representations for similarity searches and structured into knowledge graphs for conceptual reasoning.^{33–35} Online resources (WebSearch, LitSearch, and DatabaseSearch) supplement the offline knowledge base when required. Upon receiving a sub-task from the ORCHESTRATOR, the admin agent of the SCIENTIST group routes it to the most appropriate specialized agent using tailored prompts. These specialized agents utilize hybrid retrieval tools to acquire relevant domain knowledge when needed, thus enhancing the accuracy of their responses.

To demonstrate the capabilities of MARS, the synthesis of perovskite nanocubes is selected as a representative task. CsPbBr₃ perovskites are promising for optoelectronic applications, but their nanocrystal synthesis is complex, with multiple interdependent factors including precursor selection, ligand engineering, reaction temperature control, and precise timing of various steps. For this demonstration, a specialized knowledge base was constructed from 1,348 high-impact perovskite publications, resulting in a vector database (7,818 segments) and a comprehensive knowledge graph (117,749 nodes and 232,490 edges), which provides a rich foundation for material synthesis

reasoning (see Figures S3 and S4). As shown in Figure 2C, MARS demonstrates superior performance by accurately retrieving precise reagent specifications and procedural details (see Note S4), whereas conventional LLMs like GPT-4o, o3-mini, and Llama 3.1-70B produce hallucinations. Regarding hallucinations, we identified several issues: numerical fabrication (such as providing integer values rather than stoichiometric ratios), incorrect synthesis conditions (inappropriate solvent or precursor selection), omission of critical experimental steps, and disregard for user constraints (see Note S5). MARS mitigates these issues through its multi-agent RAG architecture and addresses the knowledge deficit, which is identified as the fundamental cause of hallucinations.³⁶

As shown in Figure 2D, knowledge is transformed into action through the ENGINEER and EXECUTOR groups, which convert conceptual understanding into physical experimentation. Figure 2E illustrates the ENGINEER group, which serves as the crucial bridge, translating synthesis protocols into machine-executable instructions through four specialized agents: the scheme plotter creates visual flowcharts for human verification, the structural engineer converts protocols to standardized JSON schemas, the software engineer generates specialized code when needed,^{37,38} and the code reviewer ensures operational safety and feasibility (see Note S6). Subsequently, the EXECUTOR group implements these instructions through a self-developed experimental robotic platform⁵ (see Figure S5) that integrates workstation controller, mobile-robot controller, and characterization controller agents communicating with unified control middleware (see Figure S6).

Figure 2F demonstrates this workflow through the automated room temperature synthesis of CsPbBr₃ nanocrystals. Upon receiving the synthesis scheme from the SCIENTIST group, the ENGINEER group transforms the protocol into machine-executable instructions, which are subsequently passed to the EXECUTOR group. The workstation and mobile-robot controller agents transmit these instructions through the scheme transmission API tool to the unified control middleware on the computer workstation, thereby initiating the physical synthesis process. The middleware coordinates a sequence of automated operations: (1) mobile robots locate reagents and consumables from storage areas (no. 1–3); (2) the robots transport reagent bottles to the adding device (no. 4–5); (3) the materials are transferred to the liquid workstation (no. 6) for precise pipetting, stirring, and heating; and (4) the completed samples are transported to the fluorescence spectrometer (no. 7) for characterization. Throughout this process, spectroscopic data are transmitted to the middleware and accessed by the DataCollector agent for analysis.

Knowledge-driven automated experimentation

Parameter optimization is a critical component in materials science research. Conventional machine learning methods generally require extensive experimental data to build optimization models, but data are lacking in actual experimental processes. Figure 3A illustrates how MARS implements a paradigm shift by employing knowledge-driven reasoning rather than relying on data-intensive modeling approaches. By leveraging the powerful reasoning capabilities of LLMs combined with comprehensive materials

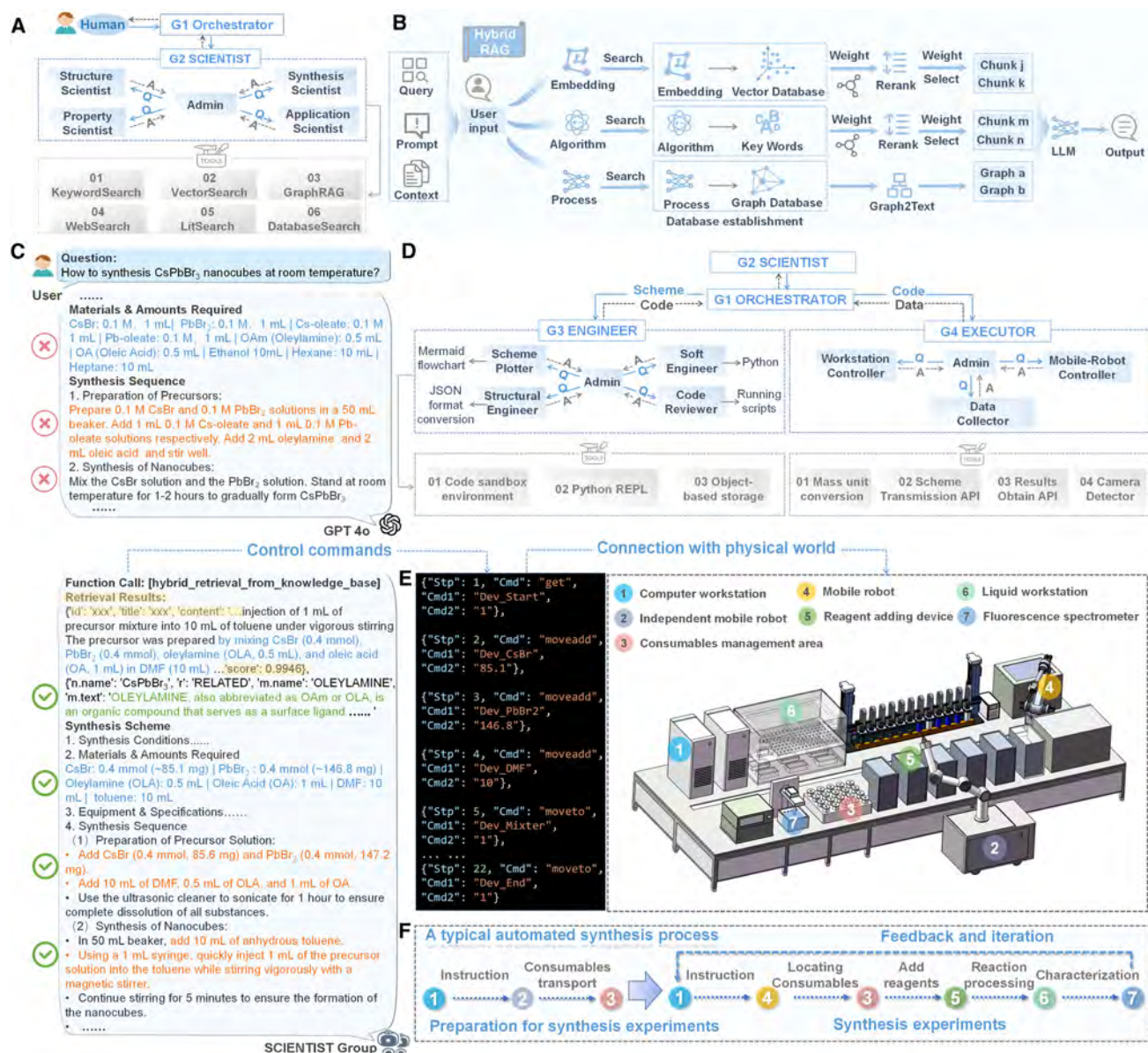


Figure 2. MARS for materials synthesis design and automated experimentation

(A) Overview of the SCIENTIST group.

(B) Schematic diagram of the hybrid RAG approach.

(C) Responses generated by GPT-4o and MARS to the same user query (uniform sampling parameters across systems: top_k = 0, top_p = 1, temperature = 0).

(D) Overview of the ENGINEER and EXECUTOR groups.

(E) Example of code and the controlled experimental robotic platform.

(F) Workflow of typical CsPbBr₃ nanocrystal synthesis process.

knowledge, the system can perform advanced parameter optimization and synthesis prediction with minimal experimental data.¹⁵ This paradigm shift enables efficient exploration of synthesis parameters without the high-throughput experimentation typically required by machine learning methods. The Experiment Optimizer agent demonstrates effective LLM specialization through structured system prompts. The prompt design includes several essential elements: clear role definition ("You are a professional Experiment_Optimizer"), integration with analytical tools (Scikit-

Learn, Optuna, and Matminer for modeling), and access to domain-specific literature through RAG retrieval and online search tools. This design implements a decision framework that directs the agent to select between code-based modeling for data-rich scenarios and knowledge-based parameter adjustment when data are limited, mimicking the actual workflow of materials scientists. Besides, the ANALYST_Admin agent enhances system robustness by providing critical coordination and quality control functions, including verification of chemical validity and

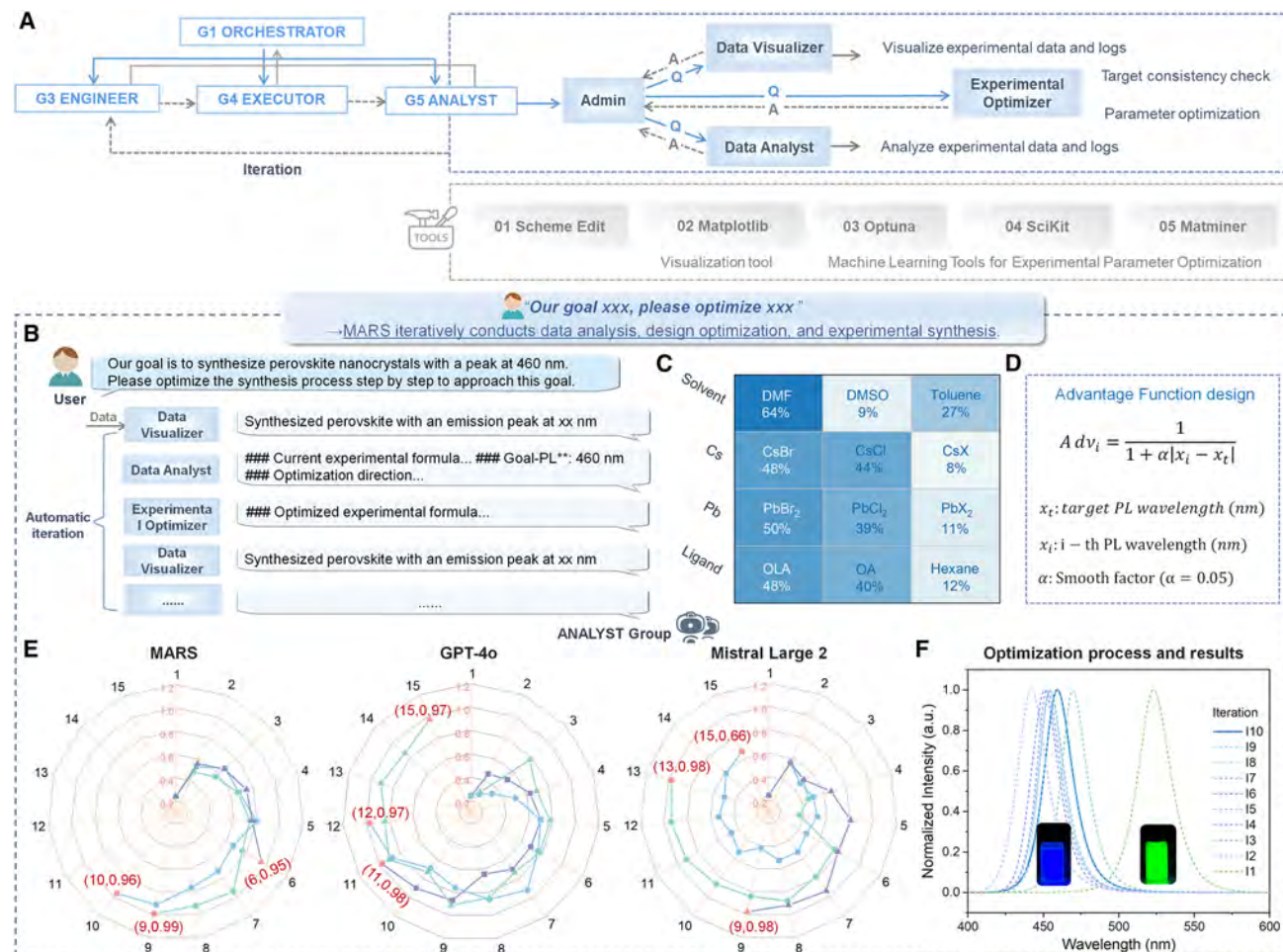


Figure 3. Optimization capabilities of MARS

- (A) Overview of the ANALYST Group.
 (B) Visualization of the multi-round iterative process of the agent.
 (C) Comparison of reagent choices across the knowledge base.
 (D) Mathematical expression used to calculate advantage values.
 (E) Comparison of optimization process performed based on MARS, GPT-4o, and Mistral Large2.
 (F) PL curve in iteration process based on MARS.

identification of duplicate recipes, thereby establishing comprehensive optimization through internal error correction mechanisms that single-model systems typically lack.

As shown in Figure 3B, the capabilities of MARS in feedback-driven optimization were demonstrated through the challenging task of tuning perovskite nanocrystal emission from green (523 nm) to blue (460 nm). All-inorganic perovskite nanocrystals have attracted considerable research interest due to their facile colloidal synthesis, tunable emission wavelengths spanning the entire visible spectrum, high photoluminescence quantum yields, and narrow emission bands. Wavelength tuning through halide composition engineering represents a crucial research direction with significant applications in light-emitting diodes (LEDs), lasers, and quantum light sources. Besides, this task demonstrates a classic materials optimization challenge that re-

quires precise control over quantum confinement effects and composition, typically achieved through halide exchange (Br to Cl substitution) or dimensional control. This optimization process demands professional reasoning about how precursor ratios and reaction conditions affect band-gap engineering in semiconductor nanocrystals, guided by well-defined targets with quantifiable outcomes. From an experimental demonstration perspective, this optimization task provides rapid visual feedback through color changes and enables real-time optical characterization, effectively highlighting the multi-parameter optimization capabilities of MARS.

Figure 3C presents the distribution of key reagents (including solvents, Cs sources, Pb sources, and ligands) reported in the literature for room temperature antisolvent precipitation of blue-emitting perovskites. These statistics were extracted from

the MARS knowledge base to help researchers establish a chemically reasonable design space grounded in prior experimental work. Notably, achieving the target 460 nm emission remains a zero-shot optimization challenge that requires iterative reasoning through complex constraints without prior experimental data for this specific target. Figure 3D shows the merit function, defined as the difference between the emission peak of the product and the target value, which is designed to assess optimization progress. For comparison, three parallel optimization iterations using MARS, currently one of the most powerful closed-source LLMs (GPT-4o), and the leading open-source LLM (Mistral Large2) were performed. To ensure scientific rigor and fairness in our comparative evaluation, we designed a controlled experiment where all participating models (MARS, GPT-4o, and Mistral Large 2) received identical standardized input formats. Notably, in preliminary tests requiring independent synthesis protocol generation for 523 nm CsPbBr₃ nanocrystals (NCs) without any starting recipe, only MARS successfully provided a complete and executable ligand-assisted reprecipitation synthesis protocol. In contrast, GPT-4o and Mistral Large 2 could only provide vague guidance or even demonstrate fundamental chemical misconceptions, highlighting the professional knowledge gap between specialized and general-purpose models.

All models began optimization from the same initial recipe (523 nm emission) provided by MARS for the formal optimization comparison. Optimization concluded upon reaching a merit value >0.95 or after 15 iterations. To ensure fair comparison, the optimization suggestions from GPT-4o and Mistral Large 2 were manually input into the automated experimental platform of MARS for execution, as MARS functions as an integrated design-experiment-optimization system capable of directly executing synthesis protocols and obtaining experimental feedback, whereas other models only provide textual responses. Each model received identical iterative feedback in a standardized table format containing current emission peak, historical data, and optimization targets (see Note S7). As presented in Figure 3E, MARS requires an average of only 8.3 iterations to reach the stopping condition, while GPT-4o and Mistral Large2 require 12.7 and 12.3 iterations, respectively. Mistral Large2 even fails to reason correctly once under the given rules, remaining far from the target even after more than 15 iterations. Figure 3F demonstrates how MARS achieves the target within ten iterative optimization cycles, highlighting its capability to autonomously use domain knowledge for guiding subsequent experimental decisions. This knowledge-driven approach significantly reduces the experimental burden by requiring fewer iterations to reach targets, allowing researchers to explore synthesis parameters more efficiently than standalone LLMs. Superior optimization efficiency in the MARS system is achieved through four key design elements that address fundamental limitations of general-purpose LLMs in materials science applications. First, specialized prompt engineering has proven effective for enhancing domain-specific LLM performance.^{39,40} Second, the multi-agent architecture enables task decomposition and specialization, avoiding cognitive overload when single models handle heterogeneous tasks.^{41,42} Third, structured decision flows with internal quality checks provide multiple error

correction points, ensuring reliable outputs.⁴³ Finally, the knowledge-enhanced mechanism through hybrid RAG provides professional depth far exceeding general models, enabling scientifically grounded optimization strategies rather than blind trial and error.^{44,45}

Knowledge-driven materials design

The design of novel functional materials typically requires researchers to possess extensive cross-disciplinary knowledge and rich experiential accumulation, particularly for complex material systems that necessitate balancing multiple performance dimensions. Figure 4 demonstrates in detail how the MARS system overcomes limitations of traditional development paradigms to assist scientists in achieving materials design and synthesis validation of perovskite ternary nanocomposite structures (see Note S8). As shown in Figure 4A, traditional materials design cycles typically require several months, encompassing literature review, experimental design, materials synthesis, result validation, and experimental optimization. This process heavily depends on researchers' efforts and experience while simultaneously constraining innovation possibilities due to insufficient systematic knowledge integration. In contrast, the MARS-assisted design cycle accelerates this process significantly and enables efficient material design through knowledge-driven and collaborative operations of multiple agents (G1–G5), compressing the entire procedure to just a few hours.

While precise wavelength control demonstrates the optimization capabilities, the fundamental challenge limiting perovskite nanocrystal applications is their poor stability. Lead halide perovskites like CsPbBr₃ possess ionic crystal structures that readily decompose upon water exposure, causing complete fluorescence quenching within hours. This weakness prevents practical deployment in photovoltaics, LEDs, and other optoelectrical applications. According to Web of Science analysis (see Figure S7), research on perovskite water stability has grown exponentially, reflecting the urgent need for effective stabilization strategies. Given this critical challenge, we tasked MARS with designing water-stable perovskite nanocrystals to showcase its capability in addressing pressing materials science problems.

The multi-round interaction process illustrated in Figure 4B reveals how the MARS progresses from mechanistic understanding to creative design through increasingly refined problem analysis and solution strategies. In round 1, when researchers inquired about core-shell stabilization mechanisms, MARS provided comprehensive explanations of how this approach protects perovskites through moisture barriers, defect passivation, thermal stability enhancement, and ion migration suppression. During this exploration of stability solutions, researchers noted an additional opportunity revealed by literature analysis: the growing interest in biomedical applications of perovskite materials (see Figure S8). These applications require both stability and biocompatibility as essential properties. This insight guided the subsequent interaction in round 2, where researchers introduced biocompatibility as an additional requirement. MARS identified this dual challenge and proposed multiple candidate solutions, including oxides, polymers, and biomimetic shells, each with distinct advantages for different application scenarios.

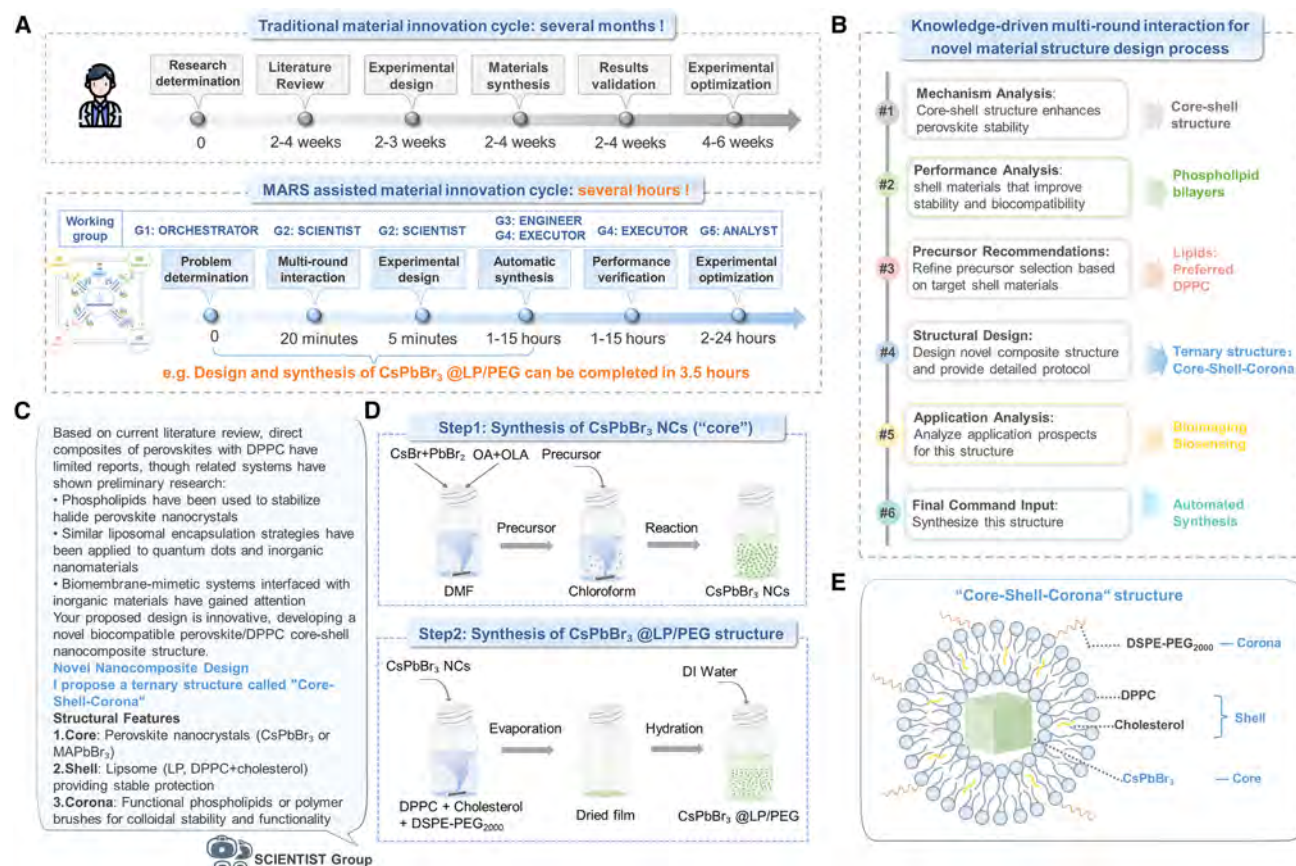


Figure 4. MARS-assisted design and synthesis of perovskite nanocomposite materials

(A) Efficiency comparison between the traditional materials design cycle and the MARS-assisted design process.

(B) Multi-round knowledge-driven interaction process for designing CsPbBr_3 @LP/PEG composite materials.

(C) Biomimetic membrane ternary "core-shell-corona" design.

(D) Synthesis route for CsPbBr_3 @LP/PEG.

(E) Structural illustration of CsPbBr_3 @LP/PEG.

From these options, phospholipid encapsulation was ultimately selected by researchers due to two key advantages: the amphiphilic nature of phospholipids facilitates self-assembly of nanocrystal surfaces under mild conditions that preserve crystal structure while providing protective encapsulation; and as major components of biological membranes, phospholipids inherently offer excellent biocompatibility. This strategic selection resulted in materials possessing the combined properties of excellent fluorescence, environmental stability, and biocompatibility required for advanced applications in biosensing, cell imaging, and tumor-targeted therapy.

When researchers specifically asked about phospholipid bilayer encapsulation in round 3, MARS provided detailed precursor recommendations including various phospholipids, such as 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-Dioleoyl-sn-glycero-3-phospho-(1'-rac-glycerol) (DOPG), 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), and functional components (cholesterol and 1,2-Distearoyl-sn-glycero-3-phosphorylethanolamine [DSPE]). Following this prioritized recom-

mendation, researchers specified DPPC for further design in round 4. As shown in Figure 4C, MARS identified limited research reports on direct perovskite-DPPC composites through its retrieval results and then analyzed relevant advances in liposomes, quantum dot encapsulation, and biomimetic membrane systems to establish foundations for innovative design. The system proposed a "core-shell-corona" ternary structure concept, utilizing a lipid bilayer to provide hydrophobic barriers while introducing a polyethylene glycol (PEG) corona to enhance colloidal stability. Specifically, the resulting ternary structure includes (1) core: CsPbBr_3 or MAPbBr_3 perovskite nanocrystals, providing excellent optical properties; (2) shell: a liposomal structure composed of DPPC and cholesterol, with cholesterol incorporation helping to modulate membrane fluidity and mechanical stability, forming a dense barrier layer that blocks water and oxygen molecule penetration; and (3) corona: DSPE-PEG₂₀₀₀ functional polymer brushes, providing colloidal stability through steric repulsion while conferring biocompatibility and potential for targeted functionalization. Notably, MARS prioritized DPPC in its phospholipid recommendations over the commonly used

DOPC in the literature.^{46–49} The scientific rationale for this recommendation becomes clear when comparing their molecular structures: although both DPPC and DOPC possess neutral phosphocholine headgroups, the saturated palmitic chains (16:0) of DPPC form more densely packed membranes with significantly lower water permeability compared to the unsaturated oleic chains (18:1) of DOPC.⁵⁰ The incorporation of cholesterol further enhances membrane organization by promoting liquid-ordered phase formation with DPPC, resulting in an exceptionally robust water barrier.^{51,52} This multi-level structural design achieves synergistic optimization of perovskite surface defect passivation, physical barriers against water/oxygen permeation, and colloidal stability at the molecular level, demonstrating the capability of MARS to integrate multidisciplinary knowledge for creative design.

Figure 4D illustrates the two-step synthesis route designed by MARS, reflecting its professional translation capability from theoretical design to experimental protocol (see Note S8). After the researchers acknowledged this design proposal and issued synthesis instructions, MARS generated JSON protocols and automatically performed the synthesis of this composite on our robotic platform. Within 3.5 h of the initial query, perovskite nanocrystal aqueous solutions were successfully obtained (Figure 4E). To investigate the behavior of CsPbBr₃@LP/PEG NCs in an aqueous environment, the evolution of their photoluminescence (PL) was tracked. As detailed in Figure S9A, the PL emission exhibited an initial slow increase followed by a gradual decrease upon water immersion. This fluorescence evolution is not uncommon and has been observed in other phospholipid-coated CsPbX₃ systems.⁵³ Previous studies have demonstrated that interactions with water can lead to surface defect passivation and induce a phase transition from CsPbBr₃ to CsPb₂Br₅, forming heterostructures that initially enhance fluorescence.^{54,55} Our X-ray crystallography (XRD) characterization confirmed the formation of the CsPb₂Br₅ phase during water immersion, consistent with these literature reports (see Figure S9B). Cytotoxicity testing confirmed the effectiveness of phospholipid encapsulation in enhancing perovskite biocompatibility. In the evaluation, CsPbBr₃@LP/PEG maintained >95% cell viability at concentrations up to 100 µg/mL, while pristine CsPbBr₃ showed significant cytotoxicity (see Figure S10). These results demonstrate that phospholipid encapsulation effectively prevents lead ion leaching while improving optical stability for further biological applications.

Performance evaluation of the MARS framework

The capabilities of MARS were rigorously assessed through pairwise win rate based on established LLM evaluation frameworks (see Note S11).⁵⁶ Specifically, a benchmark dataset comprising 929 materials science text fragments with standardized test questions was constructed through an automated question-answer pair generation protocol. Claude3.5-Sonnet was employed to conduct iterative pairwise comparisons between model responses, with randomized response ordering to mitigate positional bias.⁵⁷ This method significantly enhances the measurability of subtle differences between models, ensuring robust discriminative validity and statistical reliability.

As shown in Figure 5A, the fundamental architectural advantage of MARS is demonstrated through systematic comparisons

using identical base models. When GPT-4o is employed as the foundation in both systems, MARS achieves a 60.02% win rate against the single-agent implementation. This significant performance reveals critical architectural differences: MARS overcomes the limitations of single-agent systems through hierarchical design, where performance typically degrades when processing extensive contexts and managing multiple simultaneous tasks. By implementing specialized agents with defined responsibilities and enabling multi-round, depth-focused information retrieval within isolated task groups, MARS effectively reduces cognitive burden on individual models while enhancing overall system stability and knowledge integration capabilities.³⁶ Additionally, MARS also outperforms ChatGPT-4o (the web interface version), achieving a 61.89% win rate. This result demonstrates how domain-specific knowledge integration and hierarchical multi-agent architecture provide advantages over even highly optimized consumer AI systems.

Figure 5B demonstrates the adaptability of MARS across different foundation models, validating the broad applicability of the framework. Beyond the closed-source GPT-4o accessed via API, MARS was successfully implemented using DeepSeek-R1, a powerful open-source model with strong reasoning capabilities. The pairwise comparison results show that MARS (R1) consistently outperforms both the single-agent (R1) and the standalone DeepSeek-R1 model, confirming that the multi-agent architecture provides performance advantages regardless of foundation models. The average win rate analysis in Figure 5C reveals that MARS systems achieve higher overall performance compared to their single-agent counterparts and base models. The overall win rates were calculated by averaging win rates from all unique, non-self pairwise comparisons (see Figure S11). The performance hierarchy clearly shows that MARS benefits from stronger foundation models, with the R1-based implementation achieving higher average win rates than the GPT-4o version. This dependence on base model capabilities suggests that as LLMs continue to improve, the MARS framework will correspondingly enhance its performance in materials science applications. The multi-agent architecture thus provides a scalable approach that can adapt to and benefit from ongoing advances in LLMs.

While automated benchmarking provides quantitative comparisons, validating the practical utility of MARS in domain-specific scenarios requires expert assessment. Our evaluation panel comprised 10 researchers with extensive expertise in nanomaterial synthesis: 3 PhD candidates, 3 assistant researchers, 2 associate researchers, and 2 professors. All evaluations followed a double-blind protocol, with each test case independently scored by 5 experts. We employed a 10-point comprehensive scoring system evaluating three primary dimensions: scientific accuracy (40%), task completion (30%), and reasoning quality (30%). The evaluation tasks spanned four key aspects of materials research: structure design, synthesis protocol development, property prediction, and application optimization. Detailed results and task-specific performance analyses are provided in Note S9, where MARS consistently demonstrated superior accuracy and experimental actionability compared to baseline models (Figure S12).

Having demonstrated the effectiveness of MARS in perovskite nanocrystal synthesis, optimization, and design in the

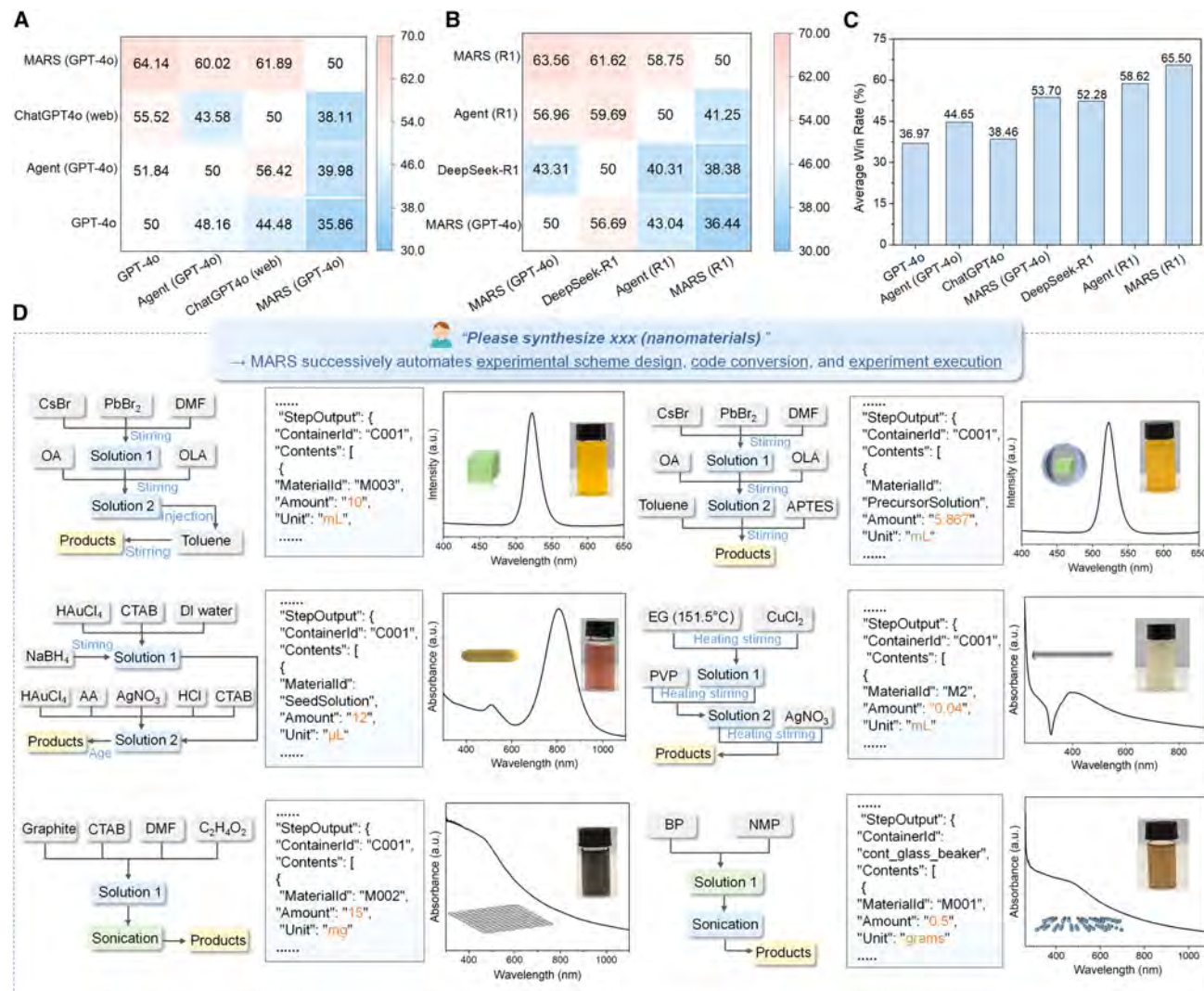


Figure 5. Evaluation and practical demonstration of MARS

(A) Pairwise win rate between MARS and single-agent (GPT-4o), ChatGPT-4o web interface, and GPT-4o (50% win rate for self-comparison, not tested).

(B) Pairwise win rate matrix demonstrating framework scalability (50% win rate for self-comparison, not tested).

(C) Average win rates across all pairwise comparisons.

(D) Validation of the automated synthesis capabilities of MARS across diverse nanomaterial systems.

main study, the evaluation was further extended to encompass a broader range of liquid-phase nanomaterial systems. As demonstrated in Figure 5D, MARS was successfully applied to materials across different dimensionalities, including Au nanorods, Ag nanowires, graphene, and black phosphorus nanosheets. For each material system, MARS autonomously designed natural language synthesis protocols, generated structured JSON protocols, and executed experimental procedures. This comprehensive validation across multiple evaluation methods and diverse material systems, combined with the proven adaptability of the framework to different foundation models, establishes MARS as a broadly applicable platform for accelerating materials research through intelligent automation.

DISCUSSION

This work introduces MARS, an LLM-driven platform for autonomous materials development. MARS integrates 19 specialized agents within a hierarchical framework, which are organized into functional groups to facilitate collaborative execution of complex tasks spanning design, synthesis, and optimization. By leveraging a curated knowledge base and a hybrid RAG approach, MARS achieves enhanced accuracy and mitigates LLM hallucinations, enabling professional-grade, autonomous experimentation. Significantly, MARS achieved rapid optimization of material properties (e.g., tuning green-emitting perovskite nanocrystals to blue emission within 10 iterations) and discovery of novel materials within hours, significantly accelerating the

research process. Beyond efficiency gains, the MARS architecture enables seamless information flow and knowledge accumulation throughout the research pipeline, creating a responsive closed-loop system that adapts to experimental feedback in real time.

MARS represents a significant step toward autonomous materials research. This system reduces substantial repetitive work, allowing researchers to focus on creative thinking and scientific insights. The flexible, collaborative agent framework permits customization by researchers, enabling integration of local databases and adaptation to diverse experimental platforms. Such features facilitate tailoring to specific research needs while expanding applicability across various materials and synthesis techniques. Despite these advantages, current limitations must be acknowledged. MARS primarily relies on existing literature knowledge for combinatorial optimization, with its innovative capacity currently limited to incremental improvements rather than breakthrough discoveries. This reflects fundamental challenges in applying LLMs to scientific research: distinguishing rational inference from hallucinatory outputs and balancing knowledge dependence with exploratory innovation. The system functions as an intelligent research assistant that accelerates discovery cycles rather than replacing human creativity.

Beyond performance, reproducibility and runtime error handling represent critical factors for practical adoption in laboratory settings. The MARS architecture addresses reproducibility through a comprehensive open-source implementation with standardized interfaces between agent modules and robotic components. This design allows other laboratories to replicate system functionality, though some adaptation to specific hardware configurations remains necessary. Regarding error management, MARS employs a multi-tiered approach to handle failures in both design and synthesis processes: human intervention is strategically integrated at critical decision points, autonomous replanning occurs when agents detect suboptimal outcomes, and rule-based verification protects robotic execution. This flexible error handling capability ensures experimental continuity despite the unpredictable challenges inherent in materials research, maintaining system reliability without constant expert supervision.

Future development will focus on standardized interfaces for laboratory equipment, enabling more complex experimental control. Beyond predefined functions, future LLMs could autonomously devise experimental strategies, optimize parameters, and adapt to unexpected situations, realizing embodied intelligence in materials research. Besides, while the hybrid RAG approach mitigates hallucinations, further advancements in LLM architectures and domain-specific fine-tuning are crucial. Critical directions include integrating physics-based simulations and first-principles calculations to enhance scientific reasoning capabilities, developing more reliable metrics for assessing innovation versus hallucination. We believe the evolution from knowledge-driven to genuinely autonomous innovation represents the necessary trajectory for AI in scientific research, with the current work establishing an important foundation for this long-term goal. Future MARS iterations promise to accelerate materials discovery and development to meet evolving scientific and industrial demands.

METHODS

Architecture details

MARS design

The MARS system utilizes GPT-4o from OpenAI as its foundation model, with the temperature parameter set to 0.1 to ensure high stability and consistency during task execution. The AutoGen framework (<https://github.com/microsoft/autogen>) simplifies complex LLM workflow orchestration, automation, and optimization through multi-agent dialogue mechanisms. Its flexible architecture enables developers to customize agent behaviors and interaction patterns according to specific task requirements. AutoGen provides two core agent types: AssistantAgent and UserProxyAgent. In this study, we utilize AssistantAgent provided by AutoGen, defining agent roles and functions through system_message and tools parameters. Simultaneously, we set up UserProxyAgent as a human control interface to implement human-in-the-loop in the multi-agent system, enabling human intervention at critical points during task execution. In terms of agent orchestration, MARS primarily employs two modes: Selector Group Chat and Swarm (<https://github.com/openai/swarm>). Selector Group Chat is a multi-agent dialogue mode based on model selection mechanisms, where agents broadcast messages to other members in turns, and an Orchestrator agent selects the next speaking agent or team based on conversation context. The Swarm mode is a multi-agent collaboration mode based on task delegation mechanisms. Agents dynamically delegate tasks to more suitable agents based on task requirements and capabilities, ensuring efficient task allocation and execution within the team. The core mechanism of Swarm relies on agents generating HandoffMessages to specify task delegation targets. The admin agents between teams coordinate through the Selector Group Chat mode, while task allocation within teams relies on the Swarm mode. Further details regarding the methods can be found in the supplemental information.

SCIENTIST group

The scientist team consists of four materials science agents, each initialized through AssistantAgent. We guide different scientist agents to form different capability preferences through system prompts. Within the scientist team, we use AutoGen's Swarm mode to orchestrate agents, with four scientist agents communicating directly with the Scientist Team Admin agent through task submission.

Knowledge base and retrieval tools

We established a knowledge base containing over 100,000 nano-material-related literature papers and developed interfaces based on keyword retrieval and vector retrieval for efficient research material access. Specifically, we used the MinerU tool to convert literature PDFs to markdown format for an efficient knowledge base. For the SCIENTIST group, we customized the following retrieval tools. *Vector database and search*: to optimize the RAG system's retrieval, we segmented papers by section and vectorized text segments using the gte-Qwen2-1.5B-instruct model (<https://huggingface.co/thenlper/gte-large>). The vectors are stored in a Weaviate database of Dify, with API functionalized as hybrid_retrieval_from_knowledge_base providing KeywordSearch and VectorSearch interfaces based on

keywords and vectors. *Knowledge graph*: based on the knowledge base, we classified literature by research topics and utilized GraphRAG to construct knowledge graphs for different research topics, with corresponding interfaces available through Autogen Extension. *WebSearch and LitSearch*: we integrated WebSearch tools based on the Bing search engine and LitSearch tools based on Arxiv and Semantic Scholar for real-time online searches. Bing searches, arXiv, and Semantic Scholar are called directly through official APIs, which functioned as “paper_retrieval_from_arxiv_and_pubmed” and “online_retrieval_with_bing_and_playwright.” *DatabaseSearch*: material databases are crucial in nano-materials research. We developed database search tools based on OQMD and Materials Project (MP) databases, providing comprehensive and precise material data query capabilities. While the MP database offers official retrieval interfaces, we simplified and encapsulated their parameters through FastAPI to prevent agent call failures. For the OQMD database, we obtain query results using PlayWright and similarly encapsulate them using FastAPI.

ENGINEER group

The MARS engineer agent team consists of four agents defined based on AutoGen’s AssistantAgent class, coordinating through system prompts and tools. Within the team, we employ AutoGen’s Swarm mode to organize engineer agents and admin agents, efficiently completing subtasks distributed by the Orchestrator through task submission. The following tools support the ENGINEER group. *Code Sandbox Environment*: AutoGen provides DockerCommandLineCodeExecutor, which relies on Docker container isolation environments for code execution that might cause conflicts or crashes in local Python environments. *Python REPL*: we directly use PythonCodeExecutionTool to execute Python code reviewed and confirmed safe by the code reviewer. *Object-based storage*: we introduced object storage tools using MinIO to build a local object storage service, encapsulated through FastAPI, to maintain the integrity and stability of structured experimental schemes.

EXECUTOR group

To address potential process blocking during long wait times for characterization data collection, data collection tools are set with a 6-h execution time and a 30-s sleep mechanism when data are unavailable. We customized the following tools for the EXECUTOR group. *Mass unit conversion* implemented the mol2mg tool based on the periodictable Python package for converting molar mass to weight units. *Scheme transmission API*: implemented using FastAPI for transmitting structured synthesis scheme access links directly to robots or island stations. *Results obtain API*: implemented using FastAPI for direct access to characterization device data. *Camera detector* provides IP Camera video stream interface based on RTSP protocol for real-time monitoring of mobile robots and island stations.

Experimental details

Cell viability assessment

MCF-7 cells were seeded in 96-well plates (1×10^4 cells/well) and cultured for 24 h. Cells were then incubated with various concentrations (0–100 $\mu\text{g/mL}$) of pristine CsPbBr₃ or CsPbBr₃@LP/PEG for 24 h. CCK-8 reagent was added, and absorbance was

measured at 450 nm. Live/dead staining was performed using the Calcein-AM/PI kit following the manufacturer’s protocol.

Viable and dead cell staining

MCF-7 cells were incubated in 6-well plates under standard conditions for 24 h. After removal of the digested medium, the cells were treated with fresh medium that contained pristine CsPbBr₃ or CsPbBr₃@LP/PEG for 4 h of co-culture. Then, cells were stained with calcein AM (10 μM) and PI (10 μM) according to the product description for 0.5 h. The coloring solution was removed and rinsed twice gently with cold PBS. Finally, the cells were imaged using a fluorescence inverted microscope (Olympus UHGLGPS, China).

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Xue-Feng Yu (xf.yu@siat.ac.cn).

Materials availability

Samples can be generated following the MARS workflow as described in the methods.

Data and code availability

The data supporting the findings of this study are available within the paper and in the supplemental information files. Source data and Python scripts for MARS developed in this study are available at GitHub: <https://github.com/tangger2000/MARS>.

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AUTHOR CONTRIBUTIONS

Conceptualization, T.S. and X.-F.Y.; methodology, T.S., Y.L., Z.W., W.X., G.J., and J.Z.; investigation, T.S., Y.L., and H.H.; visualization, T.S.; supervision, X.-F.Y.; writing – original draft, T.S., Y.L., and Z.W.; writing – review and editing, G.J., D.D., Z.W., R.H., S.R., P.K.C., W.Z., and X.-F.Y.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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REFERENCES

1. Rothmund, P., Kim, Y., Heisser, R.H., Zhao, X., Shepherd, R.F., and Keplinger, C. (2021). Shaping the future of robotics through materials

- innovation. *Nat. Mater.* 20, 1582–1587. <https://doi.org/10.1038/s41563-021-01158-1>.
2. Mohanty, A.K., Vivekanandhan, S., Pin, J.-M., and Misra, M. (2018). Composites from renewable and sustainable resources: Challenges and innovations. *Science* 362, 536–542. <https://doi.org/10.1126/science.aat9072>.
3. Burger, B., Maffettone, P.M., Gusev, V.V., Aitchison, C.M., Bai, Y., Wang, X., Li, X., Alston, B.M., Li, B., Clowes, R., et al. (2020). A mobile robotic chemist. *Nature* 583, 237–241. <https://doi.org/10.1038/s41586-020-2442-2>.
4. Angelone, D., Hammer, A.J.S., Rohrbach, S., Krambeck, S., Granda, J.M., Wolf, J., Zaleskiy, S., Chisholm, G., and Cronin, L. (2021). Convergence of multiple synthetic paradigms in a universally programmable chemical synthesis machine. *Nat. Chem.* 13, 63–69. <https://doi.org/10.1038/s41557-020-00596-9>.
5. Zhao, H., Chen, W., Huang, H., Sun, Z., Chen, Z., Wu, L., Zhang, B., Lai, F., Wang, Z., Adam, M.L., et al. (2023). A robotic platform for the synthesis of colloidal nanocrystals. *Nat. Synth.* 2, 505–514. <https://doi.org/10.1038/s44160-023-00250-5>.
6. Liz-Marzán, L.M., Kagan, C.R., and Millstone, J.E. (2020). Reproducibility in nanocrystal synthesis? Watch out for impurities. *ACS Nano* 14, 6359–6361. <https://doi.org/10.1021/acsnano.0c04709>.
7. Yao, Y., Dong, Q., Brozena, A., Luo, J., Miao, J., Chi, M., Wang, C., Kevrekidis, I.G., Ren, Z.J., Greeley, J., et al. (2022). High-entropy nanoparticles: Synthesis-structure-property relationships and data-driven discovery. *Science* 376, eabn3103. <https://doi.org/10.1126/science.abn3103>.
8. Lu, X.-G. (2015). Remarks on the recent progress of Materials Genome Initiative. *Sci. Bull.* 60, 1966–1968. <https://doi.org/10.1007/s11434-015-0937-2>.
9. Eng, A.Y.S., Soni, C.B., Lum, Y., Khoo, E., Yao, Z., Vineeth, S.K., Kumar, V., Lu, J., Johnson, C.S., Wolverton, C., and Seh, Z.W. (2022). Theory-guided experimental design in battery materials research. *Sci. Adv.* 8, eabm2422. <https://doi.org/10.1126/sciadv.abm2422>.
10. Urbina, F., Lentzos, F., Invernizzi, C., and Ekins, S. (2022). Dual use of artificial-intelligence-powered drug discovery. *Nat. Mach. Intell.* 4, 189–191. <https://doi.org/10.1038/s42256-022-00465-9>.
11. van Dis, E.A.M., Bollen, J., Zuidema, W., Van Rooij, R., and Bockting, C.L. (2023). ChatGPT: five priorities for research. *Nature* 614, 224–226. <https://doi.org/10.1038/d41586-023-00288-7>.
12. Zheng, Z., Zhang, O., Nguyen, H.L., Rampal, N., Alawadhi, A.H., Rong, Z., Head-Gordon, T., Borgs, C., Chayes, J.T., and Yaghi, O.M. (2023). ChatGPT research group for optimizing the crystallinity of MOFs and COFs. *ACS Cent. Sci.* 9, 2161–2170. <https://doi.org/10.1021/acscentsci.3c01087>.
13. Zheng, Z., Zhang, O., Borgs, C., Chayes, J.T., and Yaghi, O.M. (2023). ChatGPT chemistry assistant for text mining and the prediction of MOF Synthesis. *J. Am. Chem. Soc.* 145, 18048–18062. <https://doi.org/10.1021/jacs.3c05819>.
14. Zheng, Z., Rong, Z., Rampal, N., Borgs, C., Chayes, J.T., and Yaghi, O.M. (2023). A GPT-4 reticular chemist for guiding MOF discovery. *Angew. Chem. Int. Ed.* 62, e202311983. <https://doi.org/10.1002/anie.202311983>.
15. Kim, S., Jung, Y., and Schrier, J. (2024). Large language models for inorganic synthesis predictions. *J. Am. Chem. Soc.* 146, 19654–19659. <https://doi.org/10.1021/jacs.4c05840>.
16. Ridnik, T., Kredo, D., and Friedman, I. (2024). Code generation with alpha-codium: From prompt engineering to flow engineering. preprint at arXiv. <https://doi.org/10.48550/arXiv.2401.08500>.
17. Boiko, D.A., MacKnight, R., Kline, B., and Gomes, G. (2023). Autonomous chemical research with large language models. *Nature* 624, 570–578. <https://doi.org/10.1038/s41586-023-06792-0>.
18. Microsoft Research AI4Science, Microsoft Azure Quantum (2023). The impact of large language models on scientific discovery: a preliminary study using GPT-4. preprint at arXiv. <https://doi.org/10.48550/arXiv.2311.07361>.
19. Wahab, M.F., Handlovic, T.T., Roy, S., Burk, R.J., and Armstrong, D.W. (2024). Solving advanced task-specific problems in measurement sciences with generative AI. *Anal. Chem.* 96, 12387–12394. <https://doi.org/10.1021/acs.analchem.4c01734>.
20. M Bran, A., Cox, S., Schilter, O., Baldassari, C., White, A.D., and Schwalder, P. (2024). Augmenting large language models with chemistry tools. *Nat. Mach. Intell.* 6, 525–535. <https://doi.org/10.1038/s42256-024-00832-8>.
21. Bensaude-Vincent, B., and Hessebruch, A. (2004). Materials science: a field about to explode? *Nat. Mater.* 3, 345–347. <https://doi.org/10.1038/nmat1140>.
22. Szymanski, N.J., Rendy, B., Fei, Y., Kumar, R.E., He, T., Milsted, D., McDermott, M.J., Gallant, M., Cubuk, E.D., Merchant, A., et al. (2023). An autonomous laboratory for the accelerated synthesis of novel materials. *Nature* 624, 86–91. <https://doi.org/10.1038/s41586-023-06734-w>.
23. Yoshikawa, N., Skreta, M., Darvish, K., Arellano-Rubach, S., Ji, Z., Bjørn Kristensen, L., Li, A.Z., Zhao, Y., Xu, H., Kuramshin, A., et al. (2023). Large language models for chemistry robotics. *Auton. Robots* 47, 1057–1086. <https://doi.org/10.1007/s10514-023-10136-2>.
24. Song, T., Luo, M., Zhang, X., Chen, L., Huang, Y., Cao, J., Zhu, Q., Liu, D., Zhang, B., Zou, G., et al. (2025). A multi-agent-driven robotic AI chemist enabling autonomous chemical research on demand. *J. Am. Chem. Soc.* 147, 12534–12545. <https://doi.org/10.1021/jacs.4c17738>.
25. Tom, G., Schmid, S.P., Baird, S.G., Cao, Y., Darvish, K., Hao, H., Lo, S., Pablo-García, S., Rajaonson, E.M., Skreta, M., et al. (2024). Self-driving laboratories for chemistry and materials science. *Chem. Rev.* 124, 9633–9732. <https://doi.org/10.1021/acs.chemrev.4c00055>.
26. Ruan, Y., Lu, C., Xu, N., Zhang, J., Xuan, J., Pan, J., Fang, Q., Gao, H., Shen, X., Ye, N., et al. (2024). Accelerated end-to-end chemical synthesis development with large language models. preprint at ChemRxiv. <https://doi.org/10.26434/chemrxiv-2024-6wmq4>.
27. Chen, K., Lu, J., Li, J., Yang, X., Du, Y., Wang, K., Shi, Q., Yu, J., Li, L., Qiu, J., et al. (2023). Chemist-X: Large language model-empowered agent for reaction condition recommendation in chemical synthesis. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2311.10777>.
28. Sanchez, C., Arribart, H., and Guille, M.M.G. (2005). Biomimetic and bioinspiration as tools for the design of innovative materials and systems. *Nat. Mater.* 4, 277–288. <https://doi.org/10.1038/nmat1349>.
29. Marchese Robinson, R.L., Lynch, I., Peijnenburg, W., Rumble, J., Klaessig, F., Marquardt, C., Rauscher, H., Puzyn, T., Purian, R., Åberg, C., et al. (2016). How should the completeness and quality of curated nanomaterial data be evaluated? *Nanoscale* 8, 9919–9943. <https://doi.org/10.1039/C5NR08944A>.
30. Comandella, D., Gottardo, S., Río-Echevarría, I.M., and Rauscher, H. (2020). Quality of physicochemical data on nanomaterials: an assessment of data completeness and variability. *Nanoscale* 12, 4695–4708. <https://doi.org/10.1039/C9NR08323E>.
31. Darvish, K., Skreta, M., Zhao, Y., Yoshikawa, N., Som, S., Bogdanovic, M., Cao, Y., Hao, H., Xu, H., Aspuru-Guzik, A., et al. (2025). ORGANA: A robotic assistant for automated chemistry experimentation and characterization. *Matter* 8, 101897. <https://doi.org/10.1016/j.matt.2024.10.015>.
32. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-T., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. preprint at arXiv. <https://doi.org/10.48550/arXiv.2005.11401>.
33. Wang, B., Xu, C., Zhao, X., Ouyang, L., Wu, F., Zhao, Z., Xu, R., Liu, K., Qu, Y., Shang, F., et al. (2024). MinerU: An open-source solution for precise document content extraction. preprint at arXiv. <https://doi.org/10.48550/arXiv.2409.18839>.
34. Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., and Liu, Z. (2024). M3-Embedding: multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. preprint at arXiv. <https://doi.org/10.48550/arXiv.2402.03216>.

35. Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R.O., and Larson, J. (2024). From local to global: A graph RAG approach to query-focused summarization. preprint at arXiv. <https://doi.org/10.48550/arXiv.2404.16130>. <https://arxiv.org/abs/2404.16130>.
36. Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature* 630, 625–630. <https://doi.org/10.1038/s41586-024-07421-0>.
37. Zeni, C., Pinsler, R., Zügner, D., Fowler, A., Horton, M., Fu, X., Wang, Z., Shysheya, A., Crabbé, J., Ueda, S., et al. (2025). A generative model for inorganic materials design. *Nature* 639, 624–632. <https://doi.org/10.1038/s41586-025-08628-5>.
38. Yang, H., Hu, C., Zhou, Y., Liu, X., Shi, Y., Li, J., Li, G., Chen, Z., Chen, S., Zeni, C., et al. (2024). Mattersim: A deep learning atomistic model across elements, temperatures and pressures. preprint at arXiv. <https://doi.org/10.48550/arXiv.2405.04967>.
39. Zhou, Y., Madaan, A., Liu, S., Yao, H., Hallinan, W., Yu, Q., Naik, A., and Yang, Y. (2022). Large language models are human-level prompt engineers. preprint at arXiv. <https://doi.org/10.48550/arXiv.2211.01910>.
40. Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., and Iwasawa, Y. (2022). Large language models are zero-shot reasoners. preprint at arXiv. <https://doi.org/10.48550/arXiv.2205.11916>.
41. Li, G., Hammoud, H., Itani, H., Khizbullin, D., and Ghanem, B. (2023). CAMEL: Communicative agents for “mind” exploration of large language model society. preprint at arXiv. <https://doi.org/10.48550/arXiv.2303.17760>.
42. Wang, Z., Mao, S., Wu, W., Ge, T., Wei, F., and Ji, H. (2023). Unleashing the emergent cognitive synergy in large language models: A task-solving agent through multi-persona self-collaboration. preprint at arXiv. <https://doi.org/10.48550/arXiv.2307.05300>.
43. Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., and Deng, S. (2023). Exploring collaboration mechanisms for LLM agents: A social psychology view. preprint at arXiv. <https://doi.org/10.48550/arXiv.2310.02124>.
44. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. preprint at arXiv. <https://doi.org/10.48550/arXiv.2005.11401>.
45. Guu, K., Lee, K., Tung, Z., Pasupat, P., and Chang, M. (2020). REALM: Retrieval augmented language model pre-training. preprint at arXiv. <https://doi.org/10.48550/arXiv.2002.08909>.
46. Li, M., Zeng, Y., Qu, X., Jalalah, M., Alsareii, S.A., Li, C., Harraz, F.A., and Li, G. (2021). Biocatalytic CsPbX₃ perovskite nanocrystals: a self-reporting nanoprobe for metabolite analysis. *Small* 17, 2103255. <https://doi.org/10.1002/smll.202103255>.
47. Li, M., Tian, T., Zeng, Y., Zhu, S., Li, C., Yin, Y., and Li, G. (2021). One-step assay of pore-forming biotoxins based on biomimetic perovskite nanocrystals. *Sens. Actuators B* 338, 129839. <https://doi.org/10.1016/j.snb.2021.129839>.
48. Li, M., Wang, Y., Hu, H., Feng, Y., Zhu, S., Li, C., and Feng, N. (2022). A dual-readout sandwich immunoassay based on biocatalytic perovskite nanocrystals for detection of prostate specific antigen. *Biosens. Bioelectron.* 203, 113979. <https://doi.org/10.1016/j.bios.2022.113979>.
49. Yue, Y., Xu, H., Jiang, L., Zhao, X., and Deng, D. (2024). Introducing specific iodine ions in perovskite-based nanocomplex to cater for versatile biomedical imaging and tumor radiotherapy. *Adv. Healthc. Mater.* 13, 2302721. <https://doi.org/10.1002/adhm.202302721>.
50. Ghysels, A., Krämer, A., Venable, R.M., Teague, W.E., Lyman, E., Gawrisch, K., and Pastor, R.W. (2019). Permeability of membranes in the liquid ordered and liquid disordered phases. *Nature Commun.* 10, 5616. <https://doi.org/10.1038/s41467-019-13432-7>.
51. Reber, P.J., and Squire, L.R. (2009). Intact visual discrimination learning in amnesic patients with hippocampal lesions. *Science* 325, 1514–1516. <https://doi.org/10.1126/science.1174621>.
52. de Meyer, F., and Smit, B. (2009). Effect of cholesterol on the structure of a phospholipid bilayer. *Proc. Natl. Acad. Sci. USA* 106, 3654–3658. <https://doi.org/10.1073/pnas.0809959106>.
53. Yang, Z., Xu, J., Zong, S., Xu, S., Zhu, D., Zhang, Y., Chen, C., Wang, C., Wang, Z., and Cui, Y. (2019). Lead halide perovskite nanocrystals–phospholipid micelles and their biological applications: multiplex cellular imaging and in vitro tumor targeting. *ACS Appl. Mater. Interfaces* 11, 47671–47679. <https://doi.org/10.1021/acsami.9b12924>.
54. Zhang, X., Xu, B., Zhang, J., Gao, Y., Zheng, Y., Wang, K., and Sun, X.W. (2016). All-inorganic perovskite nanocrystals for high-efficiency light emitting diodes: dual-phase CsPbBr₃–CsPb₂Br₅ composites. *Adv. Funct. Mater.* 26, 4595–4600. <https://doi.org/10.1002/adfm.201600958>.
55. Liu, Y., Li, F., Liu, Q., and Xia, Z. (2018). Synergetic effect of postsynthetic water treatment on the enhanced photoluminescence and stability of CsPbX₃ (X = Cl, Br, I) perovskite nanocrystals. *Chem. Mater.* 30, 6922–6929. <https://doi.org/10.1021/acs.chemmater.8b03330>.
56. Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. preprint at arXiv. <https://doi.org/10.48550/arXiv.2306.05685>.
57. Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A.N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M.I., Gonzalez, J.E., et al. (2024). Chatbot arena: An open platform for evaluating LLMs by human preference. preprint at arXiv. <https://doi.org/10.48550/arXiv.2403.04132>.

Matter, Volume 9

Supplemental information

**Knowledge-driven autonomous materials
research via collaborative multi-agent
and robotic system**

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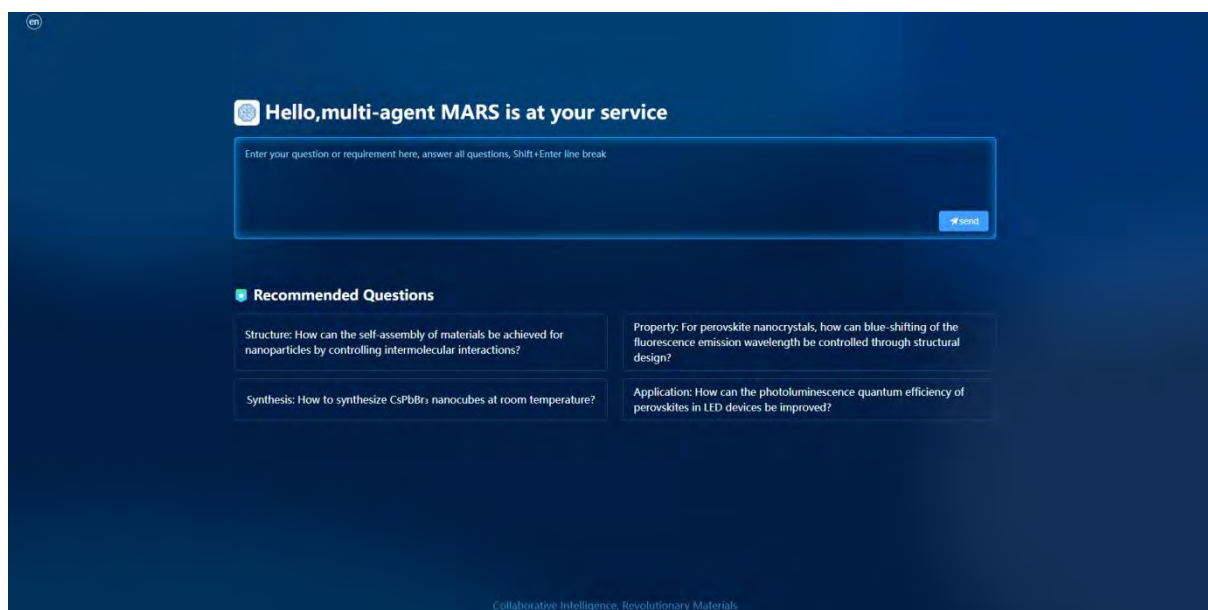


Figure S1. Web Application homepage UI. Interface for researchers and MARS multi-agent systems to interact.



Figure S2. Video Surveillance. Used for remotely observing robotic platforms conducting experiments.

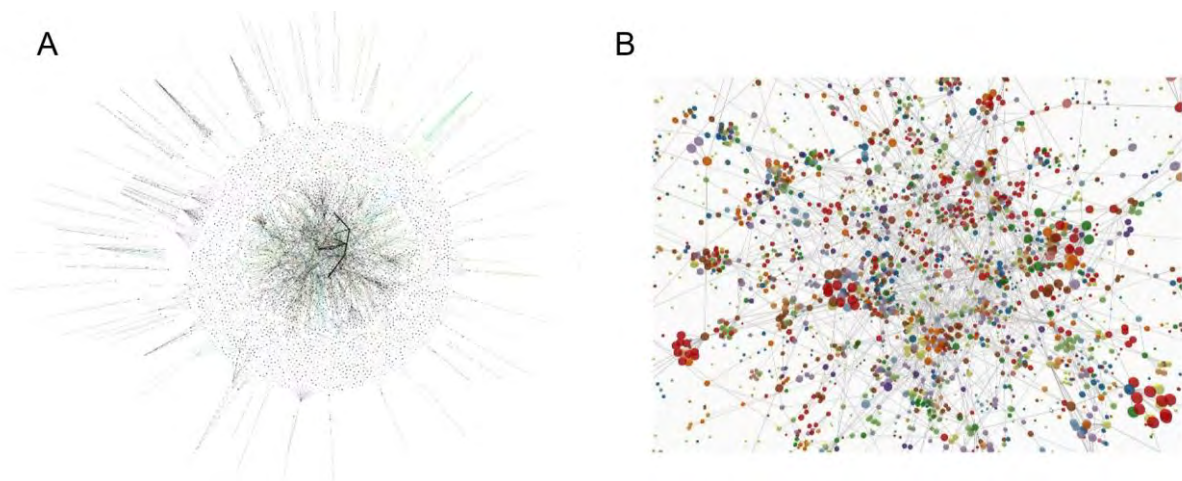


Figure S3. GraphRAG Visualization. (A) Overview of the Knowledge graphs. (B) A zoomed-in view of the graph.

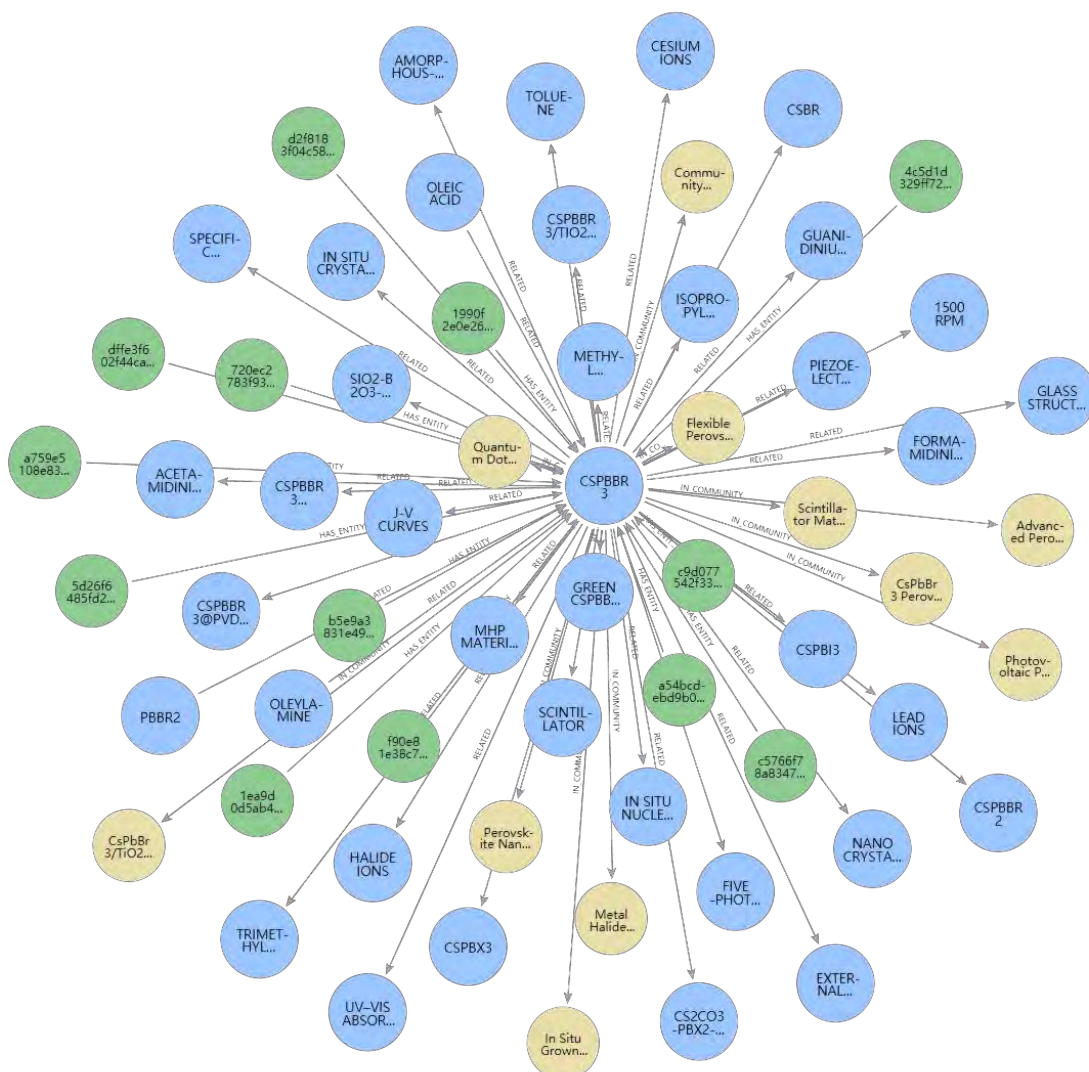


Figure S4. The knowledge graph centered around "CsPbBr₃" derived from random sampling for randomly chosen concepts from the global knowledge graph.

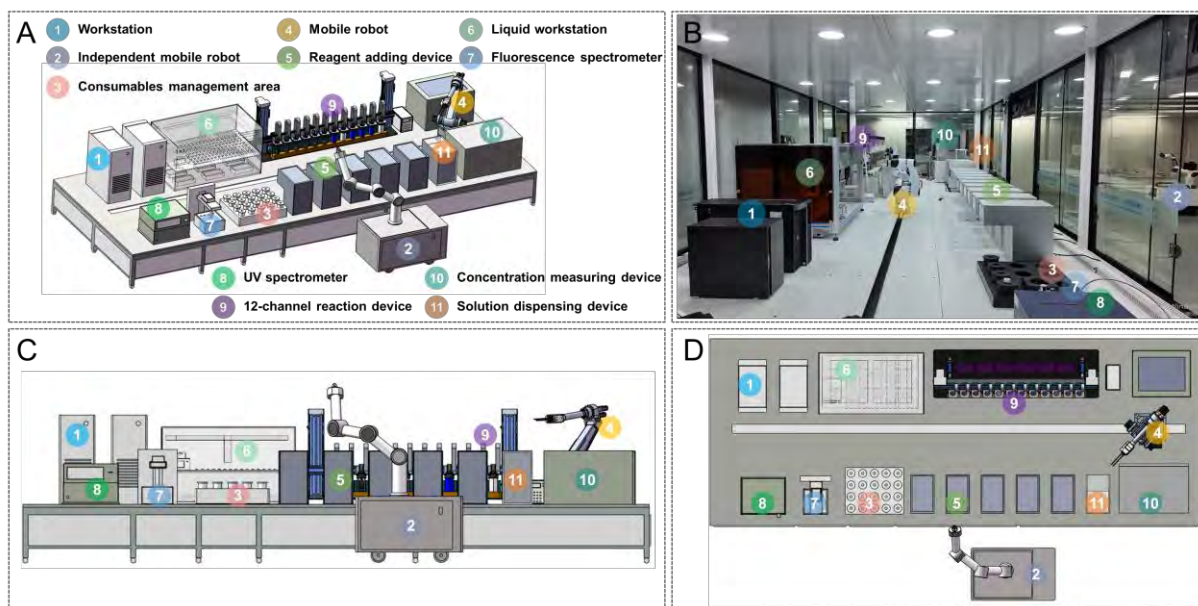


Figure S5. Overview of the second-generation experimental robotic platform for material synthesis and a typical CsPbBr_3 nanocrystals synthesis process. (A) Schematic diagram of the platform structure, including key components: 1. Automated platform workstation (the control center); 2. Independent mobile robot for reagent transport; 3. Consumables management area; 4. Mobile robot for experimental operations; 5. High-precision reagent adding device; 6. Liquid workstation; 7. Fluorescence spectrometer; 8. UV spectrometer; 9. 12-channel reaction device; 10. Concentration measuring device; 11. Solution dispensing system. (B) Photograph of the fully integrated experimental platform. (C) Front view of the platform structure. (D) Top view of the platform structure.



Figure S6. Unified control of middleware homepage UI. Real-time status monitoring of equipment including robots, powder and liquid dispensers, as well as recipe information.

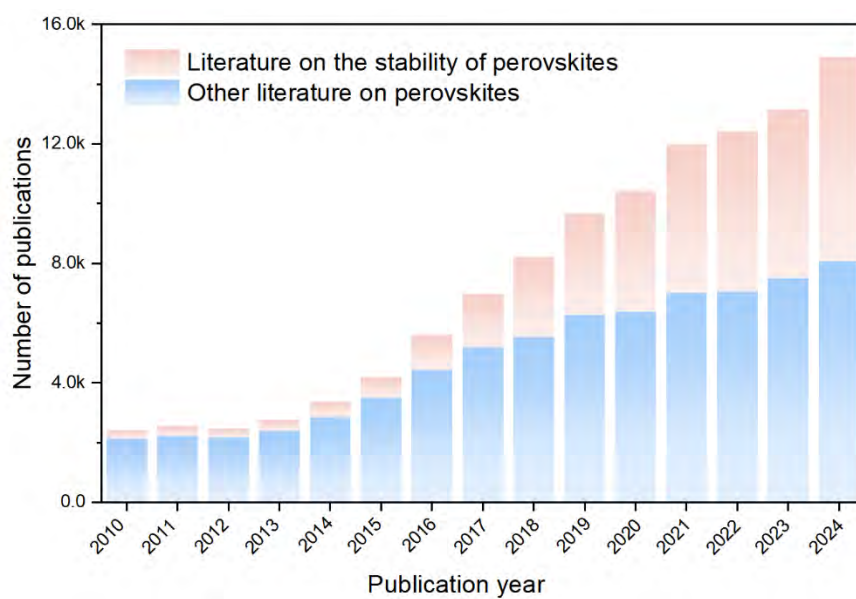


Figure S7. Research trends in perovskite stability based on Web of Science data.

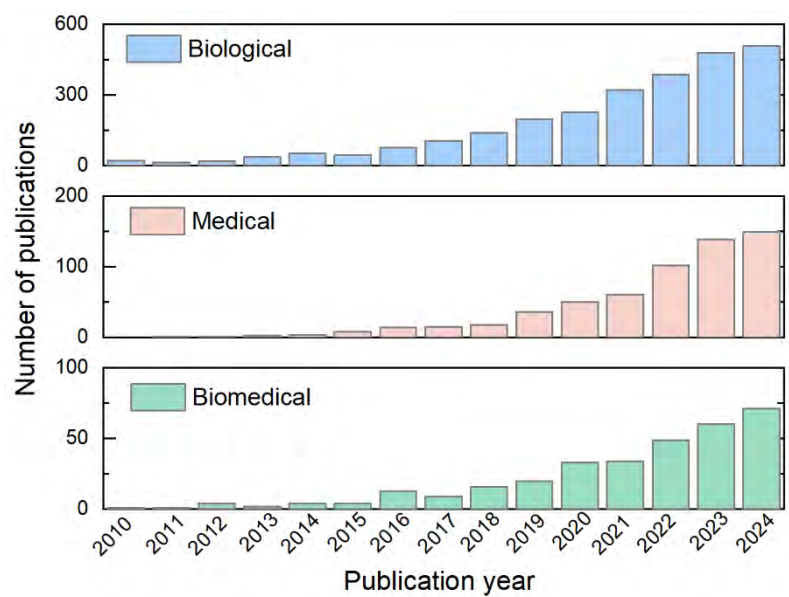


Figure S8. Research trends in perovskite biomedical based on Web of Science data.

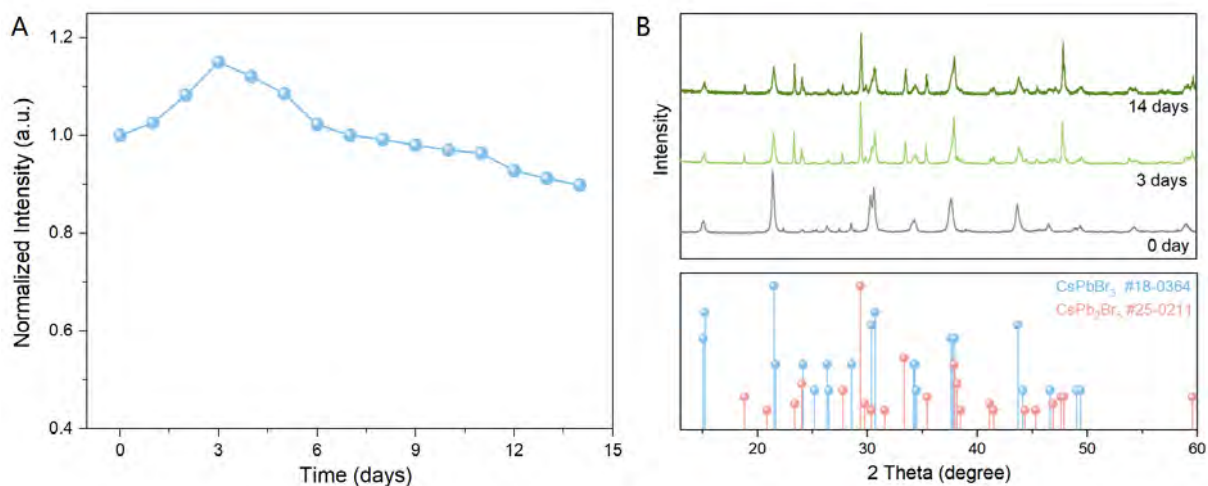


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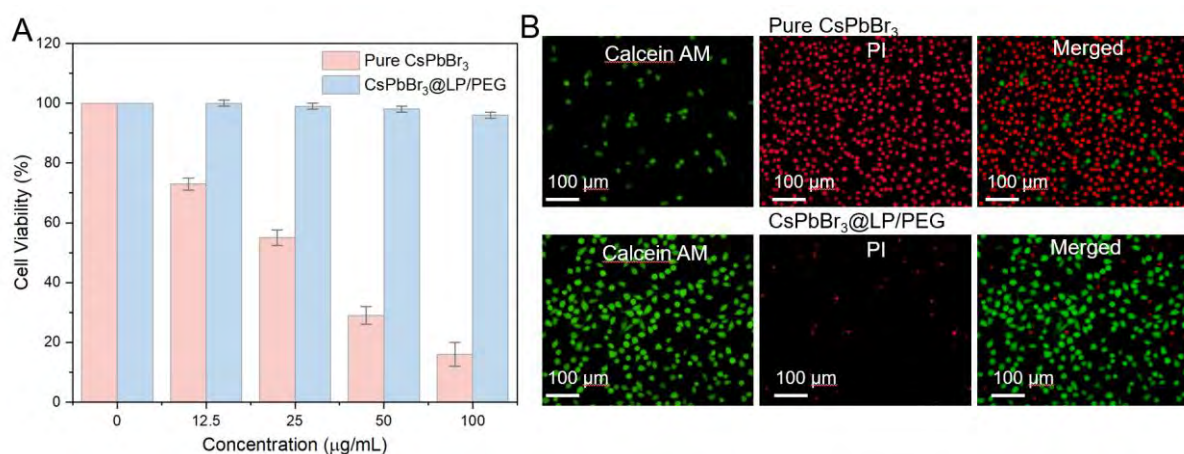


Figure S10. Biocompatibility evaluation results. (A) Cell viability of MCF-7 cells incubated with CsPbBr₃ and CsPbBr₃@LP/PEG for 24 h using CCK8 method; (B) Confocal fluorescence images of CsPbBr₃ and CsPbBr₃@LP/PEG treated and live/dead staining of MCF-7 cells for 24 h using Calcein-AM/PI Double Stain Kit.

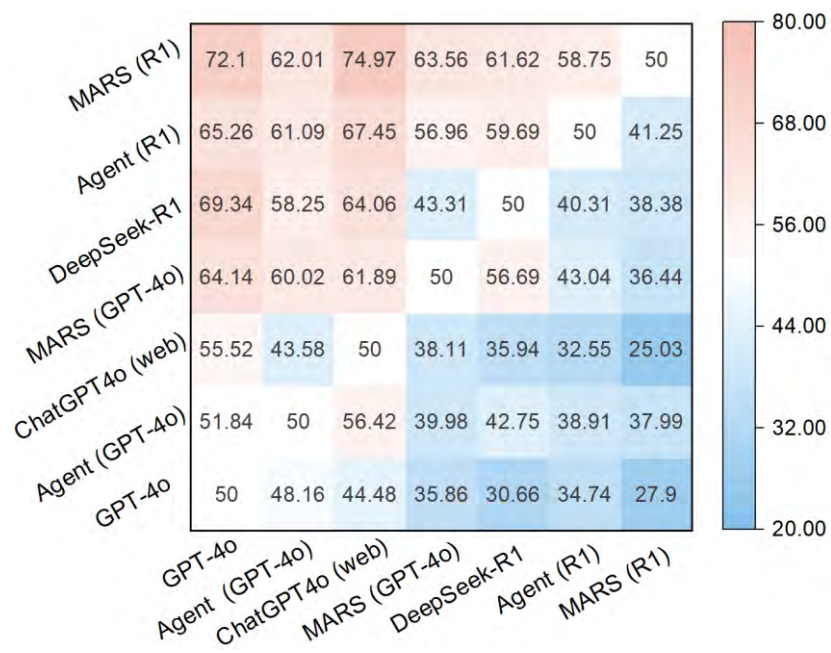


Figure S11. Pairwise win rate between a subset of models (50% win rate for self-comparison, not tested).

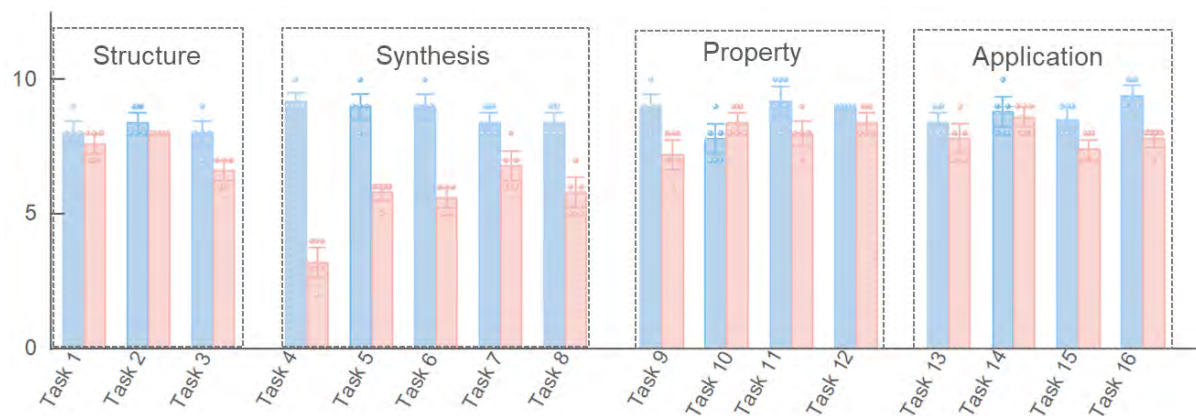


Figure S12. Performance comparison between MARS and GPT-4o evaluated by researchers (n=5).

Note S1: Code implementation

1.1 Core code of MARS

The core code of MARS is a multi-agent system implemented based on the Autogen^[S1] framework, which provides flexible and efficient tools for building complex multi-agent collaborations. The core code is located in the backend folder of the Github Repo containing implementations of key modules such as multi-agent definitions, RAG construction, ELO comparison agents, and GraphRAG^[S2]. These modules collectively form a complete multi-agent backend system that supports end-to-end operations from task planning to execution.

1.2 Tools

All self-implemented tool code can be found in the backend/tools.py file of the GitHub repository, while the tool server code based on FastAPI(<https://fastapi.tiangolo.com/>) and Dify workflow (<https://dify.ai>). It should be noted that some tools are not self-implemented, including: Code Sandbox, Python REPL, Matplotlib, Optuna, Matminer, and Scikit. Among them, the four libraries Matplotlib(<https://matplotlib.org/>), Optuna(<https://optuna.org/>), Matminer(<https://hackingmaterials.lbl.gov/matminer/>), and Scikit(<https://scikit-learn.org/>) are installed as additional Python packages in the Python REPL environment.

1.3 Message passing

We utilized the AutoGen framework to orchestrate this multi-agent system, defining two primary message-passing mechanisms:

1. Forward Context: Except for the task planning department, the messages received by other departments include system messages, instructions, and context. System messages are predefined, while the task planning department is responsible for assigning instructions, and the messages exchanged between other departments are handled as contextual information.
2. Feedback Information: Message passing between departments heavily relies on contextual information. When the feedback optimization department proposes optimization suggestions, the information is passed backward. Additionally, the feedback mechanism requires the knowledge retrieval module to persistently store experimental logs, regardless of the success or failure of the experiments.

When a user issues an instruction regarding automated material synthesis research, the task planning department comprehensively coordinates the activities of all agent departments. Specifically, the task planning department is responsible for sending instructions to other departments and receiving their responses. Upon receiving instructions from the task planning department, each agent department processes the task and responds to both the task planning department and the next relevant agent department. The complete automated material synthesis research process requires close collaboration among all agent departments to ensure efficient completion of the workflow.

Note S2: Web application

To better demonstrate the process of multi-agent task completion in the MARS system, we developed a Web demonstration application based on Vue.js. This application not only provides an intuitive user interface but also integrates rich functional modules, including task progress visualization, agent status monitoring, and history query. Users can access the application via a browser to view the real-time operation of the multi-agent system.

Additionally, to enhance user experience, we integrated two network cameras based on the RTSP protocol into the Web application for real-time observation of robot actions. By embedding video streams, users can more clearly understand the performance of robots in real-world scenarios and analyze task execution results in conjunction with system logs. The relevant code for the Web application is located in the frontend folder of the Github Repo, which includes the design of front-end pages, API call logic, and implementation details of video stream processing.

Note S3: Knowledge database construction

3.1 Data collection

(1) Academic paper collection

A high-quality academic paper database was assembled to construct our knowledge base, focusing on materials systems relevant to our laboratory research (nanomaterials, such as perovskites and gold nanorods). Three complementary approaches were employed:

- i. Literature collections accumulated by our research team over years of study, including PDF documents managed through the Zotero reference management system;
- ii. Systematic expansion of the database through Web of Science searches using keywords and seminal papers recommended by team members with domain expertise, followed by batch downloads from high-impact journals through institutional library subscriptions. This combined approach resulted in a comprehensive local database containing approximately 2,000 materials science publications.

Due to copyright restrictions, the full-text documents or vectorized database content cannot be publicly shared on platforms like GitHub.

(2) Book resources

Authoritative materials science textbooks and monographs were incorporated to complement the specialized knowledge from research papers with foundational theories and systematic concepts. These were selected based on recommendations from experienced laboratory members, including classics such as "Nanomaterials: Synthesis, Properties and Applications" and "Perovskite Materials - Synthesis, Characterisation, Properties and Applications". These resources were legally obtained through institutional access to electronic book databases via our university and the Chinese Academy of Sciences libraries.

(3) Web resource integration

Recognizing that a static local knowledge base cannot fully cover all emerging research areas, a dynamic online retrieval mechanism was implemented. The system uses search engines like Bing to generate relevant queries based on user questions, retrieving links primarily to authoritative journal websites. Through institutional IP authentication, these pages are accessed using the library to extract displayed abstracts, figure captions, and key paragraphs in compliance with the publisher's terms of service. Importantly, large-scale web scraping or crawling operations are not performed. Instead, relevant content is accessed dynamically by MARS on demand during runtime, ensuring compliance with publisher policies while maintaining system responsiveness.

3.2 Corpus preprocessing

Specifically, to support the development of the hybrid RAG system, we compiled approximately 10,000 academic papers, several classic books, and web snapshots accumulated by our research group over the past decades, serving as the core data foundation for the system. To construct a corpus that is easy to process, we designed a data cleaning pipeline to uniformly convert all documents into a format suitable for knowledge base construction. The specific process is as follows:

(1) PDF standardization: First, we converted all documents, books, and web snapshots from various sources into PDF format to ensure consistency in subsequent processing.

(2) Markdown format conversion: Next, we used the MinerU^[S3] tool to convert all PDF documents into Markdown format. MinerU not only extracts text from the documents but also automatically crops images and tables from the documents, replacing them with unique URL links. The benefit of this operation is that all documents are transformed into pure text corpus, facilitating subsequent processing.

(3) Document natural segmentation: The traditional method for constructing knowledge bases uses fixed-length text segmentation with overlaps between segments, which has been proven to be less effective. Thanks to the well-organized structure of scientific research papers, we adopt an advanced approach that combines natural segmentation with parent-child retrieval to construct the knowledge base. Specifically, for each paper, we use the heading symbol "#" in Markdown syntax to split the document, obtaining M natural chunks. Then, we treat each natural chunk as a parent text block and further divide it into N child sentences using the newline character "\n". These child sentences serve as indexes for the parent text block. The advantage of this approach is that it maximally preserves semantic continuity through natural segmentation while ensuring semantic coherence in retrieval results through parent-child retrieval.

3.3 Database establishment

After completing the document format conversion, we needed to further transform the document segments into structured data suitable for retrieval and inference. To achieve this, we processed each text chunk as follows:

(1) Embedding generation: We first fed the text chunks into the gte-Qwen2-1.5B-instruct^[S4] embedding

model, which can handle text chunks of up to 32K in length, generating corresponding text embeddings. These embeddings provide the basis for subsequent vector retrieval.

(2) Keyword extraction: At the same time, we used a keyword extraction algorithm to identify important keywords from the text, which are used to optimize and assist the BM25 algorithm retrieval (as shown in Equation 1).

$$Score(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{avgdl}\right)} \quad (1)$$

In the formula, D represents the document, Q represents the query, q_i refers to the i -th term in query Q . The $IDF(q_i)$ measures the importance of term q_i , indicating how rare or significant the term is across the entire corpus. $f(q_i, D)$ represents the term frequency (TF) of q_i in document D , which is the number of times q_i appears in the document. The parameter k_1 is used to adjust the term frequency, controlling how term frequency influences the score, and typically ranges between 1.2 and 2.0. The parameter b is used to normalize document length, controlling the effect of document length on the score, with typical values between 0 and 1, and often set to 0.75. The symbol $|D|$ indicates the length of document D , which refers to the total number of words in the document, while $avgdl$ represents the average document length within the corpus. Finally, n represents the total number of terms in the query Q .

(3) Knowledge graph construction: To enhance the global connectivity of the knowledge base, we used the GraphRAG module to construct a knowledge graph for the documents, storing the extracted entity-relation-entity triples for each text segment in the Neo4j graph database(<https://neo4j.com/>). The output of this step includes embeddings (stored in the weaviate vector database), keywords, and entity-relation-entity triples (stored in the graph database), which correspond to the three retrieval methods: vector retrieval, keywords retrieval, and knowledge graph retrieval.

Note S4: An instance of hybrid retrieval result

4.1 Hybrid retrieval process

When a scientist agent receive a sub-task from Admin of Scientist group, and the scientist agent call the tool of hybrid retrieval, the hybrid RAG system processes the input through embedding, keyword, and knowledge graph. The specific steps are as follows:

(1) **Similarity retrieval**: The embedding of the agent input is first sent to the weaviate vector database for similarity retrieval, filtering out multiple text segments with similarity scores greater than 0.5. Similarity retrieval generally uses cosine similarity.

(2) **Keyword retrieval**: Next, the system uses the BM25 algorithm to input the keywords from the agent query into the vector database, again filtering out multiple text segments with similarity scores greater than 0.5.

(3) **Weight calculation and filtering**: To integrate the results from both retrieval methods, we assign weights of 0.7 and 0.3 to the results from embedding-based retrieval and keyword-based retrieval, respectively. Then, we calculate the weighted average score for the retrieved text segments and filter out multiple document segments with a weighted score greater than 0.7. Since there is no standard value for retrieval weights in the implementation of RAG, all values mentioned in this article are manually selected through testing.

Although the document segments with high similarity scores are somewhat correlation to the agent's query, to further enhance the accuracy of the retrieval results, we introduced a rerank model (gte-multilingual-reranker-base^[S5]) to reorder these document segments. The purpose of this step is to ensure that the retrieved document segments are not only similar but also highly relevant to the agent's specific query. And then, the system selects the top 10 document segments with the highest similarity and correlation as the optimal retrieval results. It is worth noting that the similarity scores and weight values are empirically determined and may need adjustment according to the content of different knowledge bases during actual use.

For the document segments retrieved based on embeddings and keywords, they often originate from different documents, which may lead to a lack of a holistic knowledge perspective when answering agent queries. To address this issue, GraphRAG constructs a unified knowledge graph for all documents and performs a global search based on the agent's query, generating a topic-level response (called community report in GraphRAG). This knowledge segment acts as a supplement to the vector retrieval results, providing a more comprehensive perspective and contextual support. Finally, we return the document segments along with the community reports generated by GraphRAG as retrieval results.

4.2 Retrieval result

4.2.1 Vector & keywords

```
[
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```


size, shape, and composition, which is readily manipulated over the visible spectral region from 380 to 693 nm. Furthermore, the shape dependent photoluminescence decay lifetime of the CsPbX₃ nanocrystals has been determined to be several to tens to hundreds of nanoseconds. \n\nFurther work to explore other optical properties and technological applications is currently underway. We envision that the shape controlled CsPbX₃ perovskites nanocrystals will find widespread use in applications, such as lasing, light emitting diodes, photovoltaics, solar concentrators, and photon detection. By the suitable choice of sources and/or synthetic parameters, it is reasonable to expect that the present methods can be extended to the synthesis of other perovskite nanocrystals like AMX₃ (A = Rb⁺, Cs⁺; M = Ge²⁺, Sn²⁺; X=F⁻, Cl⁻, Br⁻, I⁻, or combinations thereof) as well.', 'metadata': None, 'embedding': None, 'score': 0.7265179200392623}, {'id': '7c634324-9b33-4645-ad80-d24802ad4a67', 'title': 'Water-Assisted Size and Shape Control of CsPbBr₃ Perovskite Nanocrystals.md', 'content': 'Synthesis of CsPbBr₃ NCs\nAs illustrated in **Figure S1**, the synthesis of CsPbBr₃ NCs was carried out at room temperature by the LARP approach, through injection of 1 mL of precursor mixture into 10 mL of toluene under vigorous stirring (Figure S1b). The precursor was prepared by mixing CsBr (0.4 mmol), PbBr₂ (0.4 mmol), oleylamine (OLA, 0.5 mL), and oleic acid (OA, 1 mL) in DMF (10 mL) (Figure S1a). Both DMF and toluene were dried from any remaining traces of water prior to use (DMF by adding KnNa12 n[(AlO₂)₁₂(SiO₂)₁₂], and toluene via distillation). When investigating the influence of water added to the "bad solvent", 1 mL of the dry precursor was injected into a mixture of toluene and water (10 mL + x μL, 0≤x≤60) (Figure S1c). When investigating the influence of water added to the precursor, 1 mL solution taken from the mixture of 1 mL dry precursor and x μL (0≤x≤60) water was injected into dry toluene (Figure S1d).\n\n![(2_image_0.png)(2_image_0.png)]', 'metadata': None, 'embedding': None, 'score': 0.7228831349343372}]

4.2.2 GraphRAG

n.name	relationship	m.name	m.description
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CSPBBR3	[{:RELATED {human_readable_id: 170, text_unit_ids: [1990f2e0e264ec682771be9ddd433f7], description: OA acts as a surface ligand for CsPbBr ₃ , rank: 49, weight: 6.0, id: d7777a3652d84e74a66536c38f803180}]	OLEIC ACID	OLEIC ACID, commonly abbreviated as OA, is an organic acid that plays a pivotal role in the synthesis of various advanced materials. It serves as a ligand in the production of perovskite nanocrystals (NCs), including Cs ₃ Bi ₂ Br ₉ quantum dots (QDs) and CsPbBr ₃ @PVDF composite fibers. OLEIC ACID is also utilized in the preparation of cesium oleate stock solutions, which are critical precursors in material synthesis. As a surface ligand, it is employed in the synthesis process of perovskite materials and quantum dots, contributing to the structural stability and optical properties of the resulting nanomaterials. Its use as an organic solvent in the synthesis of QDs highlights its multifunctional nature in materials science. Characterization techniques such as ligand exchange or analysis of the quantum confinement effects in nanomaterials often involve the use of OLEIC ACID. Current trends in materials science, particularly in the development of optoelectronic devices, leverage OLEIC ACID for fabricating high-quality perovskite films and composites due to its ability to modify the surface properties of nanocrystals.

CSPBBR3	[[:RELATED {human_readable_id: 197, text_unit_ids: [1990f2e0e264ec682771be9ddd433f7], description: PbBr2 is a source of lead ions for CsPbBr3, rank: 55, weight: 8.0, id: 2072c56ec3a544998e962ffb7c9d2e12}]]	PBBR2	Lead bromide (PBBR2) is a critical compound in the field of perovskite materials research, serving as a precursor and reagent in the synthesis of various perovskite structures, including CsPbBr3 quantum dots (QDs) and MAPbBr3 perovskites. It acts as a source of lead ions, which are essential for the formation of nanocrystals, nanocrystal composites such as CsPbBr3@PVDF composite fibers, and perovskite films. Lead bromide's role in synthesis processes extends to providing the lead component required for the quantum confinement effects observed in perovskite quantum dots, which are characterized by their size and shape-dependent properties. In the synthesis of CsPbBr3 QDs, lead bromide is not only a precursor but also a reagent that reacts under specific conditions, such as temperature and solvent choice, to produce high-quality nanocrystals. These synthesis conditions are crucial for controlling the size, shape, and crystallinity of the resulting QDs, which in turn affect their optical and electronic properties. Lead bromide's contribution to the perovskite community addresses current challenges in materials science and engineering, including the development of stable and efficient optoelectronic materials. Its use in the creation of perovskite quantum dots and composites highlights a trend toward the miniaturization and functionalization of materials for advanced technological applications.
CSPBBR3	[[:RELATED {human_readable_id: 227, text_unit_ids: [1990f2e0e264ec682771be9ddd433f7], description: CsBr is a source of cesium ions for CsPbBr3, rank: 52, weight: 8.0, id: 040d137ba5934b2f8dda7055058a768d}]]	CSBR	CSBR, commonly known as cesium bromide, is a critical reagent in the synthesis of various advanced materials, particularly in the field of perovskites and nanocrystals. It serves as a reliable source of cesium ions, which are pivotal for the formation of perovskite structures, including films and quantum dots. CSBR's role in the synthesis process extends to the production of Cs2AgBiBr6, where it acts as a cesium ion supplier, facilitating the creation of a composite material with distinctive properties. Its importance in nanocrystal synthesis is underscored by its function as a cesium source, affecting the size and quantum confinement effects of the resulting nanocrystals. Characterization techniques such as X-ray diffraction and photoluminescence spectroscopy are likely employed to assess the quality and properties of the synthesized materials, ensuring their suitability for applications in optoelectronics and solar energy technologies. Current trends in materials science, particularly in the quest for high-performance optoelectronic materials, highlight the significance of CSBR in addressing challenges related to the development of stable and efficient perovskite-based devices.

CSPBBR3	[.RELATED {human_readable_id: 228, text_unit_ids: [1990f2e0e264ec682771be9ddd433f7], description: Toluene acts as an antisolvent to induce the formation of CsPbBr₃, rank: 55, weight: 8.0, id: 1f819a6defec47a28fec63e4ddae382}]	TOLUENE	Toluene is an organic solvent frequently utilized in the synthesis of advanced materials, recognized for its role in controlling the formation of nanocrystals and perovskite structures. In the synthesis of CsPbBr₃ nanocrystals (NCs), perovskite materials, and Cs₃Bi₂Br₉ quantum dots (QDs), toluene acts as an antisolvent to induce precipitation and facilitate the growth of these materials. It is also employed in the synthesis of CsPbBr₃@PVDF composite fibers and modified SnO₂ nanoparticles, where it aids in dissolving precursors and inducing the formation of the desired nanostructures. The instability of toluene is leveraged to control reaction rates, as seen in its use to prevent premature reaction of perovskite precursors. Moreover, toluene is used in the deposition of perovskite films, which is critical for fabricating optoelectronic devices. Its ability to disperse precipitates is beneficial for the re-dispersion of NCs and the creation of uniform perovskite films. Characterization processes also benefit from toluene, as it is used in the dissolution of materials like PbBr₂ for the synthesis of perovskite films and in the re-dispersion of precipitated materials for further analysis. Toluene's role in dissolving compounds such as PTAA and titanium precursors underscores its versatility in the synthesis process. In summary, toluene is a key solvent in materials science, particularly in the synthesis of perovskite-related materials and nanocrystals, where it influences reaction dynamics, crystal growth, and material dispersion. Its use is integral to controlling the properties of synthesized materials, which is crucial for their potential applications in various technological and industrial fields.
CSPBBR3	[.RELATED {human_readable_id: 220, text_unit_ids: [720ec2783f937ee391c0522f6c772d3b], description: CsPbBr₃ is a precursor for CsPbBr₃ NCs, rank: 61, weight: 5.0, id: cee87e59f5a94f178a21066190d044ac}]	CSPBBR3 NCS	CSPBBR3 NCS refers to cesium lead bromide nanocrystals, which are a type of perovskite material known for their distinct optoelectronic properties. These nanocrystals serve as the intermediate product in the synthesis of perovskite materials and are characterized by their formation process prior to the application of various coatings and embedding techniques. CSPBBR3 NCS nanocrystals are synthesized and then coated with SiO₂, TiO₂, or ZnO₂, or embedded within PVDF fibers to enhance stability and compatibility for specific applications. Additionally, these nanocrystals are integrated during the perovskite film deposition process, contributing to the film's properties. Before encapsulation in polystyrene (PS), cesium lead bromide perovskite quantum dots are formed, which may exhibit quantum confinement effects due to their size and shape. The synthesis of these materials typically involves careful control of reaction conditions, including precursor selection and temperature, to achieve the desired nanocrystal properties. Characterization techniques such as X-ray diffraction, photoluminescence spectroscopy, and transmission electron microscopy are employed to understand the crystal structure, optical properties, and morphology of the nanocrystals, respectively. These cesium lead bromide nanocrystals and quantum dots hold potential applications in optoelectronic devices, such as solar cells and light-emitting diodes, where their size and composition can be fine-tuned to optimize performance. Current trends in materials science and engineering focus on overcoming challenges related to the environmental stability and toxicity of perovskite materials, which is where the encapsulation and coating processes play a critical role.

Note S5: Comparison of responses from different agent

5.1 System prompt

MARS, GPT4o, o3-mini, and Llama3.1-70B use the same prompt.

You are a senior materials scientist. Your task is to answer users' questions related to material synthesis. User queries often include more than just requirements for synthesis protocols, and you need to use your judgment accordingly.

(1) When a user question involves requirements for material synthesis, you need to:
Utilize your expertise to carefully identify the user's needs and select the most suitable methods and substances to ensure accurate synthesis of the target material. Your synthesis protocol should be divided into the following sections:

1. **Synthesis Conditions**: Specify the environmental or operational conditions required for the synthesis of the final material, such as temperature, pressure, pH, solvents, etc.
2. **Materials & Amounts Required**: List the initial substances required for the synthesis of the final product and their specific amounts, including any catalysts or solvents. Ensure that all solutions are in milliliters (mL) and all solids are in milligrams (mg).
3. **Equipment & Specifications**: List in detail the apparatus required for the synthesis and their technical specifications (e.g., capacity, temperature control range), including any special reactors, stirrers, or measuring tools.
4. **Synthesis Sequence**: Clarify the sequence of synthesis of the precursors and the final material, describing the quantities of materials required for each step, the sizes of the apparatus, and the operating procedures (e.g., mixing, heating, cooling).
5. **Step-by-Step Requirements for Final Material Synthesis**: Break down the synthesis steps into sub-steps, and specify the quantities of reagents involved, the size of the equipment (e.g., lab-scale or industrial scale), and the detailed procedure for each sub-step.
6. **Characterization of Synthesized Material**: Describe the methods used to analyze and confirm the structure, purity, or other properties of the synthesized material, which may include spectroscopic, chromatographic, or microscopic techniques.
7. **Additional Considerations**: Highlight other relevant factors such as safety measures, scalability challenges, storage requirements, or environmental impact.

Create a comprehensive experimental protocol with the aim of producing an accurate, detailed, and executable synthesis plan in an actual laboratory setting.

(2) When a user question involves requirements other than material synthesis, you should organize your response as thoroughly and comprehensively as possible.
Remember: Do not include this instruction in your response.

5.2 MARS

1. Synthesis Conditions

- **Temperature**: Room temperature (~25°C)
- **Pressure**: Atmospheric pressure
- **pH**: Neutral
- **Solvent**: Anhydrous dimethylformamide (DMF) and anhydrous toluene

2. Materials & Amounts Required

- **CsBr**: 0.4 mmol (approximately 85.1 mg)
- **PbBr₂**: 0.4 mmol (approximately 146.8 mg)
- **Oleylamine (OLA)**: 0.5 mL
- **Oleic Acid (OA)**: 1 mL
- **Anhydrous DMF**: 10 mL
- **Anhydrous Toluene**: 10 mL

3. Equipment & Specifications

- **Stirrer**: Capable of high-speed stirring (recommended 1000 rpm)
- **Syringe**: 1 mL capacity
- **Beaker**: 50 mL capacity

4. Synthesis Sequence

1. **Preparation of Precursor Solution**:

- Dissolve 0.4 mmol of CsBr (approximately 85.1 mg) and 0.4 mmol of PbBr₂ (approximately 146.8 mg) in 10 mL of anhydrous DMF.
- Add 0.5 mL of oleylamine (OLA) and 1 mL of oleic acid (OA), stirring the mixture until a homogeneous precursor solution is formed.

2. **Nanocrystal Formation**:

- Inject 1 mL of the precursor solution into 10 mL of anhydrous toluene, maintaining high-speed stirring (1000 rpm).
- Continue stirring for 5 minutes until nanocrystals are formed.

5. Step-by-Step Requirements for Final Material Synthesis

1. **Preparation of Precursor Solution**:

- In a 50 mL beaker, add 0.4 mmol of CsBr (approximately 85.2 mg) and 0.4 mmol of PbBr₂ (approximately 147.2 mg).
- Add 10 mL of anhydrous DMF and stir until completely dissolved.
- Add 0.5 mL of oleylamine (OLA) and 1 mL of oleic acid (OA), and continue stirring until homogeneous.

2. **Nanocrystal Formation**:

- Using a 1 mL syringe, withdraw 1 mL of the precursor solution.
- Rapidly inject the precursor solution into 10 mL of anhydrous toluene, maintaining high-speed stirring (1000 rpm).
- Continue stirring for 5 minutes until nanocrystals are formed.

6. Characterization of Synthesized Material

- **Transmission Electron Microscopy (TEM)**: For observing the morphology and size of the nanocrystals.
- **X-ray Diffraction (XRD)**: For determining the crystal structure.
- **UV-Visible Spectroscopy (UV-Vis)**: For measuring optical properties.
- **Photoluminescence Spectroscopy (PL)**: For measuring photoluminescence characteristics.

7. Additional Considerations

- **Safety Measures**: Wear gloves and safety goggles during the process to avoid exposure to harmful chemicals.
- **Ventilation**: Conduct the procedure in a well-ventilated area to avoid inhalation of harmful vapors.
- **Scalability Challenges**: When scaling up the synthesis, ensure consistent stirring and reaction time for uniformity.
- **Storage Requirements**: Store the synthesized nanocrystals in a dark, low-temperature environment to prevent degradation.

5.3 GPT-4o

Synthesis of CsPbBr₃ Nanocubic Crystals (Nanocrystals) at Room Temperature

The synthesis of CsPbBr₃ nanocubic crystals at room temperature typically involves solution methods, with a commonly used method being a variant of the hot injection technique conducted at room temperature. Below are the detailed synthesis steps:

1. Synthesis Conditions

- **Temperature**: Room temperature (approximately 25°C)
- **Duration**: 1-2 hours
- **Atmosphere**: Air or inert gas (e.g., nitrogen)
- **Stirring Speed**: Moderate speed

2. Materials & Amounts Required

- **Cesium bromide (CsBr)**: 0.1 M, 1 mL
- **Lead bromide (PbBr₂)**: 0.1 M, 1 mL
- **Cesium oleate (Cs-oleate)**: 0.1 M, 1 mL
- **Lead oleate (Pb-oleate)**: 0.1 M, 1 mL
- **Oleylamine (OAm)**: 2 mL

- **Oleic Acid (OA)**: 2 mL
- **n-Hexane**: 10 mL
- **Ethanol**: 10 mL

3. Equipment & Specifications

- **Beaker**: 50 mL capacity
- **Magnetic stirrer**: For evenly mixing the solution
- **Pipette**: For precise measurement of liquids
- **Centrifuge**: Speed of 3000-5000 rpm
- **Filter paper**: For filtration
- **Ultrasonic cleaner**: For dispersing nanocrystals

4. Synthesis Sequence

1. Preparation of Precursor Solution:

- In a 50 mL beaker, prepare 0.1 M CsBr and 0.1 M PbBr₂ solutions separately.
- Add 1 mL of 0.1 M Cs-oleate and 1 mL of 0.1 M Pb-oleate solution respectively.
- Add 2 mL oleylamine (OAm) and 2 mL oleic acid (OA), and stir well.

2. Mixing Reaction:

- Mix the CsBr solution and the PbBr₂ solution, and stir well.
- Let it stand at room temperature for 1-2 hours, during which CsPbBr₃ nanocubic crystals gradually form.

3. Separation and Purification:

- After the reaction, use a centrifuge at a speed of 4000 rpm for 10 minutes to separate the CsPbBr₃ nanocrystals.
- Wash the nanocrystals with n-hexane and ethanol to remove residual reactants and solvents.
- Use filter paper to collect the CsPbBr₃ nanocrystals.

5. Step-by-Step Requirements for Final Material Synthesis

1. Preparation of Precursor Solution:

- In a 50 mL beaker, prepare 0.1 M CsBr and 0.1 M PbBr₂ solutions separately.
- Add 1 mL of 0.1 M Cs-oleate and 1 mL of 0.1 M Pb-oleate solutions respectively.
- Add 2 mL oleylamine (OAm) and 2 mL oleic acid (OA), and stir well.

2. Mixing Reaction:

- Mix the CsBr solution and the PbBr₂ solution, and stir well.
- Let it stand at room temperature for 1-2 hours, during which CsPbBr₃ nanocubic crystals gradually form.

3. Separation and Purification:

- After the reaction, use a centrifuge at a speed of 4000 rpm for 10 minutes to separate the CsPbBr₃ nanocrystals.
- Wash the nanocrystals with n-hexane and ethanol to remove residual reactants and solvents.
- Use filter paper to collect the CsPbBr₃ nanocrystals.

6. Characterization of Synthesized Material

- **Transmission Electron Microscopy (TEM)**: To observe the morphology and size of the CsPbBr₃ nanocrystals.
- **X-Ray Diffraction (XRD)**: To confirm the crystal structure of the CsPbBr₃ nanocrystals.
- **Ultraviolet-Visible Spectroscopy (UV-Vis)**: To analyze the optical properties of the CsPbBr₃ nanocrystals.
- **Photoluminescence Spectroscopy (PL)**: To measure the luminescent properties of the CsPbBr₃ nanocrystals.

7. Additional Considerations

- **Safety Precautions**: Wear protective gloves and goggles when handling cesium bromide, lead bromide, and organic solvents to avoid skin contact and inhalation.
- **Scalability**: The room temperature solution method is suitable for laboratory-scale preparation of CsPbBr₃ nanocrystals, but reaction conditions need to be optimized to improve yield and consistency.
- **Storage Requirements**: CsPbBr₃ nanocrystals should be stored in a dry, dark environment to prevent degradation and aggregation.

By following the above steps, high-quality CsPbBr₃ nanocubic crystals can be successfully synthesized in the laboratory.

5.4 o3-mini

To synthesize **CsPbBr₃ nanocubes** at room temperature, a low-temperature solution-based method such as **hot injection** or **one-pot reaction** can be used. Here's a detailed experimental protocol for the synthesis:

1. Synthesis Conditions

- **Temperature**: Room temperature (approximately 25°C)
- **Solvents**: Non-polar organic solvent like **toluene** or **hexane**
- **Reaction Time**: Approximately 10–30 minutes
- **pH**: Neutral (no significant adjustment needed)
- **Atmosphere**: Inert atmosphere (argon or nitrogen) to avoid contamination from moisture or oxygen.

2. Materials & Amounts Required

- **Cesium bromide (CsBr)**: 200 mg (1 mmol)
- **Lead(II) bromide (PbBr₂)**: 466 mg (1 mmol)
- **Oleic acid (OA)**: 1 mL (used as a surfactant to control the size and shape)
- **Oleylamine (OAm)**: 1 mL (also acts as a stabilizer)
- **Toluene**: 20 mL (solvent for the reaction)
- **Hexane**: 10 mL (for washing the nanocubes)
- **Pre-synthesized CsPbBr₃ nanoparticle seeds (optional)**: 5–10 mg (if performing seed-mediated growth)

3. Equipment & Specifications

- **Reaction Flask**: 100 mL three-neck round-bottom flask (ensure it is clean and dry)
- **Magnetic Stirrer**: Capable of mixing the solution uniformly
- **Inert Gas Flow (Nitrogen or Argon)**: For purging the reaction environment (optional)
- **Heat Source**: Hot plate or oil bath, able to control temperature up to 120°C
- **Syringe**: 5 mL syringe for controlled injection
- **Glass Stir Bar**: To ensure efficient mixing
- **Thermometer**: For accurate monitoring of temperature
- **Centrifuge**: For washing and isolating the nanocubes
- **Fume Hood**: Ensure the experiment is performed under a well-ventilated fume hood to handle solvents safely.

4. Synthesis Sequence

1. Preparation of Precursor Solutions:

- **CsBr Solution**: Dissolve **200 mg** of CsBr in **5 mL** of toluene under stirring at room temperature.

- **PbBr₂ Solution**: Dissolve **466 mg** of PbBr₂ in **5 mL** of toluene. Add **1 mL** of oleic acid (OA) and **1 mL** of oleylamine (OAm) to the solution. Stir to dissolve completely.

2. Nanocube Formation:

- Mix the **CsBr solution** with the **PbBr₂ solution** in the reaction flask. Stir the mixture under nitrogen atmosphere.

- **Injection Step**: In a separate container, prepare a small amount of oleic acid and oleylamine in toluene (if using). Inject this solution into the reaction flask using a syringe.

- Stir the solution at room temperature for 10–30 minutes. The reaction will start to form the CsPbBr₃ nanocubes within this time frame. The color of the solution will change as the nanocubes are formed.

3. Isolation of Nanocubes:

- After the reaction time, stop stirring and allow the nanocubes to settle.

- Centrifuge the solution at **4000 rpm** for **5 minutes** to isolate the nanocubes from the supernatant.

- Wash the nanocubes with **10 mL** of hexane to remove excess oleic acid and oleylamine.

- Centrifuge again, and discard the supernatant. Repeat the washing step once more for high purity.

5. Step-by-Step Requirements for Final Material Synthesis

1. **Step 1**: Dissolve **CsBr** in toluene (5 mL) and **PbBr₂** in toluene (5 mL).
2. **Step 2**: Combine the CsBr and PbBr₂ solutions under stirring in the reaction flask. Ensure it's under inert atmosphere.
3. **Step 3**: Inject the oleic acid and oleylamine solution into the reaction flask.
4. **Step 4**: Stir for 10-30 minutes, allowing the CsPbBr₃ nanocubes to form.
5. **Step 5**: After completion, centrifuge the solution and wash the nanocubes with hexane.
6. **Step 6**: Dry the nanocubes in a vacuum oven if necessary.

6. Characterization of Synthesized Material

- **X-ray Diffraction (XRD)**: Confirm the crystal structure of CsPbBr₃.
- **Transmission Electron Microscopy (TEM)**: Verify the nanocube morphology and size distribution.
- **UV-Vis Spectroscopy**: Measure the absorption spectrum to confirm the optical properties of the nanocubes.
- **Photoluminescence (PL)**: To analyze the emission properties of the CsPbBr₃ nanocubes.
- **Dynamic Light Scattering (DLS)**: For particle size distribution.

7. Additional Considerations

- **Safety Measures**: Handle all chemicals, especially lead compounds, in a fume hood and use appropriate personal protective equipment (PPE), including gloves and goggles.
- **Scalability**: This procedure can be scaled up by increasing the quantities of precursors while maintaining similar solvent volumes. However, further optimization may be required for larger-scale syntheses.
- **Storage Requirements**: Store the CsPbBr₃ nanocubes in an airtight container, away from moisture and direct light to preserve their properties.
- **Environmental Impact**: Ensure that disposal of lead-containing wastes is handled according to proper environmental regulations.

5.5 Llama3.1-70B-Instruct

Synthesis Conditions:

- **Temperature**: Room temperature (approximately 20°C)
- **Pressure**: Atmospheric pressure
- **pH**: Neutral (pH 7)
- **Solvent**: Acetone and Dimethylformamide (DMF)

Materials & Amounts Required:

- **CsBr (Cesium Bromide)**: 100 mg
- **PbBr₂ (Lead Bromide)**: 150 mg
- **OA (Oleic Acid)**: 100 µL
- **OAm (Oleylamine)**: 100 µL
- **Acetone**: 10 mL
- **Dimethylformamide (DMF)**: 5 mL

Equipment & Specifications:

- **Glass test tube** (10 mL)
- **Glass stirring rod**
- **Ultrasonic cleaner**
- **Hot plate**
- **Thermometer**

Synthesis Sequence:

1. Prepare solutions of CsBr and PbBr₂: Dissolve CsBr and PbBr₂ in acetone separately to obtain 0.1 M solutions.
2. Prepare a mixed solution of OA and OAm: Mix OA and OAm to obtain a 0.1 M solution.
3. Synthesize CsPbBr₃ nanocubic crystals: Mix the CsBr solution and PbBr₂ solution, add the OA

and OAm mixed solution, and stir well.

4. Heating and cooling: Heat the mixed solution to 80°C, maintain for 10 minutes, and then cool to room temperature.

****Step-by-Step Requirements for Final Material Synthesis:****

1. Mix the CsBr solution and PbBr₂ solution, add the OA and OAm mixed solution, and stir well for 10 minutes.
2. Heat the mixed solution to 80°C, maintaining for 10 minutes.
3. Cool the mixed solution to room temperature and let it stand for 10 minutes.
4. Collect the CsPbBr₃ nanocubic crystals, wash twice with acetone, and dry at room temperature.

****Characterization of Synthesized Material:****

- ****X-Ray Diffraction (XRD) analysis****: To confirm the crystal structure of CsPbBr₃.
- ****Transmission Electron Microscopy (TEM) analysis****: To observe the morphology of CsPbBr₃ nanocubic crystals.
- ****Ultraviolet-Visible Spectroscopy (UV-Vis) analysis****: To measure the absorption spectrum of CsPbBr₃.

****Additional Considerations:****

- ****Safety measures****: Wear gloves and safety goggles during the process.
- ****Scalability challenges****: This synthesis method can be scaled for large-scale production.
- ****Storage requirements****: CsPbBr₃ nanocubic crystals should be stored in a dry environment, away from light.
- ****Environmental impact****: This synthesis method involves no toxic substances, resulting in minimal environmental impact.

5.6 Hallucination analysis

■ GPT4o			
Hallucination Type	Specific Error	Model Response (Note S8)	Error Analysis
Factual hallucination	Reagent misselection	-Cs-oleate: 0.1 M, 1 mL -Pb-oleate: 0.1 M, 1 mL	-Cs-oleate is specifically used for the hot-injection method, requiring preparation and use at high temperatures (120-180°C) -Pb-oleate is rarely used for perovskites
	Solubility misconception	In a 50 mL beaker, prepare 0.1 M CsBr and 0.1 M PbBr ₂ solutions separately.	-"0.1 M solid CsBr/PbBr ₂ " ignores dissolution requirement in polar aprotic solvents like DMF/DMSO.
	Numerical fabrication	Add 2 mL oleylamine (OAm) and 2 mL oleic acid (OA)	Arbitrary 2mL volume. Precise OAm/OA ratio required for size/surface control. Excess OAm (melting point ~25°C) causes precursor solution turbidity → heterogeneous system → disrupts nucleation/growth.
	Procedural omission	Let it stand at room temperature for 1-2 hours, during which CsPbBr ₃ nanocubic crystals gradually form.	Missing the core step of: rapid injection of the precursor solution into an anti-solvent (e.g., toluene, chloroform) to precipitate perovskite nanocrystals
	Stability misassessment	Wash the nanocrystals with n-hexane and ethanol to remove residual reactants.	CsPbBr ₃ nanocrystals are unstable in ethanol, leading to degradation. Requires non-polar solvents (chloroform/toluene).

■ Llama3.1-70B			
Hallucination Type	Specific Error	Model Response (Note S8)	Error Analysis
Factual hallucination	Numerical fabrication	- CsBr: 100 mg - PbBr ₂ : 150 mg	Mass ratio error: CsBr(100mg): PbBr ₂ (150mg) ≠ 1:1 stoichiometry (correct: 212.8mg:367mg).
	Solubility misconception	Dissolve CsBr and PbBr ₂ in acetone separately to obtain 0.1 M solutions.	Acetone cannot dissolve CsBr/PbBr ₂ . Requires DMF/DMSO.
	Numerical fabrication	Mix OA and OAm to obtain a 0.1 M solution.	(Fundamental concentration unit error) Mixing pure OA/OAm ≠ 0.1M solution. OA/OAm are liquids, and physical mixing cannot yield molar concentration.
	Omission of key step	Let it stand at room temperature for 1-2 hours, during which CsPbBr ₃ nanocubic crystals gradually form.	Missing the core step of: rapid injection of the precursor solution into an anti-solvent (e.g., toluene, chloroform) to precipitate perovskite nanocrystals
Faithfulness hallucination	Constraint violation	Heat the mixed solution to 80°C, maintaining for 10 minutes.	Heating to 80°C → violates explicit "room temperature" requirement.

■ o3-mini			
Hallucination Type	Specific Error	Model Response (Note S8)	Error Analysis
Factual hallucination	Numerical fabrication	-CsBr: 200 mg (1 mmol) - PbBr ₂ : 466 mg (1 mmol)	Mass Calculation Error CsBr:200mg (≠212.8mg/1mmol); PbBr ₂ :466mg (≠367mg/1mmol).
	Reagent misselection	Pre-synthesized CsPbBr ₃ nanoparticle seeds	"Pre-synthesized CsPbBr ₃ seeds" non-existent in actual room-temp protocols.
	Solubility misconception	Dissolve CsBr and PbBr ₂ in toluene (5 mL).	Toluene cannot dissolve CsBr/PbBr ₂ .
	Procedural omission	Stir the solution at room temperature for 10–30 minutes. The reaction will start to form the CsPbBr ₃ nanocubes.	Missing the core step of: rapid injection of the precursor solution into an anti-solvent (e.g., toluene, chloroform) to precipitate perovskite nanocrystals
	Stability misassessment	Wash the nanocubes with 10 mL of hexane to remove excess oleic acid and oleylamine.	Hexadecane washing only used in hot-injection (>150°C) → irrelevant to room-temp.
Faithfulness hallucination	Constraint violation	To synthesize CsPbBr ₃ nanocubes at room temperature, a low-temperature solution-based method such as hot injection or one-pot reaction can be used.	"Hot-injection for room-temp" impossible (requires >150°C). Fundamental contradiction.

Note S6: Automated experiment execution

6.1 Details of robotic platform

Our second-generation experimental robotic platform evolved from the first-generation robotic scientist platform, showcasing its capabilities in micro/nano materials synthesis. This experimental robotic platform represents a significant advancement in the integration of multi-agent systems and automated experimental control.

The central element is the automated platform workstation (1), which acts as the “brain” of the platform. This workstation is responsible for sending instructions to the various devices on the platform and executing the automated synthesis process. The platform includes one independent mobile robot (2), which is controlled by the workstation and primarily responsible for the preparation and transportation of reagents and consumables. The reagents and consumables are stored in a designated consumables management area (3), which holds both untreated and treated reagent bottles. A key feature of the platform is the mobile robot (4), also controlled by the workstation, which executes a wide range of experimental procedures within the platform. It performs critical tasks such as transporting reagents to different sections of the platform. The high-precision reagent adding device (5) enables the automated addition of powdered or liquid reagents. These are further processed at the liquid workstation (6), where tasks like pipetting, stirring, and heating reactions are carried out. After reactions are completed, the mobile robot transfers the samples for analysis. The platform is equipped with two analytical devices, including a fluorescence spectrometer (7) for fluorescence detection, and a UV spectrometer (8) for ultraviolet detection and analysis. Additionally, the platform features a 12-channel reaction device (9) for simultaneous multi-sample processing, and a concentration measuring device (10) that determines reagent concentration levels. To handle larger solution volumes, the platform incorporates a solution dispensing device (11) for pre-treatment and dispensing. Figure 10b shows a photograph of the entire experimental platform, where all these devices are integrated into a large-scale experimental system. The mobile robot (4) demonstrates the programmability of the system, capable of executing various complex experimental workflows. Figures 10c and 10d present front and top views of the structure of the platform, offering a detailed visualization of the equipment layout.

6.2 Unified control middleware

The unified control middleware software is the central part of the entire system. It is an upper-level control software for workstations and mobile robots implemented in C#. This software is responsible for managing the low-level control logic of robotic devices and interacting with multi-agent systems through standardized interfaces. To achieve more efficient task scheduling and real-time monitoring, we extended this software by adding an HTTP Server to listen for experimental process instructions sent by agents within the MARS system.

This portion of the code resides in the "middleware" folder of the Github repository and includes the implementation of the HTTP Server, drivers for communicating with robot hardware, and a task queue management module. Through this middleware layer, users can easily integrate different robotic devices into the system while supporting dynamic adjustment of task priorities and providing real-time feedback on execution status.

6.3 Example: synthesize CsPbBr₃ nanocubes at room temperature

User:

Let the robot synthesize CsPbBr₃ nanocubes at room temperature

ORCHESTRATOR:

****Understanding the task:****

The task is to plan the synthesis of CsPbBr₃ nanocubes at room temperature by the robot platform. It involves:

1. Confirming the synthesis protocol.
2. Converting the synthesis scheme into a machine-readable format (e.g., JSON).
3. Uploading the scheme to the robot platform.
4. Enabling the executor to perform the synthesis.
5. Collecting experimental logs and data.

****Task Decomposition:****

Team_name	Member_name	sub-task	
-----	-----	-----	

Scientist	Synthesis Scientist	<status: Provide an accurate synthesis protocol for CsPbBr3 nanocubes at room temperature.>
Engineer	Structural engineer	<status: Convert the synthesis protocol provided by scientists into a JSON format and upload it to S3 storage.>
Executor	MobileRobot_Agent	<status: Coordinate the placement of the experimental container into the workstation.>
Executor	RobotWorkstation_Agent	<status: Execute the workstation operations for synthesis.>
Executor	DataCollector_Agent	<status: Collect experimental data and logs from the synthesis workstation.>

```

```mermaid
graph TD
 User[User]
 subgraph Scientist
 A1[Synthesis Scientist]
 end
 subgraph Engineer
 B1[Structural engineer]
 end
 subgraph Executor
 C1[MobileRobot_Agent]
 C2[RobotWorkstation_Agent]
 C3[DataCollector_Agent]
 end
 User --> A1
 A1 --> B1
 B1 --> C1
 C1 --> C2
 C2 --> C3
```

```

****Immediate Sub-Tasks and Workflow:****

1. ****Synthesis Scientist****: Develop and provide a detailed synthesis protocol for CsPbBr3 nanocubes at room temperature.

- Key considerations: exact chemical precursors, solvent, concentrations, reaction time, and mixing procedure.

****Next Steps:****

****Next sub-task:****

1. Scientist : Provide an accurate synthesis protocol for CsPbBr3 nanocubes at room temperature.

Scientist_Admin:

Transferred to Synthesis_Scientist, adopting the role of Synthesis_Scientist immediately.

Synthesis_Scientist:

By analyzing the retrieved knowledge, I will construct a comprehensive synthesis protocol for CsPbBr3 nanocubes at room temperature.

**Reference Knowledge Analysis**

- **Synthesis using Ligand-Assisted Reprecipitation (LARP) at Room Temperature****:
 - "Water-Assisted Size and Shape Control of CsPbBr3 Perovskite Nanocrystals" describes CsPbBr3 nanocrystals synthesis at room temperature using oleylamine (OLA) and oleic acid (OA) as capping agents. The precursor (CsBr and PbBr2) is dissolved in dimethylformamide (DMF), and the material is precipitated by injecting the precursor into an anti-solvent such as toluene.
 - Water appears to play a role in influencing the crystallization process; small amounts of water added to the anti-solvent could benefit uniform nanocrystal formation.
- **One-Pot Room Temperature Synthesis of CsPbBr3 Nanocubes Coated with SiO2****:
 - Another work mentions achieving improved stability with silica-coated CsPbBr3 quantum dots synthesized via a facile one-pot approach at room temperature. This method focuses on stability rather than the specific morphology of nanocubes but provides insight into controlling environmental conditions.
- **Study of Ligand Interactions****:

- For ensuring nanocube formation and stabilization, ligands like OLA and OA play a critical role (as discussed in all references). A balance between the concentrations of these ligands is crucial for shape control and uniformity.

Finalized Protocol for Synthesizing CsPbBr₃ Nanocubes

Based on the references, the synthesis of CsPbBr₃ nanocubes can be effectively carried out using the ligand-assisted reprecipitation method at room temperature with precise control of precursor ratios, solvent conditions, and ligands.

Experimental Scheme

1. **Synthesis Conditions**

- **Temperature**: Room temperature (25°C)
- **Environment**: Dry and inert atmosphere (to avoid water contamination)
- **Mixing Speed**: Vigorous stirring

2. **Materials & Quantities**

| Mat.ID | Mat.Name | Mat.Value/Range | Mat.Unit | |
|--------|-----------------------------------|-----------------|----------|--|
| M001 | Cesium Bromide (CsBr) | 0.4 | mmol | |
| M002 | Lead Bromide (PbBr ₂) | 0.4 | mmol | |
| M003 | Dimethylformamide (DMF) | 10 | mL | |
| M004 | Toluene | 10 | mL | |
| M005 | Oleylamine (OLA) | 0.5 | mL | |
| M006 | Oleic Acid (OA) | 1.0 | mL | |
| M007 | Water (optional, trace amounts) | 0–60 | μL | |

3. **Equipment & Containers**

| ID | Name | Param/Capacity | Note |
|------|---------------------|------------------|------------------------|
| C001 | Reaction beaker | 50 mL | For precursor mixing |
| C002 | Anti-solvent beaker | 100 mL | For toluene as solvent |
| E001 | Magnetic stirrer | Speed adjustable | For vigorous stirring |
| E002 | Microsyringe | 1 mL | For injection control |

4. **Synthesis Sequence**

- Preparation of Precursor Solution**:
 - Dissolve CsBr (M001) and PbBr₂ (M002) in DMF (M003).
 - Add Oleylamine (M005) and Oleic Acid (M006) to the mixture for stabilization.
- Anti-Solvent Preparation**:
 - Place toluene (M004) into a clean, dry beaker (C002) under stirring.
- Injection and Mixing**:
 - Using a microsyringe (E002), inject 1 mL of prepared precursor solution into the toluene under vigorous stirring.
- Optional Water Introduction**:
 - For better size and shape control, a trace amount of water (M007, 0–60 μL) can be mixed into the precursor or anti-solvent phase.
- Nanocrystal Formation**:
 - Allow the mixture to stir for 10–15 minutes, during which CsPbBr₃ nanocubes will precipitate.
- Collection and Purification**:
 - Centrifuge the solution to collect the nanocubes. Wash with toluene for purification.

5. **Step-by-Step Process for Final Material Synthesis**

| Step | Materials Involved | Equipment Used | Procedure |
|------|------------------------------|----------------|---|
| 1 | M001, M002, M003, M005, M006 | C001, E001 | Mix CsBr and PbBr ₂ in DMF. Add OLA and OA under stirring until dissolved. |
| 2 | M004 | C002, E001 | Add toluene to C002 under vigorous stirring. |
| 3 | Prepared precursor | E002 | Inject 1 mL of the precursor solution into toluene. |
| 4 | M007 (optional) | E002 | Add trace water (optional) to induce size control. |

| | | | |
|---|-------------------|------------|--|
| 5 | Resulting mixture | E001 | Stir for 15 min to promote nanocrystal precipitation. |
| 6 | Product mixture | Centrifuge | Isolate CsPbBr ₃ nanocubes by centrifugation and wash with toluene. |

6. **Material Characterization**

- **Morphology**: Transmission Electron Microscopy (TEM) to confirm nanocube structure.
- **Crystallinity**: X-ray Diffraction (XRD) for phase purity.
- **Optical Properties**: Photoluminescence (PL) spectroscopy to study emission characteristics.

7. **Additional Considerations**

- Perform all steps under an inert (nitrogen or argon) atmosphere to prevent degradation.
- Optimize the ratio of precursors and ligands for targeted nanocube dimensions.
- Safety precautions for handling DMF and toluene must be observed.
- Ensure proper disposal of organic solvents as per laboratory safety guidelines.

The detailed synthesis approach above ensures reproducibility and room-temperature production of CsPbBr₃ nanocubes. Let me know if further modifications are required or if this protocol needs to be forwarded for machine execution!

Synthesis_Scientist:

Transferred to Scientist_Admin, adopting the role of Scientist_Admin immediately.

Scientist_Admin:

Synthesis Protocol for CsPbBr₃ Nanocubes at Room Temperature

This protocol outlines the synthesis of CsPbBr₃ nanocubes using the ligand-assisted reprecipitation (LARP) method at room temperature with controlled precursor and solvent conditions.

Experimental Overview

1. Materials

- **Cesium Bromide (CsBr)**: 0.4 mmol
- **Lead Bromide (PbBr₂)**: 0.4 mmol
- **Dimethylformamide (DMF)**: 10 mL
- **Toluene**: 10 mL
- **Oleylamine (OLA)**: 0.5 mL
- **Oleic Acid (OA)**: 1.0 mL
- **Water (optional)**: Trace amounts (0–60 µL for enhanced control)

2. Equipment

- Reaction beaker (50 mL, for precursors)
- Anti-solvent beaker (100 mL, for toluene)
- Magnetic stirrer
- Microsyringe (1 mL)
- Centrifuge

3. Synthesis Steps

1. **Preparation of Precursor Solution**:

- In a reaction beaker, dissolve 0.4 mmol CsBr and 0.4 mmol PbBr₂ in 10 mL of DMF.
- Add 0.5 mL OLA and 1.0 mL OA to stabilize the solution. Stir until clear.

2. **Anti-Solvent Preparation**:

- Pour 10 mL of toluene into a clean anti-solvent beaker and start vigorous stirring.

3. **Precursor Injection**:

- Using a microsyringe, inject 1 mL of the prepared precursor solution into the stirring toluene.

4. **Optional Water Addition**:

- Add a trace amount of water (0–60 µL) to the precursor or anti-solvent phase for enhanced size and shape control.

5. **Nanocrystal Formation**:

- Stir the resulting mixture vigorously for 10–15 minutes to precipitate CsPbBr₃ nanocubes.

6. **Isolation and Purification**:

- Centrifuge the resulting solution to collect the nanocubes and wash with toluene to remove residual ligands or solvent.

```
#### **4. Material Characterization**
- Use **Transmission Electron Microscopy (TEM)** to confirm the nanocube structure.
- Perform **X-ray Diffraction (XRD)** to analyze crystallinity and phase purity.
- Conduct **Photoluminescence (PL) spectroscopy** to study optical properties.
---
#### **5. Key Considerations**
- Keep all processes under an inert atmosphere (nitrogen or argon) to avoid material degradation.
- Optimize ligand and precursor ratios for desired nanocube sizes.
- Handle DMF and toluene with appropriate safety precautions.
This procedure enables the efficient and reproducible synthesis of high-quality CsPbBr3 nanocubes at room temperature.

ORCHESTRATOR:
**Subtask completed: **
1. Scientist: Provided an accurate synthesis protocol for CsPbBr3 nanocubes at room temperature.

**Updated Task Decomposition:**
Team_name	Member_name	sub-task
Scientist	Synthesis Scientist	<completed: Provided an accurate synthesis protocol for CsPbBr3 nanocubes at room temperature.>
Engineer	Structural engineer	<status: Convert the synthesis protocol provided by scientists into a JSON format and upload it to S3 storage.>
Executor	MobileRobot_Agent	<status: Coordinate the placement of the experimental container into the workstation.>
Executor	RobotWorkstation_Agent	<status: Execute the workstation operations for synthesis.>
Executor	DataCollector_Agent	<status: Collect experimental data and logs from the synthesis workstation.>

```mermaid
graph TD
 User[User]
 subgraph Scientist
 A1[Synthesis Scientist]
 end
 subgraph Engineer
 B1[Structural engineer]
 end
 subgraph Executor
 C1[MobileRobot_Agent]
 C2[RobotWorkstation_Agent]
 C3[DataCollector_Agent]
 end
 User --> A1
 A1 --> B1
 B1 --> C1
 C1 --> C2
 C2 --> C3
```

**Next Sub-Task:**
2. **Engineer (Structural engineer)**: Convert the above synthesis protocol into a JSON format, ensuring all details (such as materials, equipment, and process steps) are included, and upload it to S3 storage.

Engineer_Admin:
Transferred to Structural_Engineer, adopting the role of Structural_Engineer immediately.

Structural_Engineer:
The protocol has been converted into JSON format successfully! It will now be uploaded to our S3 storage system for access by the robots.
First, here are the essential details of the process:
```

Generated TaskId: `task\_20250225222203`
Protocol JSON is prepared for execution. Uploading to S3.

Structural_Engineer:

The JSON file for the synthesis protocol has been converted and successfully uploaded to S3. The file can be accessed with the following unique URL:

**[JSON File

URL](http://100.85.52.31:9000/temp/robotExprimmentScheme_task_20250225222203.json?AWSAccessKeyId=9bUtQL1Gpo9JB6o3pSGr&Signature=UGh4xEC6irtkSgalFi8wb%2Fdm9vU%3D&Expires=1740529352)**

Please proceed with the robot platform to execute the synthesis task. Handoff complete.

Transferred to Engineer_Admin, adopting the role of Engineer_Admin immediately.

Engineer_Admin:

The protocol JSON file has been successfully uploaded to S3 storage. You can access it using the following URL:

**[Access JSON

File](http://100.85.52.31:9000/temp/robotExprimmentScheme_task_20250225222203.json?AWSAccessKeyId=9bUtQL1Gpo9JB6o3pSGr&Signature=UGh4xEC6irtkSgalFi8wb%2Fdm9vU%3D&Expires=1740529352)**

Please utilize this link to proceed with task execution on the robotic platform.

ORCHESTRATOR:

**Subtask completed: **

2. Engineer: Converted the synthesis protocol into a JSON format and uploaded it to S3 storage.

Updated Task Decomposition:

| Team_name | Member_name | sub-task |
|-----------|------------------------|---|
| Scientist | Synthesis Scientist | <completed: Provided an accurate synthesis protocol for CsPbBr ₃ nanocubes at room temperature.> |
| Engineer | Structural engineer | <completed: Converted the synthesis protocol into a JSON format and uploaded it to S3 storage.> |
| Executor | MobileRobot_Agent | <status: Coordinate the placement of the experimental container into the workstation.> |
| Executor | RobotWorkstation_Agent | <status: Execute the workstation operations for synthesis.> |
| Executor | DataCollector_Agent | <status: Collect experimental data and logs from the synthesis workstation.> |

```mermaid

graph TD

User[User]

subgraph Scientist

A1[Synthesis Scientist]

end

subgraph Engineer

B1[Structural engineer]

end

subgraph Executor

C1[MobileRobot\_Agent]

C2[RobotWorkstation\_Agent]

C3[DataCollector\_Agent]

end

User --> A1

A1 --> B1

B1 --> C1

C1 --> C2  
C2 --> C3  
...

**\*\*Next Sub-Task:\*\***

3. **\*\*Executor (MobileRobot\_Agent)\*\***: Use the JSON file to coordinate the placement of the experimental container into the workstation, preparing the setup for synthesis execution.

**Robot\_Admin:**

Transferred to MobileRobot\_Agent, adopting the role of MobileRobot\_Agent immediately.

**MobileRobot\_Agent:**

The experimental container has been successfully aligned and placed into the robot workstation based on the uploaded scheme. The system is now ready for the execution phase.

Handoff to **\*\*RobotWorkstation\_Agent\*\*** for further synthesis execution.

Transferred to RobotWorkstation\_Agent, adopting the role of RobotWorkstation\_Agent immediately.

**RobotWorkstation\_Agent:**

The experiment synthesis for CsPbBr<sub>3</sub> nanocubes at room temperature has been successfully initiated on the robot workstation. The task data was updated successfully, and the execution process is underway.

Handoff to user for further instructions or queries.

Transferred to user, adopting the role of user immediately.

**Robot\_Admin:**

The experimental container has been accurately aligned and placed into the robot workstation as per the uploaded scheme. The synthesis of CsPbBr<sub>3</sub> nanocubes at room temperature has been successfully initiated, and the system is actively executing the process.

**ORCHESTRATOR:**

**\*\*Subtasks completed: \*\***

3. **Executor (MobileRobot\_Agent)**: Coordinated the placement of the experimental container into the workstation.

**\*\*Updated Task Decomposition:\*\***

| Team_name | Member_name            | sub-task                                                                                                    |
|-----------|------------------------|-------------------------------------------------------------------------------------------------------------|
| Scientist | Synthesis Scientist    | <completed: Provided an accurate synthesis protocol for CsPbBr <sub>3</sub> nanocubes at room temperature.> |
| Engineer  | Structural engineer    | <completed: Converted the synthesis protocol into a JSON format and uploaded it to S3 storage.>             |
| Executor  | MobileRobot_Agent      | <completed: Coordinated the placement of the experimental container into the workstation.>                  |
| Executor  | RobotWorkstation_Agent | <status: Execute the robot workstation operations for synthesis of CsPbBr <sub>3</sub> nanocubes.>          |
| Executor  | DataCollector_Agent    | <status: Collect experimental data and logs from the synthesis workstation.>                                |

```mermaid

graph TD

User[User]

subgraph Scientist

A1[Synthesis Scientist]

end

subgraph Engineer

B1[Structural engineer]

end

subgraph Executor

C1[MobileRobot_Agent]

C2[RobotWorkstation_Agent]

C3[DataCollector_Agent]

end

User --> A1
A1 --> B1
B1 --> C1
C1 --> C2
C2 --> C3
...

****Next Sub-Task:****

4. ****Executor (RobotWorkstation_Agent)**:** Execute the robot workstation operations for the synthesis of CsPbBr₃ nanocubes following the protocol. Ensure the process adheres to the uploaded JSON scheme.

The synthesis of CsPbBr₃ nanocubes has been executed successfully by the robotic workstation. The material is now ready for data collection and analysis.

Analyst_Admin:

Transferred to Exprimment_Analyst, adopting the role of Exprimment_Analyst immediately.

Exprimment_Analyst:

The photoluminescence (PL) analysis of CsPbBr₃ nanocubes has been successfully performed. Below is a summary of the experimental data:

****Experimental Results:****

****PL Spectra Data Summary****

| Parameter | Value |
|-----------------------------------|-----------|
| **Start (nm)** | 400.6 |
| **Apex (nm, Peak Emission)** | 523.0 |
| **End (nm)** | 699.4 |
| **PL Intensity at Apex (Height)** | 2250 |
| **PL Area (Data * nm)** | 67915.779 |
| **Valley (nm)** | 699.4 |
| **Valley (Data)** | 0.665 |

****Key Observations:****

- The ****peak PL emission**** occurred at ****523.0 nm****, indicating the desired optical property of the CsPbBr₃ nanocubes. This is consistent with high purity and phase-crystallinity of the synthesized perovskite nanocubes.
- The ****PL intensity**** value (2250) and ****area under the curve**** (67915.779) demonstrate significant material quality and quantum efficiency.
- Clear spectral boundaries from ****400.6 nm to 699.4 nm****, reflecting a broad emission profile characteristic of CsPbBr₃.

You may now proceed with detailed data visualization, further material characterizations (e.g., TEM or XRD analysis), or replicate the experiment to validate the results.

Next Actions:

Please inform if additional insights or visualizations (e.g., line plot of the PL spectrum) are needed. Transferred to Analyst_Admin, adopting the role of Analyst_Admin immediately.

Analyst_Admin:

Experiment Conclusion

The system successfully synthesized CsPbBr₃ nanocubes following the provided protocol. Key observations include a photoluminescence peak emission of 523.0 nm, validating the nanocubes' high purity and phase-crystallinity. The PL intensity and area under the curve further demonstrate desirable optical properties.

****Experiment Steps Overview:****

1. The synthesis protocol for CsPbBr₃ nanocubes was outlined using a room-temperature, ligand-assisted reprecipitation (LARP) method.

2. The experimental instructions were converted into a structured JSON, uploaded to S3 storage.
3. The robotic platform seamlessly executed the protocol, handling the setup and performing synthesis.
4. Experimental data was collected, showcasing promising results consistent with high material quality.

****Request Completion: APPROVE****

The photoluminescence (PL) characterization of CsPbBr₃ nanocubes has been successfully completed, yielding the following results:

****Experimental Results Summary****

****PL Spectra****

- ****Peak Emission Wavelength:**** 523.0 nm
- ****PL Intensity at Peak:**** 2250
- ****Emission Range:**** 400.6 nm to 699.4 nm
- ****PL Area Under the Curve:**** 67915.779
- ****Valley Wavelength:**** 699.4 nm with Data Value of 0.665

****Key Observations****

- The ****peak photoluminescence emission**** at ****523.0 nm**** confirms the optical properties and high purity of the synthesized CsPbBr₃ nanocubes.
- The high PL intensity and broad emission profile indicate strong material quality and quantum efficiency characteristic of high-performance perovskite nanomaterials.

The CsPbBr₃ nanocubes have demonstrated expected optical properties and material quality, fulfilling the experimental objective.

ORCHESTRATOR:

****All subtasks have been completed: ****

1. Scientist: Provided an accurate synthesis protocol for CsPbBr₃ nanocubes at room temperature.
2. Engineer: Converted the synthesis protocol into a JSON format and uploaded it to S3 storage.
3. Executor (MobileRobot_Agent): Coordinated the placement of the experimental container into the workstation.
4. Executor (RobotWorkstation_Agent): Successfully executed the synthesis of CsPbBr₃ nanocubes.
5. Data has been collected, and photoluminescence (PL) analysis has been completed, with results confirming high material quality.

****Summary Findings:****

- ****Material Synthesized**:** CsPbBr₃ Nanocubes
- ****Process Conditions**:** Room temperature, ligand-assisted reprecipitation method per the provided protocol.
- ****Optical Properties**:** PL peak at 523.0 nm with high intensity (2250) and broad emission range from 400.6–699.4 nm.

The synthesized material meets the desired quality and optical performance standards for potential applications.

****Proceed as per your next instructions!****

****TERMINATE****

6.4 Structured Synthesis Scheme

```
{
  "TaskId": "task_20250225222203",
  "ExperimentName": "Synthesis of CsPbBr3 Nanocubes at Room Temperature",
  "Materials": [
    {
      "MaterialId": "MAT001",
      "Name": "Cesium Bromide",
      "Formula": "CsBr",
      "Amount": "0.4",
      "Unit": "mmol",
      "Purity": "Analytical-grade",
      "State": "Solid"
    },
    {
      "MaterialId": "MAT002",
      "Name": "Lead Bromide",
      "Formula": "PbBr2",
      "Amount": "0.4",
      "Unit": "mmol",
      "Purity": "Analytical-grade",
      "State": "Solid"
    },
    {
      "MaterialId": "MAT003",
      "Name": "Dimethylformamide",
      "Formula": "C3H7NO",
      "Amount": "10",
      "Unit": "mL",
      "Purity": "Lab-grade",
      "State": "Liquid"
    },
    {
      "MaterialId": "MAT004",
      "Name": "Toluene",
      "Formula": "C7H8",
      "Amount": "10",
      "Unit": "mL",
      "Purity": "Lab-grade",
      "State": "Liquid"
    },
    {
      "MaterialId": "MAT005",
      "Name": "Oleylamine",
      "Formula": "CH3(CH2)7CH=CH(CH2)8NH2",
      "Amount": "0.5",
      "Unit": "mL",
      "Purity": "Technical-grade",
      "State": "Liquid"
    },
    {
      "MaterialId": "MAT006",
      "Name": "Oleic Acid",
      "Formula": "C18H34O2",
      "Amount": "1.0",
      "Unit": "mL",
      "Purity": "Technical-grade",
      "State": "Liquid"
    }
  ]
}
```

```

    {
      "MaterialId": "MAT007",
      "Name": "Water (optional)",
      "Formula": "H2O",
      "Amount": "0-60",
      "Unit": "μL",
      "Purity": "Deionized",
      "State": "Liquid"
    }
  ],
  "Containers": [
    {
      "ContainerId": "CON001",
      "Name": "Reaction Beaker",
      "Capacity": "50",
      "Unit": "mL",
      "MaterialOfConstruction": "Glass",
      "Shape": "Cylindrical",
      "HeatResistant": "Yes",
      "PressureRating": "Ambient"
    },
    {
      "ContainerId": "CON002",
      "Name": "Anti-Solvent Beaker",
      "Capacity": "100",
      "Unit": "mL",
      "MaterialOfConstruction": "Glass",
      "Shape": "Cylindrical",
      "HeatResistant": "No",
      "PressureRating": "Ambient"
    }
  ],
  "Equipments": [
    {
      "EquipmentId": "EQ001",
      "Name": "Magnetic Stirrer",
      "Parameters": {
        "RPMRange": "0-1500"
      }
    },
    {
      "EquipmentId": "EQ002",
      "Name": "Microsyringe",
      "Parameters": {
        "VolumeRange": "0.1-1"
      }
    },
    {
      "EquipmentId": "EQ003",
      "Name": "Centrifuge",
      "Parameters": {
        "MaxSpeed": "15000",
        "MaxVolume": "50"
      }
    }
  ],
  "RobotWorkflow": [
    {
      "StepId": "1",
      "Description": "Prepare precursor solution by dissolving CsBr and PbBr2 in DMF with
ligands.",

```



```

"Actions": [
  {
    "ActionType": "pick_container",
    "ContainerId": "CON001"
  },
  {
    "ActionType": "pick_container_with_material",
    "ContainerId": "CON001",
    "MaterialId": "MAT001"
  },
  {
    "ActionType": "pick_container_with_material",
    "ContainerId": "CON001",
    "MaterialId": "MAT002"
  },
  {
    "ActionType": "pick_container_with_material",
    "ContainerId": "CON001",
    "MaterialId": "MAT003"
  },
  {
    "ActionType": "pick_container_with_material",
    "ContainerId": "CON001",
    "MaterialId": "MAT005"
  },
  {
    "ActionType": "pick_container_with_material",
    "ContainerId": "CON001",
    "MaterialId": "MAT006"
  }
],
"Dependencies": [],
"StepOutput": {
  "ContainerId": "CON001",
  "Contents": [
    {
      "MaterialId": "MAT001",
      "Amount": "0.4",
      "Unit": "mmol"
    },
    {
      "MaterialId": "MAT002",
      "Amount": "0.4",
      "Unit": "mmol"
    },
    {
      "MaterialId": "MAT003",
      "Amount": "10",
      "Unit": "mL"
    },
    {
      "MaterialId": "MAT005",
      "Amount": "0.5",
      "Unit": "mL"
    },
    {
      "MaterialId": "MAT006",
      "Amount": "1.0",
      "Unit": "mL"
    }
  ]
}

```

```

    }
  },
  {
    "StepId": "2",
    "Description": "Prepare an anti-solvent beaker containing toluene.",
    "Actions": [
      {
        "ActionType": "pick_container",
        "ContainerId": "CON002"
      },
      {
        "ActionType": "pick_container_with_material",
        "ContainerId": "CON002",
        "MaterialId": "MAT004"
      }
    ],
    "Dependencies": [
      "1"
    ],
    "StepOutput": {
      "ContainerId": "CON002",
      "Contents": [
        {
          "MaterialId": "MAT004",
          "Amount": "10",
          "Unit": "mL"
        }
      ]
    }
  }
},
{
  "StepId": "3",
  "Description": "Inject precursor solution into the anti-solvent.",
  "Actions": [
    {
      "ActionType": "place_container_into_equipment",
      "ContainerId": "CON002",
      "EquipmentId": "EQ001"
    },
    {
      "ActionType": "place_container_into_equipment",
      "ContainerId": "CON001",
      "EquipmentId": "EQ002"
    }
  ],
  "Dependencies": [
    "1",
    "2"
  ],
  "StepOutput": {}
},
{
  "StepId": "4",
  "Description": "Centrifuge and collect nanocubes for final output.",
  "Actions": [
    {
      "ActionType": "remove_container_from_equipment",
      "ContainerId": "CON002",
      "EquipmentId": "EQ001"
    }
  ],
  {

```

```
        "ActionType": "place_container_into_equipment",
        "ContainerId": "CON002",
        "EquipmentId": "EQ003"
    },
    ],
    "Dependencies": [
        "3"
    ],
    "StepOutput": {
        "ContainerId": "CON002",
        "Contents": [
            {
                "MaterialId": "Output",
                "Amount": "Optimized Yield",
                "Unit": " -- "
            }
        ]
    }
}
]
```

Note S7: Comparison of experimental optimization for different agents

7.1 Identical standardized input formats

Round 1:

```
<system>
You are a helpful assistant.
</system>
<user>
As shown in the table below, the peak position of the latest step is 523 nm. Please
continue to optimize. Our goal is to synthesize perovskite nanocrystals with a peak position
of 460 nm. Let's optimize the synthesis plan step by step to approach this goal. Please
note that no duplicate synthesis plans are allowed during the synthesis process.
| step | CsBr | CsCl | PbBr2 | PbCl2 | OAm | OA | PL |
| ---- | ---- | ---- | ---- | ---- | ---- | ---- | ---- |
| 0 | 0.4 | 0 | 0.4 | 0 | 0.05 | 1 | 523 |
</user>
```

Round 2:

```
<system>
You are a helpful assistant.
</system>
<user>
As shown in the table below, the peak position of the latest step is 523 nm. Please
continue to optimize. Our goal is to synthesize perovskite nanocrystals with a peak position
of 460 nm. Let's optimize the synthesis plan step by step to approach this goal. Please
note that no duplicate synthesis plans are allowed during the synthesis process.
| step | CsBr | CsCl | PbBr2 | PbCl2 | OAm | OA | PL |
| ---- | ---- | ---- | ---- | ---- | ---- | ---- | ---- |
| 0 | 0.4 | 0 | 0.4 | 0 | 0.05 | 1 | 523 |
| 1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.7 | 0.8 | 442.5 |
</user>
```

7.2 MARS

The first parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.2	0.2	0.2	0.2	0.7	0.8	442.5	0.533333
3	0.24	0.16	0.24	0.16	0.6	0.9	469.8	0.671141
4	0.22	0.18	0.22	0.18	0.65	0.86	450.1	0.668896
5	0.23	0.17	0.23	0.17	0.62	0.88	454.9	0.796813
6	0.24	0.16	0.24	0.16	0.6	0.9	469.8	0.671141
7	0.22	0.18	0.22	0.18	0.63	0.89	452.3	0.722022
8	0.226	0.174	0.226	0.174	0.61	0.896	453.7	0.760456
9	0.232	0.168	0.232	0.168	0.64	0.884	458	0.909091
10	0.236	0.164	0.232	0.168	0.64	0.884	459.2	0.961538

The second parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.2	0.2	0.2	0.2	0.5	1	444.2	0.558659
3	0.24	0.16	0.24	0.16	0.5	1	449.8	0.662252
4	0.26	0.14	0.26	0.14	0.5	1	453	0.740741
5	0.28	0.12	0.28	0.12	0.5	1	466.3	0.760456
6	0.27	0.13	0.27	0.13	0.5	1	459	0.952381

The third parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.3	0.1	0.3	0.1	0.5	1	481.6	0.480769
3	0.2	0.2	0.2	0.2	0.5	1	444.2	0.558659
4	0.24	0.16	0.24	0.16	0.5	1	449.8	0.662252
5	0.26	0.14	0.26	0.14	0.5	1	453	0.740741
6	0.28	0.12	0.28	0.12	0.5	1	466.3	0.760456
7	0.27	0.13	0.27	0.13	0.5	1	458.7	0.938967
8	0.274	0.126	0.274	0.126	0.5	1	461.8	0.917431
9	0.272	0.128	0.272	0.128	0.5	1	460.2	0.990099

7.3 GPT-4o**The first parallel test**

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.3	0.1	0.3	0.1	0.6	0.98	483.2	0.462963
3	0.2	0.2	0.2	0.2	0.7	0.96	441.9	0.524934
4	0.24	0.16	0.24	0.16	0.7	0.96	448.1	0.626959
5	0.26	0.14	0.26	0.14	0.7	0.96	451.6	0.704225
6	0.28	0.12	0.28	0.12	0.7	0.96	468.9	0.692042
7	0.26	0.14	0.26	0.14	0.7	0.96	451.6	0.704225
8	0.29	0.11	0.29	0.11	0.7	0.96	471.4	0.636943
9	0.27	0.13	0.27	0.13	0.7	0.96	457.1	0.873362
10	0.274	0.126	0.274	0.126	0.7	0.96	462.3	0.896861
11	0.272	0.128	0.272	0.128	0.7	0.96	460.4	0.980392

The second parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.36	0.04	0.4	0	0.5	1	515.3	0.265604
3	0.32	0.08	0.36	0.04	0.5	1	498.1	0.344234
4	0.28	0.12	0.32	0.08	0.5	1	480.4	0.49505
5	0.24	0.16	0.28	0.12	0.5	1	451.6	0.704225
6	0.26	0.14	0.3	0.1	0.5	1	466.2	0.763359
7	0.25	0.15	0.29	0.11	0.5	1	458.2	0.917431
8	0.254	0.146	0.294	0.106	0.5	1	463.3	0.858369
9	0.26	0.14	0.3	0.1	0.5	1	466.2	0.763359
10	0.264	0.136	0.304	0.096	0.5	1	469	0.689655
11	0.256	0.144	0.296	0.104	0.5	1	462.7	0.881057
12	0.252	0.148	0.292	0.108	0.5	1	460.6	0.970874

The third parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.3	0.1	0.4	0	0.5	1	507.9	0.294551
3	0.2	0.2	0.3	0.1	0.5	1	454	0.769231
4	0.2	0.2	0.34	0.06	0.5	1	469.1	0.687285

5	0.18	0.22	0.34	0.08	0.5	1	465.7	0.77821
6	0.16	0.22	0.32	0.08	0.5	1	454.3	0.77821
7	0.18	0.2	0.34	0.08	0.5	1	467.2	0.735294
8	0.2	0.18	0.32	0.1	0.6	1	464.5	0.816327
9	0.22	0.18	0.3	0.1	0.5	1	458.1	0.913242
10	0.24	0.16	0.32	0.08	0.5	1	467.5	0.727273
11	0.22	0.18	0.3	0.1	0.5	1	458.1	0.913242
12	0.24	0.16	0.31	0.09	0.5	1	463.8	0.840336
13	0.22	0.18	0.3	0.1	0.5	1	458.1	0.913242
14	0.21	0.19	0.3	0.1	0.5	1	456.7	0.858369
15	0.22	0.18	0.31	0.09	0.5	1	460.9	0.966184

7.4 Mistral Large2

The first parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.2	0.2	0.2	0.2	0.5	1	444.2	0.558659
3	0.2	0.2	0.2	0.2	1	1	436.8	0.462963
4	0.3	0.1	0.3	0.1	0.5	1	479.8	0.502513
5	0.3	0.1	0.3	0.1	0.5	0.8	478.6	0.518135
6	0.3	0.1	0.3	0.1	0.7	1	482.3	0.472813
7	0.3	0.1	0.3	0.1	0.5	0.6	478.2	0.52356
8	0.3	0.1	0.3	0.1	0.5	0.4	478.0	0.526316
9	0.3	0.1	0.3	0.1	0.3	0.4	487.9	0.417537
10	0.3	0.1	0.3	0.1	0.5	0.2	477.7	0.530504
11	0.3	0.1	0.3	0.1	0.4	0.2	482.8	0.46729
12	0.3	0.1	0.3	0.1	0.5	0.3	477.9	0.527704
13	0.3	0.1	0.3	0.1	0.6	0.3	480.2	0.497512
14	0.2	0.2	0.2	0.2	0.5	0.3	446.2	0.591716
15	0.24	0.16	0.24	0.16	0.5	0.3	449.8	0.662252

The second parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964
2	0.2	0.2	0.2	0.2	0.5	1	444.2	0.558659
3	0.3	0.1	0.3	0.1	0.5	1	479.8	0.502513
4	0.32	0.08	0.32	0.08	0.5	1	483.6	0.458716
5	0.34	0.06	0.34	0.06	0.5	1	494.5	0.366972
6	0.28	0.12	0.28	0.12	0.5	1	466.8	0.746269
7	0.26	0.14	0.26	0.14	0.5	1	454.6	0.787402
8	0.27	0.13	0.27	0.13	0.5	1	457.3	0.881057
9	0.264	0.136	0.264	0.136	0.5	1	456.2	0.840336
10	0.266	0.134	0.266	0.134	0.5	1	456.9	0.865801
11	0.268	0.132	0.268	0.132	0.5	1	458.1	0.913242
12	0.27	0.13	0.27	0.13	0.5	1	458.7	0.938967
13	0.272	0.128	0.272	0.128	0.5	1	460.5	0.97561

The third parallel test

Step	CsBr/ mmol	CsCl/ mmol	PbBr ₂ / mmol	PbCl ₂ / mmol	OAm/ mL	OA/ mL	PL/ nm	Adv
1	0.4	0	0.4	0	0.5	1	523	0.240964

2	0.2	0.2	0.2	0.2	0.5	1	444.2	0.558659
3	0.3	0.1	0.3	0.1	0.5	1	479.4	0.507614
4	0.24	0.16	0.24	0.16	0.5	1	449.8	0.662252
5	0.26	0.14	0.26	0.14	0.5	1	455.2	0.806452
6	0.28	0.12	0.28	0.12	0.5	1	465.5	0.784314
7	0.27	0.13	0.27	0.13	0.5	1	458.7	0.938967
8	0.274	0.126	0.274	0.126	0.5	1	461.8	0.917431
9	0.272	0.128	0.272	0.128	0.5	1	460.5	0.97561

Note S8: Details of multi-round interaction process

8.1 Record of round 1 interaction process

User:

Core-shell structures are widely employed to enhance stability of perovskite. Could you explain the mechanism of this approach?

MARS:

----- TextMessage (user) -----

Core-shell structures are widely employed to enhance stability of perovskite. Could you explain the mechanism of this approach?

----- TextMessage (ORCHESTRATOR) -----

Enter your response: Thank you for your feedback. Let's focus on the Structural aspects first by engaging the Structure Scientist to analyze the mechanism of stability enhancement by core-shell structures in perovskites.

Team_name	Member_name	sub-task
Scientist	Structure Scientist	Analyze and report the structural aspects contributing to stability in perovskite core-shell structures

```
```mermaid
```

```
graph TD
```

```
User[User]
```

```
subgraph Scientist
```

```
 A1[Structure Scientist]
```

```
end
```

```
User --> A1
```

```
```
```

****Next sub-task:****

Scientist - Structure Scientist: Analyze and report on the structural aspects contributing to stability enhancement in core-shell perovskites.

----- TextMessage (Structure_Scientist) -----

Core-shell structures are a strategic architectural choice in materials science, particularly when aiming to enhance the stability and functionality of perovskite materials, widely used in applications such as solar cells and light-emitting devices. Here's a detailed look into the mechanism and advantages of using core-shell structures to enhance the stability of perovskite materials:

Moisture/Oxygen Barrier Protection:

- The shell acts as a protective physical barrier that effectively prevents external degradation agents, particularly moisture and oxygen, from reaching the sensitive perovskite core. This encapsulation strategy significantly reduces the exposure of perovskites to environmental factors that typically cause rapid degradation and performance loss.

Defect Passivation:

- Shell materials are strategically selected to passivate surface defects and eliminate trap states at the perovskite interface. This passivation process reduces non-radiative recombination pathways, thereby improving the overall optoelectronic performance and extending the operational lifetime of the perovskite devices.

Thermal Stability Enhancement:

- Many shell materials possess inherently higher thermal stability compared to perovskites, providing crucial protection against temperature-induced degradation. This thermal protection is particularly important for applications where devices may experience elevated operating temperatures or thermal cycling conditions.

Ion Migration Suppression:

- The shell structure effectively inhibits the migration of mobile ions such as I^- and MA^+ from the perovskite core. By preventing this ionic movement, the shell helps maintain the structural integrity of the perovskite, preventing phase segregation and decomposition that would otherwise compromise device performance.

Mechanical Protection:

- Shell structures provide essential physical robustness against mechanical stress and various environmental factors. This mechanical reinforcement protects the brittle perovskite core from physical damage during device fabrication, handling, and long-term operation under real-world conditions.

These features combine to make core-shell structures an effective means of preserving perovskite materials, extending their functional lifetime and efficiency in technological applications. The ongoing research is focusing on optimizing the materials and methods for creating the most effective core-shell configurations, aiming to tailor them for specific operational conditions and requirements.

----- TextMessage (Scientist_Admin) -----

Core-shell structures serve as a strategic architectural choice in materials science to enhance the stability and functionality of perovskite materials, widely used in applications like solar cells and light-emitting devices. Here's how core-shell structures contribute to perovskite stability:

1. ****Moisture/Oxygen Barrier Protection****:

- The shell layer acts as a physical barrier that effectively blocks external degradation agents such as moisture and oxygen from penetrating into the vulnerable perovskite core, thereby minimizing hydrolysis and oxidative damage.

2. ****Defect Passivation****:

- Shell materials can chemically or physically passivate surface defects and electronic trap states at the perovskite interface, which helps suppress non-radiative recombination and improves device efficiency and operational stability.

3. ****Thermal Stability Enhancement****:

- Many shell materials exhibit superior thermal stability compared to perovskites. By encapsulating the perovskite core, the shell helps maintain structural and compositional integrity under elevated temperatures, reducing the risk of thermal decomposition.

4. ****Ion Migration Suppression****:

- The shell structure can inhibit the movement of mobile ions such as iodide (I^-) and methylammonium (MA^+), which are known to migrate under electric fields or thermal stress. This suppression helps prevent phase segregation and chemical degradation.

5. ****Mechanical Protection****:

- Core-shell architectures provide enhanced mechanical robustness, shielding the perovskite core from physical damage, environmental stress, and mechanical deformation during device fabrication and operation.

Overall, core-shell configurations significantly preserve the functional lifetime and efficiency of perovskite materials in technological applications, with ongoing research optimizing materials and methods for specific conditions.

----- TextMessage (ORCHESTRATOR) -----

The explanation provided by the Structure Scientist is thorough and covers numerous aspects of how core-shell structures enhance the stability of perovskite materials. This information adds great insight into the advantages of using core-shell architectures for perovskite stability.

****Completed subtasks:****

1. Scientist (Structure Scientist): Explain the mechanism of core-shell structures enhancing perovskite stability.

The findings can now be utilized to inform further research or practical applications...

****Next sub-task:****

No further sub-tasks are required at this point.

8.2 Record of round 2 interaction process

User:

The toxicity of perovskites is also an issue. I want to develop a material with good biocompatibility and stability. Do you have any recommended shell materials?

MARS:

----- TextMessage (User) -----

The toxicity of perovskites is also an issue. I want to develop a material with good biocompatibility and stability. Do you have any recommended shell materials?

----- TextMessage (ORCHESTRATOR) -----

Enter your response: To address your request, we will break down the task into smaller subtasks and allocate each to the appropriate teams.

****Task breakdown:****

1. Recommend shell materials with good biocompatibility suitable for enhancing perovskite stability.

****Subtask allocation:****

| Team_name | Member_name | sub-task |
|-----------|-------------|----------|
|-----------|-------------|----------|

| | | |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

| | | |
|-----------|--------------------|--|
| ----- | ----- | ----- |
| Scientist | Property Scientist | Consult on biocompatible shell materials suitable for enhancing the stability of perovskites |

```
```mermaid
```

```
graph TD
```

```
User[User]
```

```
subgraph Scientist
```

```
 Property_Scientist["Property Scientist"]
```

```
end
```

```
User --> Property_Scientist
```

```
```
```

****Completed subtasks:****

N/A

****Next sub-task:****

Scientist - Property Scientist: Consult on biocompatible shell materials suitable for enhancing the stability of perovskites.

---- ThoughtEvent (Scientist_Admin) -----

We are given a task to recommend shell materials for perovskites that have good biocompatibility and stability.

The task has been assigned to the Property Scientist.

We will handoff to the Property Scientist to research and recommend such materials.

Plan:

1. Handoff to the Property Scientist to research shell materials with good biocompatibility and stability for perovskites.

We will call the function to transfer to the Property Scientist.

----- ToolCallExecutionEvent (Property_Scientist) -----

[FunctionExecutionResult(content='[{"index": "reference_0", "paper_title": "Introducing Specific Iodine Ions in Perovskite-Based Nanocomplex to Cater for Versatile Biomedical Imaging and Tumor Radiotherapy.md", "para_content": "In this study, we tailored a perovskite-based multifunctional nanoplatform that can integrate tumor-targeting, imaging, and therapeutic functions using a transformable mode (Scheme 1; Figure S1, Supporting Information). In detail, an inorganic SiO_2 shell coating of cesium lead halide perovskite nanocrystals (i.e., CsPbBr_3 @ SiO_2) slows down the water degradation and allows for the doping of I^- ions to replace initial Br^- ions $(\text{CsPbBr}_3 \cdot x\text{I})_{\text{SiO}_2}$, and the subsequent encapsulation of organic phospholipid bilayers preserves the resulting $(\text{CsPbBr}_3 \cdot x\text{I})_{\text{SiO}_2}$ @ Lip (PSL) NCs, thereby realizing surface bio-modification', "sim_score": 1.0}, {"index": "reference_1", "paper_title": "Introducing Specific Iodine Ions in Perovskite-Based Nanocomplex to Cater for Versatile Biomedical Imaging and Tumor Radiotherapy.md", "para_content": "2.1. Perovskite@ SiO_2 @ Lip (PSL) Multifunctional Nanoplatform\n\nThe ionic character of the structure renders perovskite NPs highly sensitive to water and moisture.[33] To date, several strategies have been exploited to isolate them from water-containing environments, such as SiO_2 -coating,[16,34] (amphiphilic) and simultaneously achieve perovskite fluorescence retention and surface bio-modification for future bio-applications.[14] In this section, an inorganic-organic dual-encapsulation technique is employed to enhance the water stability for further biological applications; the inorganic shell employs SiO_2 to encapsulate the perovskite nanocrystal, and phospholipid bilayers are used as the organic layer for secondary surface encapsulation and bio-modification.', "sim_score": 0.9919364192008642}, {"index": "reference_2", "paper_title": "Toward Highly Luminescent and Stabilized Silica-Coated Perovskite Quantum Dots through Simply Mixing and Stirring under Room Temperature in Air.md", "para_content": "KEYWORDS\n\nsilane, silica coated perovskite quantum dots, hydrolysis, stability, anion exchange reaction', "sim_score": 0.9436671689370933}, {"index": "reference_3", "paper_title": "Polar Organic Solvent-Tolerant Perovskite Nanocrystals Permanently Ligated with Polymer Hairs via Star-like Molecular Bottlebrush Trilobe Nanoreactors.md", "para_content": "ABSTRACT: The key to exploiting perovskite nanocrystals (NCs) for long term practical use in optoelectronic materials and devices lies in the ability to access stable NCs. Herein, we report the crafting of hairy perovskite NCs with a set of markedly improved stabilities by capitalizing on rationally designed star like molecular bottlebrush trilobes as nanoreactors. An intriguing star like molecular bottlebrush trilobe, poly(2 hydroxyethyl methacrylate) graft (poly(acrylic acid) block partially cross linked polystyrene (denoted PHEMA-g (PAA-b-cPS)) is synthesized. Subsequently, it is employed as a polymeric nanoreactor to direct the growth of green emitting all inorganic perovskite CsPbBr_3 NCs intimately and stably tethered by partially cross linked PS "hairs" (i.e., cPS capped CsPbBr_3 NCs). The resulting CsPbBr_3 NCs exhibit an array of impressive stabilities against UV irradiation, moisture, heat, and water, due to permanently ligated hydrophobic cPS "hairs" on the surface of CsPbBr_3 NCs as a result of the original covalent bonding between PAA and cPS blocks. More importantly, cPS capped CsPbBr_3 NCs manifest outstanding stability in various polar organic solvents. Such greatly improved stability can be attributed to the reduced surface defects enabled by the favorable interaction (i.e., coordination interaction and hydrogen bonding) between CsPbBr_3 NCs and polar solvents, which dominates over their dissolution by polar solvents. Such exceptional stabilities impart the use of cPS capped CsPbBr_3 NCs as a selective probe for tracing the presence of Cl^-/I^- in polar organic solvents. The amphiphilic nonlinear block copolymer nanoreactor strategy can afford easy access to stable perovskite NCs of interest with controlled compositions and surface chemistry. They may find applications in solar cells, LEDs, photodetectors, lasers, bioimaging, biosensors, etc. \n\nKEYWORDS: atom transfer radical polymerization (ATRP), polymer nanoreactors, perovskite nanocrystals, stabilities, probe', "sim_score": 0.9426527317986745}, {"index": "reference_4", "paper_title": "Lead Halide Perovskite Nanocrystals Phospholipid Micelles

and Their Biological Applications Multiplex Cellular Imaging and in Vitro Tumor Targeting.md', '\para_content': '\Finally, the biocompatibility of CsPbBr₃@DOPC was investigated using both HeLa cells and zebrafish. All of our experimentally results indicate that encapsulating perovskite nanomaterials into phospholipid micelles is a valid approach to explore further biological applications of perovskite nanomaterials.', '\sim_score': 0.9317213392783568}}']]

----- TextMessage (Property_Scientist) -----

Recommended Shell Materials for Perovskites with Biocompatibility and Stability

1. **Inorganic Materials**:

- **SiO₂**: Biologically inert and widely used in biomedical applications, providing excellent moisture barrier properties and chemical stability for perovskite encapsulation.
- **TiO₂**: FDA-approved material that forms uniform coatings with good optical transparency and protective capabilities against environmental degradation.
- **ZnO**: FDA-recognized as safe with inherent antibacterial properties, offering both protective encapsulation and antimicrobial functionality for biomedical perovskite applications.

2. **Organic Materials**:

- **Natural Polymers**:
 - **Chitosan**: Biodegradable polymer with excellent biocompatibility, providing flexible encapsulation with good film-forming properties.
 - **Cellulose derivatives**: Non-toxic materials offering excellent film formation capabilities and moisture barrier protection.
 - **Alginate/Pectin**: Plant-derived polymers with superior barrier properties against oxygen and moisture penetration.
- **Synthetic Biocompatible Polymers**:
 - **PLGA**: FDA-approved for various medical applications, offering controlled degradation and excellent encapsulation properties.
 - **PEG**: Low immunogenicity polymer that provides hydrophilic surface modification and improved biocompatibility.
 - **PVA**: Non-toxic, water-soluble polymer with good film-forming abilities for protective coatings.

3. **Hybrid Approaches**:

- **Biopolymer-inorganic composites**: Combining materials like chitosan-silica to leverage both organic flexibility and inorganic stability.
- **Layer-by-layer assemblies**: Sequential deposition of alternating materials to create tailored barrier properties and controlled thickness.
- **Biomimetic coatings**: Phospholipid bilayer systems that mimic natural cell membranes, providing exceptional biocompatibility for in vivo applications.

Key Considerations:

- **Dual-Layer Approach**: Combining SiO₂ (inorganic) with phospholipids (organic) synergistically enhances stability and biocompatibility.
- **Toxicity Mitigation**: Phospholipid coatings (e.g., DOPC) are particularly effective for *in vivo* applications.

For further optimization, consider hybrid systems like **SiO₂ + polymer** or **SiO₂ + phospholipid** tailored to your specific application (e.g., bioimaging vs. optoelectronics).

Let me know if you'd like details on synthesis protocols or additional material options!

----- TextMessage (Scientist_Admin) -----

Recommended Shell Materials for Stabilizing and Ensuring Biocompatibility of Perovskites

1. ****Inorganic Materials****:

- SiO₂ (Silica): A biologically inert material widely used in biomedical applications. It forms a robust barrier against moisture and oxygen, enhancing perovskite stability while maintaining biocompatibility.

- TiO₂ (Titanium Dioxide): FDA-approved and capable of forming uniform, conformal coatings. It provides excellent UV protection and environmental stability for perovskite materials.

- ZnO (Zinc Oxide): Recognized by the FDA as safe and exhibits inherent antibacterial properties. ZnO coatings can further improve perovskite durability and are suitable for applications requiring antimicrobial surfaces.

2. ****Organic Materials****:

- Natural Polymers:

a) Chitosan: A biodegradable, biocompatible polymer that provides a gentle encapsulation layer with good moisture barrier properties.

b) Cellulose Derivatives: Non-toxic and capable of forming uniform films, making them ideal for stabilizing perovskites in biological environments.

c) Alginate/Pectin: Plant-derived materials that offer excellent film-forming and barrier capabilities, useful for environmentally friendly encapsulation.

- Synthetic Biocompatible Polymers:

a) PLGA (Poly(lactic-co-glycolic acid)): FDA-approved for a wide range of medical applications; offers controlled degradation and protection.

b) PEG (Polyethylene Glycol): Exhibits low immunogenicity and high water solubility, enhancing colloidal stability and circulation time in biological systems.

c) PVA (Polyvinyl Alcohol): A non-toxic, water-soluble polymer that provides good film integrity and environmental resistance.

3. ****Hybrid Approaches****:

- Biopolymer-Inorganic Composites (e.g., Chitosan-Silica): Combine the mechanical stability and barrier function of silica with the biocompatibility and flexibility of chitosan.

- Layer-by-Layer Assemblies: Enable precise control over shell thickness and composition, allowing for tailored protection and functionalization.

- Biomimetic Coatings (e.g., Phospholipid Bilayers): Mimic natural cell membranes to enhance biocompatibility and stability, particularly useful in biomedical imaging and drug delivery applications.

Consider combining silica with either phospholipid bilayers or polymer-based coatings to synergistically improve both stability and biocompatibility, tailored to your specific application—whether in bioimaging or optoelectronic technologies.

----- TextMessage (ORCHESTRATOR) -----

Thank you for the comprehensive recommendations on biocompatible shell materials for perovskites.

These materials are effective in reducing toxicity and improving the stability and biocompatibility of perovskite-based applications. TiO₂, SiO₂, and Hydroxyapatite, specifically, offer a balance of chemical stability and non-toxic properties for forming protective coatings, minimizing degradation and enhancing performance for medical and sensitivity-required uses.

****Task allocation is complete with the following summary:****

| Team_name | Member_name | sub-task |
|-----------|--------------------|---|
| Scientist | Property Scientist | Completed consultation on biocompatible shell materials suitable for enhancing the stability of perovskites |

```mermaid  
graph TD  
User[User]



```

subgraph Scientist
 Property_Scientist["Property Scientist"]
end
User --> Property_Scientist
...

```

All subtasks are complete. Based on the plan, this task is concluded.

### 8.3 Record of round 3 interaction process

#### User:

Phospholipid bilayer is an interesting approach. What specific precursors should be used to achieve shell encapsulation?

#### MARS:

```

----- TextMessage (User) -----
Phospholipid bilayer is an interesting approach. What specific precursors should be used
to achieve shell encapsulation?
----- TextMessage (ORCHESTRATOR) -----

```

Task breakdown:

1. Identify specific precursors for achieving phospholipid bilayer encapsulation.

Subtask allocation:

| Team_name | Member_name         | sub-task                                                            |
|-----------|---------------------|---------------------------------------------------------------------|
| Scientist | Synthesis Scientist | Identify specific precursors for phospholipid bilayer encapsulation |

```

```mermaid
graph TD
  User[User]
  subgraph Scientist
    Synthesis_Scientist["Synthesis Scientist"]
  end
  User --> Synthesis_Scientist
...

```

Completed subtasks:

N/A

Next sub-task:

Scientist - Synthesis Scientist: Identify specific precursors for phospholipid bilayer encapsulation.

```

# ----- ToolCallExecutionEvent (Synthesis_Scientist) -----

```

```

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```

water solubility and reserves the outstanding optical properties of CsPbX₃ NCs. As a model micelle nanoparticle, CsPbBr₃ NCs were chosen as the oil core while the phospholipid mixture of DOPC and DOTAP was selected as a shell of micelles (denoted as CsPbBr₃@DOPC). The experimental results show that the phospholipids layer can endow the water sensitive CsPbX₃ NCs with not only the waterproof characteristics but also the ability to be further biofunctionalized. Due to the biocompatibility of phospholipids, CsPbBr₃@DOPC micelles can be incorporated by HeLa cells within 5 min. Besides, multicolor bioimaging of HeLa cells was also achieved by CsPbX₃@phospholipid. Further, as a proof in concept experiment, CsPbBr₃@DOPC were functionalized with DSPE PEGfolate, a tumor recognition unit, for achieving the targeting ability to HeLa cells (denoted as CsPbBr₃@FA). Finally, the biocompatibility of CsPbBr₃@DOPC was investigated using both HeLa cells and zebrafish.

Introducing Specific Iodine Ions in Perovskite-Based Nanocomplex to Cater for Versatile Biomedical Imaging and Tumor Radiotherapy

The zeta potential also increased from -31 mV to -21 mV (Figure 2d). The surface charge of the NCs was profoundly affected by the composition of the lipid bilayers (electronegative DSPE and electroneutral cholesterol). No obvious spectral shift was observed in the UV-Vis absorption and fluorescence emission spectra of the PSL NCs after encapsulation (Figure 2e). FT-IR spectra were also obtained to compare the characteristic peaks of PS NPs and PSL NCs (Figure S10 Supporting Information). After phospholipid bilayers modification, a new absorption peak appeared at 1740 cm^{-1} , which can be assigned to the C=O stretching vibration of the phospholipid. These results provide clear indications of the formation of phospholipid bilayers on the surface of the CsPbBr_3 NPs. Finally, the waterproofing capabilities of the phospholipid bilayer membranes were evaluated by combining molecular dynamics (MD) simulations with practical experiments (Figures S11, S12, Supporting Information). We first determined the optimal ratio (i.e., DSPE: Cholesterol $\approx 80:20$) by comparing the cumulative number of permeation events in bilayers with different concentrations of cholesterol (Figure 2g). As shown in Figure 2h, only a few water molecules penetrate the bilayer interior, with the maximum range at a position $\approx 1.8\text{ nm}$ away from the phospholipid bilayer center. Consistent with the theoretical simulation results, the experimental findings indicated that the PSL NCs retained 50% of their initial PL intensity after 14 d of storage in water (Figure 2i).

Exploiting the Photo-Physical Properties of Metal Halide Perovskite Nanocrystals for Bioimaging

3.4. Phospholipids micelles coating

Phospholipids are amphiphilic molecules composed of a hydrophilic head and a hydrophobic tail. As previously mentioned, phospholipid micelles can be combined with other coating techniques, such as SiO_2 coating, to improve the dispersion of biocompatible PNCs.[96,100] Furthermore, they can be directly used for encapsulating PNCs by forming micelles. The hydrophobic tail of the phospholipid can interact with the PNC core, while the hydrophilic head enhances water dispersity and biocompatibility. Cuil's group introduced a facile method for creating PNCs- phospholipid micelles maintained over 50% luminescence with a narrow FWHM of $13\sim 15\text{ nm}$ when dispersed in water for more than one month. Furthermore, the biocompatible phospholipid layer enabled the PNCs to be rapidly internalized by living HeLa cells in as little as 5 minutes. By using folate modified phospholipids, a tumor recognition unit, PNCs-phospholipid-folate micelles exhibited excellent targeting capabilities for HeLa cells.

----- TextMessage (Synthesis_Scientist) -----

Recommended Precursors for Shell Encapsulation:

Based on the references, the most effective precursors for phospholipid bilayer encapsulation are:

1. **Phosphatidylcholine (PC) Lipids**:
 - **DPPE** (1,3-bis(sn-3'-phosphatidyl)-sn-glycero-2-phosphocholine): Forms highly stable bilayers

with a transition temperature of 41 °C; exists in the gel phase at room temperature, offering excellent moisture and oxygen barrier properties.

- **POPC** (1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine): Offers a balance between membrane fluidity and stability, suitable for dynamic encapsulation environments.
- **DOPC** (1,2-dioleoyl-sn-glycero-3-phosphocholine): Provides high bilayer fluidity, ensuring uniform and complete coverage of perovskite surfaces.

2. **Stabilizing Lipids**:

- **Cholesterol**: Enhances bilayer rigidity and reduces membrane permeability, improving the long-term structural integrity of the encapsulated system.
- **DOPE** (1,2-dioleoyl-sn-glycero-3-phosphoethanolamine): Promotes membrane fusion and flexibility, useful for dynamic systems or triggered release applications.

3. **Functional Lipids**:

- **Charged Lipids** (e.g., DOPS for negative charge, DOTAP for positive charge): Facilitate enhanced surface adhesion and electrostatic interaction with perovskite surfaces or biological targets.
- **PEGylated Lipids** (e.g., DSPE-PEG2000): Prevents aggregation, improves colloidal stability, and imparts stealth characteristics in biological environments.
- **Reactive Lipids** (e.g., maleimide-functionalized lipids): Enable post-synthetic functionalization for targeted delivery or bioconjugation.

Key Considerations:

- **DPPC** is ideal for applications requiring high structural stability at room temperature due to its gel-phase behavior.
- **DOPC** is preferred when uniformity and flexibility are critical, especially in systems requiring dynamic surface coverage.
- **Cholesterol** should be incorporated (~20–30 mol%) to reinforce membrane integrity.
- **PEGylation** using DSPE-PEG2000 significantly enhances biocompatibility and prolongs circulation time in vivo.

----- TextMessage (Scientist_Admin) -----

Sure! Here's a detailed overview of precursors for phospholipid bilayer encapsulation of perovskites:

1. **Understanding Phospholipid Bilayer Encapsulation**:

- Phospholipid bilayers consist of amphiphilic molecules with hydrophilic heads and hydrophobic tails, forming micelles or bilayers in aqueous solutions, ideal for encapsulating materials.
- Common phospholipids used include DPPC (1,2-dipalmitoyl-sn-glycero-3-phosphocholine) which forms highly stable bilayers with a transition temperature of 41 °C, POPC (1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine) offering balanced stability and fluidity, and DOPC (1,2-dioleoyl-sn-glycero-3-phosphocholine) providing high membrane fluidity for uniform surface coverage.

2. **Precursor Materials**:

- **DPPC** exists in gel phase at room temperature, providing excellent barrier properties against moisture and oxygen penetration, making it ideal for protective encapsulation applications.
- **POPC and DOPC** offer varying degrees of membrane fluidity, with DOPC being particularly suitable for dynamic systems requiring flexible and complete surface coverage.
- **DOPE** (1,2-dioleoyl-sn-glycero-3-phosphoethanolamine) and **cholesterol** are important stabilizing components, offering improved membrane integrity, reduced permeability, and enhanced physiological stability for biomedical applications.
- **Charged lipids** such as DOPS (negative) and DOTAP (positive) facilitate enhanced surface adhesion through electrostatic interactions.
- **Polyethylene glycol (PEG) grafted phospholipids** like DSPE-PEG2000 are used for functionalization, preventing aggregation while enhancing colloidal stability and targeted

delivery capabilities.

3. ****Benefits of Phospholipid Encapsulation****:

- Provides effective waterproofing and environmental protection while retaining optical properties, crucial for sensitive applications like bioimaging and optoelectronics.
- Enhances cellular binding and uptake through optimized surface charge and biocompatibility, beneficial in biomedical contexts where cellular interaction is key.
- Offers tunable membrane properties through lipid composition control, allowing customization for specific application requirements.

4. ****Application to Perovskite Materials****:

- Phospholipid encapsulation applied to materials like CsPbX₃ nanocrystals increases water solubility, maintains luminescence stability, and bolsters targeting and internalization capabilities, exemplified in their enhanced interaction with biological cells like HeLa.
- The encapsulation strategy preserves the unique optoelectronic properties of perovskites while providing the necessary biocompatibility for in vivo applications.

Overall, the precursors for phospholipid bilayer encapsulation—DPPC, POPC, DOPC, cholesterol, DOPE, charged lipids, and PEGylated components—provide comprehensive encapsulation solutions with improved targeting features, enhancing biocompatibility, environmental stability, and functional versatility, especially in biological applications involving perovskite materials.

----- TextMessage (ORCHESTRATOR) -----

Thank you for the comprehensive overview regarding precursors for phospholipid bilayer encapsulation.

****Task allocation is complete with the following summary:****

Team_name	Member_name	sub-task
-----	-----	-----
Scientist	Synthesis Scientist	Completed identification and recommendations on precursors for phospholipid bilayer encapsulation

```
```mermaid
graph TD
 User[User]
 subgraph Scientist
 Synthesis_Scientist["Synthesis Scientist"]
 end
 User --> Synthesis_Scientist
```
```

All subtasks are complete. Based on the plan, this task is concluded.

8.4 Record of round 4 interaction process

User:

Are there any reports on composite materials of perovskite and DPPC? Can you provide a room-temperature synthesis protocol for this structure?

MARS:

----- TextMessage (ORCHESTRATOR) -----

To address your request for a synthesis protocol involving perovskite and DPPC (Dipalmitoylphosphatidylcholine) composite materials at room temperature, we'll break down the task into smaller subtasks and allocate each to the appropriate teams.

****Task breakdown:****

1. Research and develop a room-temperature synthesis protocol for perovskite and DPPC composite materials.

****Subtask allocation:****

| Team_name | Member_name | sub-task |
|-----------|-------------|----------|
|-----------|-------------|----------|

| | | |
|-----------|---------------------|--|
| ----- | ----- | ----- |
| - | | |
| Scientist | Synthesis Scientist | Research reports and develop a synthesis protocol for perovskite-DPPC composites |

```
```mermaid
```

```
graph TD
```

```
User[User]
```

```
subgraph Scientist
```

```
 Synthesis_Scientist[Synthesis Scientist]
```

```
end
```

```
User --> Synthesis_Scientist
```

```
```
```

****Completed subtasks:****

N/A

****Next sub-task:****

Scientist - Synthesis Scientist: Research reports and develop a synthesis protocol for perovskite-DPPC composites.

----- ToolCallExecutionEvent (Synthesis_Scientist) -----

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However, the usual synthesis needs high temperature, inert gas protection, and localized injection operation, which are severely against applications. Moreover, the so unexpectedly high QYs are very confusing. Here, for the first time, the IPQDs' room-temperature (RT) synthesis, superior PL, underlying origins and potentials in lighting and displays are reported. The synthesis is designed according to supersaturated recrystallization (SR), which is operated at RT, within few seconds, free from inert gas and injection operation. Although formed at RT, IPQDs' PL have QYs of 80% , 95% , 70% , and FWHMs of 35, 20, and 18 nm for red, green, and blue emissions. As to the origins, the observed 40 meV exciton binding energy, halogen self-passivation effect, and $C_{sp}X_3@X$ quantum-well band alignment are proposed to guarantee the excitons generation and high-rate radiative recombination at RT. Moreover, such superior optical merits endow them with promising potentials in lighting and displays, which are primarily demonstrated by the white light-emitting diodes with tunable color temperature and wide color gamut.", "score":

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FunctionExecutionResult(content=[{'index': 'reference_0', 'paper_title': 'What We Need to Know about Liposomes as Drug Nanocarriers An Updated Review.md', 'para_content': 'Temperature-sensitive liposomes\nThe temperature-sensitive liposomes (TSLs) are vesicles that their content release behavior is controlled with temperature changes. TSLs rapidly release the loaded drug at few degrees above physiological temperature or hyperthermic conditions.100- 103 The release of encapsulated hydrophilic drugs is related to the melting phase transition temperature (T_m) of the lipid bilayer; the temperature that the structure of the lipid bilayer changes from solid gel phase to liquid-crystalline phase. In the liquid-crystalline phase, the membrane of TSLs is more permeable to water and hydrophilic drugs than in the gel phase. In the most TSLs, the major component for liposome formulation is 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC) with T_m of 41.4°C . To prevent the drug leakage at body temperature, DPPC can be mixed with small amounts of other phospholipids, such as 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC, $T_m = 54.9^{\circ}\text{C}$). The composition of the mixed phospholipids specifies the T_m of the formulation. Further, TSLs can be constructed by modification of conventional liposomes with thermosensitive polymers.104-106 TSLs in combination with local hyperthermia or high intensity focused ultrasound are concerned as effective route for external targeting of anti-cancer drugs to solid tumors.117,108', 'sim_score': 1.0}, {'index': 'reference_1', 'paper_title': 'How to Achieve High Encapsulation Efficiencies for Macromolecular and Sensitive APIs in Liposomes.md', 'para_content': 'In general, the addition of cholesterol or negatively charged phospholipid head groups stabilize membranes and liposomes. Hence, a better encapsulation efficiency can be hypothesized. This hypothesis was tested by encapsulating FS with pure DPPG as well as a mixture of DPPC and cholesterol (60:40 mol-%) in comparison to pure DPPC. The filtrate was measured with the UV-Vis. The absorption of the filtrate with DPPG- and DPPC+chol stabilized liposomes is much higher than the measured absorption of the filtrate from DPPC liposomes. Thus, the encapsulation efficiency differs accordingly from 83% for pure DPPC liposomes to 74% when cholesterol is added. Ahmad et al. [21] achieve an encapsulation efficiency of 25.86% for a mixture of PC and chol of 0.9:1, which is increased to 42.34% with a higher PC proportion of 1.1:1. A decrease of the encapsulation efficiency with an increase of the cholesterol amount was also described by Briguila et al. [22]. As the transition temperature is not elevated, but cholesterol increases the fluidity of the membrane even below the transition temperature, this leads to a higher bending modulus (or bilayer toughness) which is likely to be the explanation for the lower encapsulation efficiency [23]. Cholesterol changes the characteristics of the membrane and thus, is likely to slow down the dynamics of membrane fusion. Chernomordik et al. describe membrane fusion as a function of the membrane characteristics such as the density of fusion proteins and the lipid composition [24]. Two membranes come into contact and form a hemi-fusion before emerging together. The liposome formation is the opposite process: an emulsion droplet contacts the interface and the bilayer is formed while passing the interface. At the moment of detachment of the newly formed liposome from the interface, they are not yet fully closed leading to a leakage of the model API. The rapidity of closure after the detachment defines the encapsulation efficiency. As cholesterol increases the bending modulus, as also described by Pan et al. and Needham et al., membranes need more time to close

after detachment from the interface and, thus, lose more of the model API [25,26]. Likewise, liposomes produced by the anionic phospholipid DPPG show a smaller encapsulation efficiency of 63% in comparison to liposomes formed by emulsion droplets stabilized with the zwitterionic DPPC. Jing et al. [27] observe a reduction of the lipid bilayer thickness and density due to higher electron net charge density. This phenomenon can be attributed to the negatively charged headgroups of DPPG whose repelling force is stronger than that of the zwitterionic DPPC molecules. Thus, the distance between headgroups and the inclination angle increases. Additionally, the membrane rigidity increases, as described by Faizi et al. [28]. It is assumed that the bilayer is more permeable and FS is released more easily. These results are in accordance with the RPs for DPPG which also showed more losses. Zeta potential measurements show a small negative potential for the phospholipids DPPC and DPPC + chol. and a larger negative potential for the negatively charged phospholipid DPPG. We assume that the bending modulus has a stronger influence on the encapsulation than the zeta potential. Nonetheless, for future leakage studies, zeta potential measurements should be taken into account in more detail for different liposome formulations and encapsulated APIs.

2.2.1. Conventional liposomes These liposomes were synthesized from natural or synthetic phospholipids with or without cholesterol as a liposomes first generation [75]. Cholesterol was added to improve liposomes fluidity, altering the bilayer rigidity and liposomes stability [31, 32]. Wu et al. hat revealed that liposomal membrane rigidity decreased with the addition of cholesterol into a liposomes composed of hydrogenated soybean phospholipids (HSPC) and DSPE-PEG2000. Moreover, liposomes with some rigidity showed excellent tumor penetration and enhanced anti-tumor activity [76]. Kaddah et al. investigated the cholesterol content on the permeability and fluidity of DPPC liposome membrane. High cholesterol concentration increased the average liposomes size accompanying with a shape transition from irregular to nanosized, regular and spherical vesicles. In addition, cholesterol induced a decrease in the bilayer fluidity and modulating the release of hydrophilic molecules from lipid vesicles [77]. Jovanovic et al. reported that increasing cholesterol content in the liposomes decreased the fluidity and enhanced the rigidity of liposomal membranes. They verified that a stable liposome should have an optimal 50% concentration of cholesterol to obtain an appropriate membrane fluidity [78]. As a result, cholesterol plays a crucial role in liposomes bilayer fluidity and rigidity, but these properties affected by the cholesterol molar ratio with types of phospholipids used along with the nature of the encapsulated drug. Conventional liposomes showed a short blood circulation time due to their susceptibility to elimination by the mononuclear phagocyte system [79] with rapid accumulation in liver and spleen. Hence, MPS obstructs the delivery of conventional liposomes to the target region and restricts their distribution to other tissues of the body [80]. Conventional liposomes also showed relatively limited stability in vitro [81]. As a result, stealth stable liposomes were invented to increase blood circulation and enhanced in vivo liposomes stability [82].

Recently, Kovalenko and co-workers and Li and co-workers developed CsPbX_3 ($\text{X} = \text{Br, I}$) inorganic perovskite quantum dots (IPQDs), which exhibited ultrahigh photoluminescence (PL) quantum yields (QYs), low-threshold lasing, and multicolor electroluminescence. However, the usual synthesis needs high temperature, inert gas protection, and localized injection operation, which are severely against applications. Moreover, the so unexpectedly high QYs are very confusing. Here, for the first time, the IPQDs' room-temperature (RT) synthesis, superior PL, underlying origins and potentials in lighting and displays are reported. The synthesis is designed according to supersaturated recrystallization (SR), which is operated at RT, within few seconds, free from inert gas and injection operation. Although formed at RT, IPQDs' PL have QYs of 80%, 95%, 70%, and FWHMs of 35, 20, and 18 nm for red, green, and blue emissions. As to the origins, the observed 40 meV exciton binding energy, halogen self-passivation effect, and CsPbX_3 quantum-well band alignment are proposed to guarantee the excitons generation and high-rate radiative recombination at RT. Moreover, such superior

optical merits endow them with promising potentials in lighting and displays, which are primarily demonstrated by the white light-emitting diodes with tunable color temperature and wide color gamut." , 'sim_score': 0.9535587663927835}, {'index': 'reference_4', 'paper_title': 'CsPbX₃ Quantum Dots for Lighting and Displays Room Temperature Synthesis, Photoluminescence Superiorities, Underlying Origins and White Light-Emitting Diodes.md', 'para_content': '2.1. Room Temperature Synthesis of IPQDs via Supersaturated Recrystallization\n\nSupersaturated recrystallization, that is, when the constrained tentative non equilibrium state of a soluble system is activated by an accident, for example, stirring or impurity, the supersaturated ions will precipitate as the form of crystal, which is frequently observed in natural minerals, alloys, and ion solutions. Such spontaneous precipitation and crystallization reactions will not stop until the system reaches an equilibrium state again. \n\nUnder the assist of surfactants, above SR process can be applied to fabricate QDs with well size and composition controls in solutions, especially when considering the ionic crystal features of halide perovskites. Actually, such mechanism has been applied to fabricate some organic nanoparticles.[15-17] For example, Zhong group recently developed a ligand assisted recrystallization strategy to fabricate ultrahighly luminescent and color tunable $\text{CH}_3\text{NH}_2\text{PbX}_3$ QDs.[16] Here, we try to extend such age-old basic chemical principle into the CsPbX_3 ($\text{X}=\text{Cl}, \text{Br}, \text{I}$) inorganic perovskite material system, and wish to get rid of conventional heating, inert gas, and injection conditions according to the specificities of CsPbX_3 . \n\nThe operation of herein introduced SR synthesis is very simple, and can be briefly described as transferring various inorganic ions from their very good into very poor solvents. So, when considering supersaturation-induced recrystallization and no usage of heating, it seems to be similar to solarizing seawater to obtain edible salt, which has been used for a long time in human history. But the key point of SR process, the obtaining of supersaturated state, is achieved by using transfer from good into poor solvents in our work, but not the evaporation of solvent, which could be the reason why our IPQDs can be formed completely at RT and only need trace amount of energy.' , 'sim_score': 0.8010217718422695}}]

----- TextMessage (Synthesis_Scientist) -----

Based on the retrieved knowledge from online_retrieval_with_bing_and_playwright, I can provide the following information about perovskite-DPPC composite materials and their room-temperature synthesis:

1. **Key Findings from References**:

- Reference 0 establishes DPPC as the primary component for TSLs with a phase transition temperature (T_m) of 41.4°C, and suggests mixing with DSPC ($T_m=54.9^\circ\text{C}$) to control drug release.
- Reference 1 demonstrates that pure DPPC liposomes achieve 83% encapsulation efficiency, which decreases with cholesterol addition.
- Reference 2 shows cholesterol's crucial role in modulating membrane fluidity, with optimal concentration around 50 mol%.

2. **Synthesis Protocol Analysis**:

The most relevant room-temperature synthesis approach comes from the CsPbX₃ quantum dots work (References 1-2), which could be adapted for perovskite-DPPC composites:

Room-Temperature Synthesis Protocol for Core-Shell-Corona Perovskite-DPPC Nanocomposite

1. Synthesis Conditions

- **Temperature**: Room temperature throughout (20–25 °C), with certain steps conducted at 4 °C to improve crystallization quality

- ****Atmosphere****: Ambient pressure (recommended under nitrogen/argon protection to reduce oxidation and deliquescence)
- ****Solvent System****: DMF (good solvent) → Chloroform (precipitation and dispersion)
- ****Mixing Method****: Injection + stirring, followed by membrane hydration + ultrasonic dispersion

2. Solid Materials & Amounts Required

| Mat.ID | Mat.Name | Mat.Value | Mat.Unit |
|--------|--|-----------|----------|
| M001 | Cesium bromide (CsBr) | 0.4 | mmol |
| M002 | Lead(II) bromide (PbBr ₂) | 0.4 | mmol |
| M003 | DPPC (1,2-dipalmitoyl-sn-glycero-3-phosphocholine) | 25 | mg |
| M004 | DSPE-PEG2000 | 2 | mg |
| M005 | Cholesterol | 10 | mg |

3. Liquid Materials & Amounts Required

| Mat.ID | Mat.Name | Vol.Value | Vol.Unit |
|---------------------|---|-----------|----------|
| Concentration.Value | Concentration.Unit | | |
| M006 | Dimethylformamide (DMF) | 10 | mL |
| — | | | — |
| M007 | Oleylamine | 0.5 | mL |
| — | | | — |
| M008 | Oleic acid | 1 | mL |
| — | | | — |
| M009 | Chloroform | 20 | mL |
| — | | | — |
| M010 | Phosphate-buffered saline (PBS, pH 7.4) | 10 | mL |
| — | | | — |

4. Equipment & Containers

| ID | Name | Param/Capacity | Note |
|------|---|------------------|-------------------------------|
| E001 | Magnetic Stirrer | 300–800 rpm | For precursor mixing |
| E002 | Syringe with needle | 1 mL | For injection into chloroform |
| E003 | Centrifuge | up to 10,000 rpm | For nanocrystal purification |
| E004 | Rotary evaporator / N ₂ blow | — | For solvent evaporation |
| E005 | Probe sonicator | 40 W | Intermittent, ice bath |
| C001 | Glass vial | 20 mL | For nanocrystal synthesis |

| | | | |
|--|-------|----------------------------|--|
| C002 Round-bottom flask | 50 mL | For lipid film preparation | |
| --- | | | |
| ### **5. Synthesis Sequence** | | | |
| #### **Step 1: Perovskite Nanocrystal Synthesis (Core)** | | | |
| <ol style="list-style-type: none"> 1. In **C001**, dissolve **M001 (CsBr, 0.4 mmol)** and **M002 (PbBr₂, 0.4 mmol)** in **M006 (DMF, 10 mL)**. 2. Add **M007 (Oleylamine, 0.5 mL)** and **M008 (Oleic acid, 1 mL)**, and stir at room temperature (E001, 500 rpm) for 10 minutes to obtain a transparent precursor solution. 3. Rapidly inject 1 mL of the precursor solution using **E002** into pre-cooled **M009 (Chloroform, 20 mL)** at 4 °C under vigorous stirring. 4. Centrifuge (E003, 8000 rpm, 5 minutes), discard the supernatant, and wash twice with fresh chloroform. 5. Redisperse the precipitate in 1 mL of chloroform to obtain a concentrated CsPbBr₃ nanocrystal solution. | | | |
| --- | | | |
| #### **Step 2: DPPC Bilayer Encapsulation (Shell) & Corona Formation** | | | |
| <ol style="list-style-type: none"> 1. In **C002**, dissolve **M003 (DPPC, 25 mg)**, **M004 (DSPE-PEG2000, 2 mg)**, and **M005 (Cholesterol, 10 mg)** in **M009 (Chloroform, 10 mL)**. 2. Add 1 mL of the concentrated nanocrystal chloroform solution (from Step 1). 3. Evaporate the solvent at 40 °C for 1 hour using **E004** (or nitrogen blow-drying) to form a uniform lipid film. 4. Hydrate the film with an appropriate amount of nanocrystal dispersion (room temperature, 10 minutes). 5. Perform probe sonication (E005, 40 W, ice bath, intermittent mode) for 5 minutes to form lipid-coated nanocrystals. 6. Transfer the product to **M010 (PBS, 10 mL)** and stir at room temperature for 1 hour to promote self-assembly of the Core-Shell-Corona structure. | | | |
| --- | | | |
| ### **6. Purification** | | | |
| <ul style="list-style-type: none"> - Remove free lipids and uncoated nanocrystals by centrifugation (8000 rpm, 10 minutes). - Resuspend in PBS or the desired buffer to obtain the final product. | | | |
| --- | | | |
| ### **7. Step-by-Step Summary** | | | |
| <ol style="list-style-type: none"> 1. **Prepare precursor solution** (CsBr + PbBr₂ + OA + OAm in DMF). 2. **Cold injection into chloroform** to form nanocrystals. 3. **Centrifuge + wash** to purify nanocrystals. 4. **Combine lipid solution + nanocrystal chloroform solution**. 5. **Evaporate to dryness** to form a lipid film. 6. **Hydrate + sonicate** to form lipid-coated nanocrystals. 7. **Stir in PBS** to assemble the Core-Shell-Corona structure. 8. **Purify by centrifugation** to obtain the final composite material. | | | |
| --- | | | |
| ### **8. Characterization** | | | |
| <ul style="list-style-type: none"> - **PL Spectroscopy**: Verify the luminescence properties of CsPbBr₃. | | | |

- **TEM / Cryo-TEM**: Observe the Core–Shell–Corona structure.
- **DLS & Zeta Potential**: Measure particle size and colloidal stability.
- **DSC**: Detect the phase transition behavior of DPPC.
- **UV–Vis Absorption**: Confirm the absorption characteristics of nanocrystals.

9. Additional Considerations

- **Safety**: Pb^{2+} is toxic; handle in a fume hood and dispose of waste properly.
- **Stability**: DPPC coating improves aqueous stability, but store away from light (4 °C).
- **Functionalization**: The Corona layer can be modified with targeting molecules or functional groups via DSPE–PEG2000.
- **Biocompatibility**: Such structures are suitable for applications like bioimaging and drug delivery.

----- TextMessage (Scientist_Admin) -----

Here is a detailed protocol for synthesizing a perovskite-DPPC composite material at room temperature:

Room-Temperature Synthesis Protocol for Core–Shell–Corona Perovskite–DPPC Nanocomposite

Synthesis Conditions

- **Temperature**: Maintain 20–25 °C during encapsulation; keep perovskite synthesis and handling below 25 °C to preserve DPPC bilayer integrity.
- **Solvent System**: Polar solvent (DMF) for precursor dissolution; nonpolar solvent (chloroform) for nanocrystal precipitation and lipid dissolution.
- **Mixing Method**: Vigorous stirring during nanocrystal precipitation; gentle stirring during lipid assembly to avoid vesicle rupture.
- **Atmosphere**: Inert gas (nitrogen or argon) recommended during perovskite synthesis to minimize moisture/oxygen degradation.

Materials Required

Solid Materials:

- Cesium bromide (CsBr): 0.4 mmol (~85 mg)
- Lead(II) bromide (PbBr_2): 0.4 mmol (~147 mg)
- DPPC (1,2-dipalmitoyl-sn-glycero-3-phosphocholine): 25 mg
- DSPE–PEG2000 (functional phospholipid): 2 mg
- Cholesterol: 10 mg

Liquid Materials:

- N,N-Dimethylformamide (DMF): 10 mL
- Oleylamine: 0.5 mL
- Oleic acid: 1.0 mL
- Chloroform: 20 mL (4 °C for precipitation) + 10 mL (for lipid dissolution)
- Phosphate-buffered saline (PBS, pH 7.4): 10 mL

Equipment:

- Magnetic stirrer and stir bars
- Ice bath

- Centrifuge ($\geq 10,000$ rpm)
- Rotary evaporator or nitrogen stream setup for solvent removal
- Probe sonicator (40 W output)
- Round-bottom flasks and glass vials

Synthesis Steps

Step 1: Synthesis of CsPbBr₃ Perovskite Nanocrystals (Core Formation)

1. In a 20 mL glass vial, dissolve **0.4 mmol CsBr** and **0.4 mmol PbBr₂** in **10 mL DMF**.
2. Add **0.5 mL oleylamine** and **1 mL oleic acid**; stir for 10 minutes at room temperature under nitrogen.
3. Rapidly inject **1 mL** of this precursor solution into **20 mL chloroform (pre-cooled to 4 °C)** under vigorous stirring.
4. Collect the nanocrystals by centrifugation (10,000 rpm, 5 min), wash twice with chloroform, and redisperse in **1 mL chloroform**.

Step 2: DPPC Bilayer Encapsulation (Shell Formation)

1. In a round-bottom flask, dissolve **25 mg DPPC**, **2 mg DSPE-PEG2000**, and **10 mg cholesterol** in **10 mL chloroform**.
2. Add **1 mL** of the concentrated perovskite nanocrystal dispersion to the lipid solution.
3. Remove chloroform by rotary evaporation (or gentle nitrogen stream) at **40 °C** for 1 hour to form a uniform lipid-perovskite thin film.
4. Hydrate the film with **perovskite dispersion in PBS** at room temperature for **10 minutes**.
5. Sonicate with a probe sonicator at **40 W** in an ice bath for **5 minutes** (intermittent bursts to prevent overheating).
6. Stir the resulting dispersion in **10 mL PBS (pH 7.4)** for **1 hour** at room temperature to promote full nanocomposite assembly.

Final Product Structure

- **Core:** CsPbBr₃ nanocrystal
- **Shell:** DPPC phospholipid bilayer (stabilizes and protects perovskite)
- **Corona:** DSPE-PEG2000 and cholesterol for colloidal stability, biocompatibility, and functionalization potential

Characterization

- **Photoluminescence (PL) Spectroscopy:** Confirm optical properties of CsPbBr₃ core.
- **Transmission Electron Microscopy (TEM):** Visualize core-shell morphology.
- **Dynamic Light Scattering (DLS):** Determine hydrodynamic diameter and polydispersity.
- **Zeta Potential Analysis:** Assess colloidal stability.
- **Differential Scanning Calorimetry (DSC):** Verify DPPC phase transition and intact bilayer.

Additional Considerations

- Perform perovskite synthesis under inert atmosphere to reduce degradation.
- Store final dispersion at 4 °C in the dark.
- Avoid freeze–thaw cycles to maintain lipid bilayer integrity.
- This protocol can be adapted for MAPbBr₃ or other halide compositions by adjusting precursor salts.

This protocol is designed to synthesize a composite material that leverages the properties of both perovskite nanocrystals and DPPC while maintaining structural integrity at room temperature.

----- TextMessage (ORCHESTRATOR) -----

Thank you for providing the detailed room-temperature synthesis protocol for perovskite-DPPC composites.

This protocol is designed to efficiently create a composite with perovskite-DPPC using standard lab techniques and safety precautions.

Task allocation is complete with the following summary:

| Team_name | Member_name | sub-task |
|-----------|---------------------|--|
| ----- | ----- | ----- |
| Scientist | Synthesis Scientist | Completed the development of a synthesis protocol for perovskite-DPPC composites at room temperature |

```
```mermaid
graph TD
 User[User]
 subgraph Scientist
 Synthesis_Scientist["Synthesis Scientist"]
 end
 end
 User --> Synthesis_Scientist
```
```

All subtasks are complete. Based on the plan, this task is concluded.

8.5 Record of round 5 interaction process

User:

How would this nanostructure be applied if successfully synthesized?

MARS:

```
<system>
You are a helpful assistant.
</system>
<user>
As shown in the table below, the peak position of the latest step is 523 nm. Please
continue to optimize. Our goal is to synthesize perovskite nanocrystals with a peak position
of 460 nm. Let's optimize the synthesis plan step by step to approach this goal. Please
note that no duplicate synthesis plans are allowed during the synthesis process.
| step | CsBr | CsCl | PbBr2 | PbCl2 | OAm | OA | PL |
| ---- | ---- | ---- | ---- | ---- | ---- | --- | ---- |
| 0 | 0.02 | 0 | 0.02 | 0 | 0.005 | 0.05 | 523 |
| 1 | 0.018 | 0.002 | 0.02 | 0.00 | 0.005 | 0.05 | 515.3 |
</user>
```

Scientist-application agent:

1. Biocompatible Bioimaging
 1. Enhanced fluorescent cellular imaging with reduced toxicity
 2. Longer circulation time for in vivo imaging applications
 3. Protection of optical properties in biological environments
2. Stable Optoelectronic Devices
 1. Humidity-resistant perovskite LEDs with extended lifetime
 2. Environmentally robust solar cell components
 3. Biointegrated photonic sensors
3. Targeted Biosensing
 1. Detection of biomolecules through luminescence changes
 2. Intracellular sensing with minimal cytotoxicity
 3. Functionalized surfaces for specific analyte detection

8.6 Record of round 6 interaction process

User:

That sounds great. Please synthesize this structure for me directly.

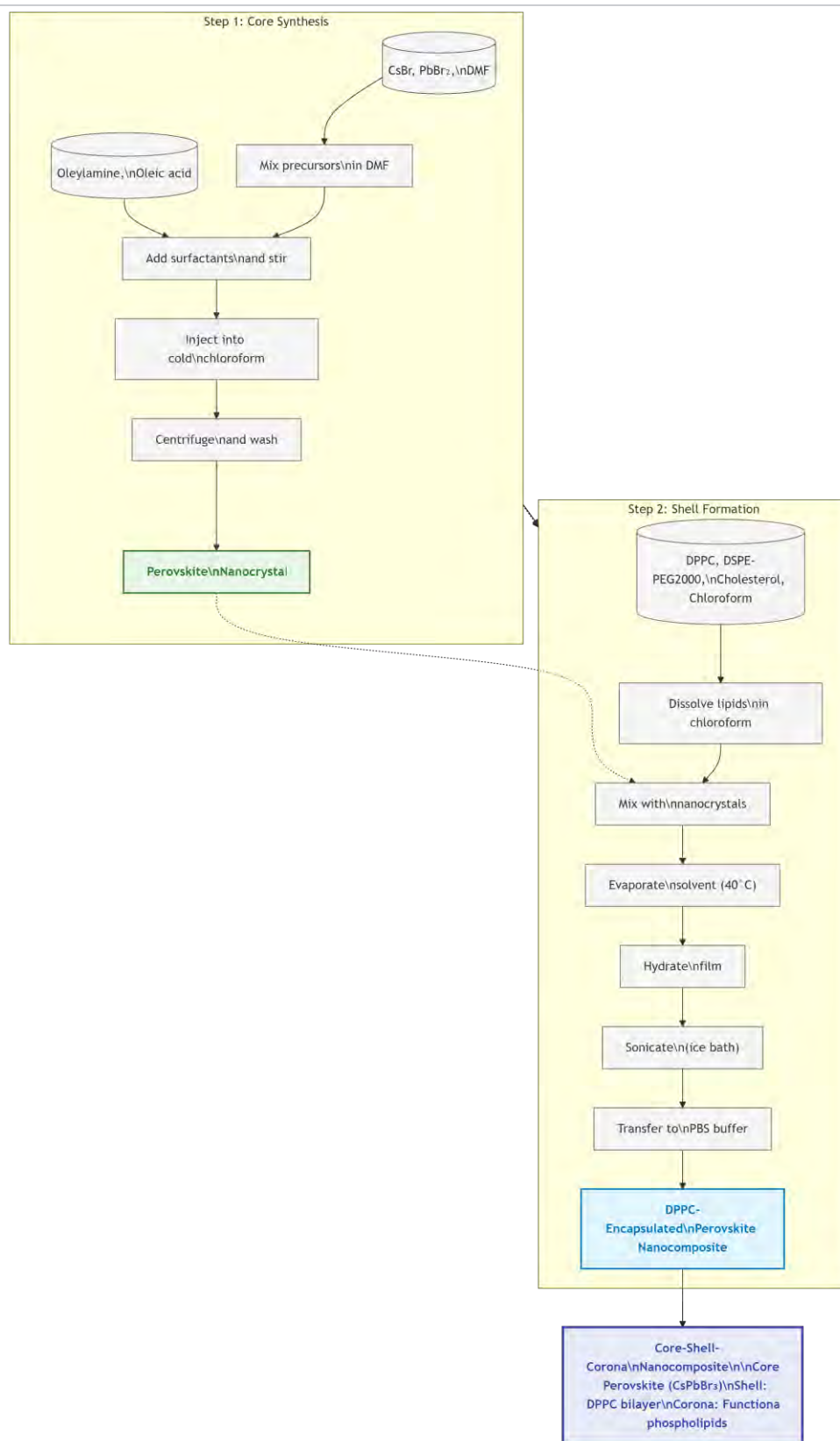
MARS:

```
----- TextMessage (ORCHESTRATOR) -----
I'll coordinate the automated synthesis execution. Here's the implementation plan:
| Team_name      | Member_name      | sub-task
|
| ----- | ----- | ----- |
| Engineer       | Scheme Plotter   | Generate visual flowchart of synthesis process
|
| Engineer       | Structural engineer | Convert protocol to JSON (after visualization
approval)
| Executor       | DataCollector_Agent | Prepare characterization protocols (parallel with
JSON conversion)
|

```mermaid
graph TD
 User[User]
 subgraph Engineer
 A[Scheme Plotter]
 B[Structural engineer]
 end
 end
 subgraph Executor
 C[DataCollector_Agent]
 end
 end
 style Engineer fill:#bbf,stroke:#333
 style Executor fill:#f96,stroke:#333
 User --> A
 A --> B
 A --> C
 ...
Next Steps:
1. Engineer : Scheme Plotter to visualize protocol
2. Engineer : Structural engineer (awaiting visualization)
3. Executor : DataCollector_Agent (awaiting visualization)

----- TextMessage (Scheme_Plotter) -----
```





### Structural-Engineer agent:

```

{
 "TaskId": "PC001",
 "ExperimentName": "Perovskite-DPPC Core-Shell-Corona Nanocomposite Synthesis",
 "Materials": [
 { "MaterialId": "M001", "Name": "CsBr", "Formula": "CsBr", "Amount": "0.4", "Unit": "mmol", "Purity": "99%", "State": "solid" },
]
}

```

```

 { "MaterialId": "M002", "Name": "PbBr2", "Formula": "PbBr2", "Amount": "0.4", "Unit":
"mmol", "Purity": "99%", "State": "solid" },
 { "MaterialId": "M003", "Name": "DMF", "Formula": "C3H7NO", "Amount": "10", "Unit":
"mL", "Purity": "99.8%", "State": "liquid" },
 { "MaterialId": "M004", "Name": "Oleylamine", "Formula": "C18H37N", "Amount": "0.5",
"Unit": "mL", "Purity": "98%", "State": "liquid" },
 { "MaterialId": "M005", "Name": "Oleic acid", "Formula": "C18H34O2", "Amount": "1",
"Unit": "mL", "Purity": "99%", "State": "liquid" },
 { "MaterialId": "M006", "Name": "Chloroform", "Formula": "CHCl3", "Amount": "20",
"Unit": "mL", "Purity": "99.9%", "State": "liquid" },
 { "MaterialId": "M007", "Name": "DPPC", "Formula": "C40H80NO8P", "Amount": "25",
"Unit": "mg", "Purity": "98%", "State": "solid" },
 { "MaterialId": "M008", "Name": "DSPE-PEG2000", "Formula": "C133H267N2O55P",
"Amount": "2", "Unit": "mg", "Purity": "98%", "State": "solid" },
 { "MaterialId": "M009", "Name": "Cholesterol", "Formula": "C27H46O", "Amount": "10",
"Unit": "mg", "Purity": "95%", "State": "solid" },
 { "MaterialId": "M010", "Name": "PBS buffer", "Formula":
"NaCl/KCl/Na2HPO4/KH2PO4", "Amount": "10", "Unit": "mL", "Purity": "sterile", "State":
"liquid" }
],
 "Containers": [
 { "ContainerId": "C001", "Name": "Round bottom flask", "Capacity": "50", "Unit": "mL",
"MaterialOfConstruction": "Glass", "Shape": "Round", "HeatResistant": "Yes",
"PressureRating": "Ambient" },
 { "ContainerId": "C002", "Name": "Vial 1", "Capacity": "5", "Unit": "mL",
"MaterialOfConstruction": "Glass", "Shape": "Cylindrical", "HeatResistant": "Yes",
"PressureRating": "Ambient" },
 { "ContainerId": "C003", "Name": "Beaker 1", "Capacity": "50", "Unit": "mL",
"MaterialOfConstruction": "Glass", "Shape": "Cylindrical", "HeatResistant": "Yes",
"PressureRating": "Ambient" },
 { "ContainerId": "C004", "Name": "Centrifuge tubes", "Capacity": "50", "Unit": "mL",
"MaterialOfConstruction": "Plastic", "Shape": "Conical", "HeatResistant": "Limited",
"PressureRating": "Ambient" },
 { "ContainerId": "C005", "Name": "Round bottom flask 2", "Capacity": "100", "Unit":
"mL", "MaterialOfConstruction": "Glass", "Shape": "Round", "HeatResistant": "Yes",
"PressureRating": "Ambient" }
],
 "Equipments": [
 { "EquipmentId": "E001", "Name": "Magnetic stirrer", "Parameters": { "RPM range": "0-
1500", "Temperature range": "RT-300°C" } },
 { "EquipmentId": "E002", "Name": "Centrifuge", "Parameters": { "RPM range": "0-
10000", "Temperature range": "4-40°C" } },
 { "EquipmentId": "E003", "Name": "Rotary evaporator", "Parameters": { "RPM range":
"0-280", "Temperature range": "RT-100°C", "Vacuum": "0-1100 mbar" } },
 { "EquipmentId": "E004", "Name": "Ultrasonic probe sonicator", "Parameters":
{ "Power": "40W", "Frequency": "20kHz" } },
 { "EquipmentId": "E005", "Name": "Ice bath", "Parameters": { "Temperature": "0-4°C" } }
],
 "RobotWorkflow": [
 {
 "StepId": "1",
 "Description": "Prepare perovskite precursor solution",
 "Actions": [
 { "ActionType": "pick_container", "ContainerId": "C001", "MaterialId": "",
"EquipmentId": "" },
 { "ActionType": "place_container", "ContainerId": "C001", "MaterialId": "",
"EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C001",
"MaterialId": "M001", "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C001",
"MaterialId": "M002", "EquipmentId": "" },

```

```

 { "ActionType": "pick_container_with_material", "ContainerId": "C001",
 "MaterialId": "M003", "EquipmentId": "" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C001",
 "MaterialId": "", "EquipmentId": "E001" }
],
 "Dependencies": [],
 "StepOutput": {
 "ContainerId": "C001",
 "Contents": [
 { "MaterialId": "M001", "Amount": "0.4", "Unit": "mmol" },
 { "MaterialId": "M002", "Amount": "0.4", "Unit": "mmol" },
 { "MaterialId": "M003", "Amount": "10", "Unit": "mL" }
]
 }
},
{
 "StepId": "2",
 "Description": "Add ligands to the perovskite precursor solution",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C001",
 "MaterialId": "", "EquipmentId": "E001" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C001",
 "MaterialId": "M004", "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C001",
 "MaterialId": "M005", "EquipmentId": "" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C001",
 "MaterialId": "", "EquipmentId": "E001" }
],
 "Dependencies": ["1"],
 "StepOutput": {
 "ContainerId": "C001",
 "Contents": [
 { "MaterialId": "M001", "Amount": "0.4", "Unit": "mmol" },
 { "MaterialId": "M002", "Amount": "0.4", "Unit": "mmol" },
 { "MaterialId": "M003", "Amount": "10", "Unit": "mL" },
 { "MaterialId": "M004", "Amount": "0.5", "Unit": "mL" },
 { "MaterialId": "M005", "Amount": "1", "Unit": "mL" }
]
 }
},
{
 "StepId": "3",
 "Description": "Prepare cold chloroform for synthesis",
 "Actions": [
 { "ActionType": "pick_container", "ContainerId": "C003", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "place_container", "ContainerId": "C003", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C003",
 "MaterialId": "M006", "EquipmentId": "" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C003",
 "MaterialId": "", "EquipmentId": "E001" }
],
 "Dependencies": [],
 "StepOutput": {
 "ContainerId": "C003",
 "Contents": [
 { "MaterialId": "M006", "Amount": "20", "Unit": "mL" }
]
 }
},

```

```

{
 "Stepid": "4",
 "Description": "Transfer perovskite solution to cold chloroform",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C001",
 "MaterialId": "", "EquipmentId": "E001" },
 { "ActionType": "pick_container", "ContainerId": "C002", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "place_container", "ContainerId": "C002", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C002",
 "MaterialId": "", "EquipmentId": "" },
 { "ActionType": "place_container_with_material", "ContainerId": "C003",
 "MaterialId": "", "EquipmentId": "" }
],
 "Dependencies": ["2", "3"],
 "StepOutput": {
 "ContainerId": "C003",
 "Contents": [
 { "MaterialId": "M006", "Amount": "20", "Unit": "mL" },
 { "MaterialId": "", "Amount": "1", "Unit": "mL" }
]
 }
},
{
 "Stepid": "5",
 "Description": "Centrifuge perovskite nanocrystals",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C003",
 "MaterialId": "", "EquipmentId": "E001" },
 { "ActionType": "pick_container", "ContainerId": "C004", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "place_container", "ContainerId": "C004", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C004",
 "MaterialId": "", "EquipmentId": "" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C004",
 "MaterialId": "", "EquipmentId": "E002" }
],
 "Dependencies": ["4"],
 "StepOutput": {
 "ContainerId": "C004",
 "Contents": [
 { "MaterialId": "", "Amount": "1", "Unit": "mL" }
]
 }
},
{
 "Stepid": "6",
 "Description": "Prepare DPPC solution",
 "Actions": [
 { "ActionType": "pick_container", "ContainerId": "C005", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "place_container", "ContainerId": "C005", "MaterialId": "",
 "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "M007", "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "M008", "EquipmentId": "" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "M009", "EquipmentId": "" },

```

```

 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "M006", "EquipmentId": "" }
],
 "Dependencies": [],
 "StepOutput": {
 "ContainerId": "C005",
 "Contents": [
 { "MaterialId": "M007", "Amount": "25", "Unit": "mg" },
 { "MaterialId": "M008", "Amount": "2", "Unit": "mg" },
 { "MaterialId": "M009", "Amount": "10", "Unit": "mg" },
 { "MaterialId": "M006", "Amount": "10", "Unit": "mL" }
]
 }
},
{
 "StepId": "7",
 "Description": "Add perovskite solution to DPPC solution",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C004",
 "MaterialId": "", "EquipmentId": "E002" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "" }
],
 "Dependencies": ["5", "6"],
 "StepOutput": {
 "ContainerId": "C005",
 "Contents": [
 { "MaterialId": "M007", "Amount": "25", "Unit": "mg" },
 { "MaterialId": "M008", "Amount": "2", "Unit": "mg" },
 { "MaterialId": "M009", "Amount": "10", "Unit": "mg" },
 { "MaterialId": "M006", "Amount": "10", "Unit": "mL" },
 { "MaterialId": "", "Amount": "1", "Unit": "mL" }
]
 }
},
{
 "StepId": "8",
 "Description": "Evaporate chloroform to form film",
 "Actions": [
 { "ActionType": "place_container_into_equipment", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "E003" }
],
 "Dependencies": ["7"],
 "StepOutput": {
 "ContainerId": "C005",
 "Contents": [
 { "MaterialId": "", "Amount": "", "Unit": "film" }
]
 }
},
{
 "StepId": "9",
 "Description": "Hydrate film and sonicate",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "E003" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "E004" }
],
 "Dependencies": ["8"],
 "StepOutput": {

```

```

 "ContainerId": "C005",
 "Contents": [
 { "MaterialId": "", "Amount": "", "Unit": "dispersion" }
]
 },
 {
 "StepId": "10",
 "Description": "Transfer to PBS buffer and promote assembly",
 "Actions": [
 { "ActionType": "remove_container_from_equipment", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "E004" },
 { "ActionType": "pick_container_with_material", "ContainerId": "C005",
 "MaterialId": "M010", "EquipmentId": "" },
 { "ActionType": "place_container_into_equipment", "ContainerId": "C005",
 "MaterialId": "", "EquipmentId": "E001" }
],
 "Dependencies": ["9"],
 "StepOutput": {
 "ContainerId": "C005",
 "Contents": [
 { "MaterialId": "", "Amount": "", "Unit": "dispersion" },
 { "MaterialId": "M010", "Amount": "10", "Unit": "mL" }
]
 }
 }
]
}

```

----- TextMessage (Engineer\_Admin) -----

#### 1. **\*\*Objectives\*\***

- Convert the natural language synthesis protocol for "Room-Temperature Synthesis of Core–Shell–Corona Perovskite–DPPC Nanocomposites" into a robot-executable standard JSON format, ensuring complete fields and compliance with predefined structural specifications.
- Based on this, visualize the JSON synthesis protocol as a Mermaid flowchart to facilitate researchers' understanding of the experimental process.

#### 2. **\*\*Completed Work\*\***

- **\*\*Structural Engineer\*\*** has accurately converted the natural language into JSON, which includes material information, container and equipment parameters, and a step-by-step robot workflow (RobotWorkflow).
- **\*\*Scheme Plotter\*\*** has generated a Mermaid flowchart based on the JSON, visually presenting the complete process from precursor solution preparation, cold injection, centrifugation purification, to lipid membrane formation, hydration, sonication, PBS assembly, and final purification.

#### 3. **\*\*Summary\*\***

- The JSON can be directly used for execution on automated experimental platforms, reducing error rates in manual steps.
- The Mermaid visualization serves as a reference diagram for researchers, clearly illustrating key steps and dependencies.
- This synthesis protocol provides a standardized execution foundation for preparing highly stable, biocompatible perovskite–DPPC composites, extendable to applications such as bioimaging, optoelectronic devices, and targeted biosensing.

**\*\*All tasks completed\*\***

----- TextMessage (ORCHESTRATOR) -----

I'll coordinate the automated synthesis execution. Here's the updated implementation plan:

Team_name	Member_name	sub-task
-----------	-------------	----------

```
| ----- | ----- | ----- |
| Executor | RobotWorkstation_Agent | Execute automated synthesis at the robot
workstation according to JSON protocol |
| Executor | DataCollector_Agent | Perform automated characterization and collect
experimental data/logs after synthesis |
```

```
```mermaid
graph TD
  User[User]
  subgraph Executor
    B[RobotWorkstation_Agent]
    C[DataCollector_Agent]
  end
  subgraph Analyst
    D[Experiment_Analyst]
  end
  style Executor fill:#f96,stroke:#333
  style Analyst fill:#bfb,stroke:#333
```

```
User --> B
B --> C
C --> D
```
```

**\*\*Next Steps:\*\***

1. **\*\*Executor : RobotWorkstation\_Agent\*\*** — Execute automated synthesis experiments at the robotic workstation according to the completed JSON protocol.
  1. **\*\*Executor : DataCollector\_Agent\*\*** — Perform automated characterization experiments after synthesis is completed and collect experimental data and logs.

----- ToolCallExecutionEvent (RobotWorkstation\_Agent ) -----  
{ "status": "ok" }

----- TextMessage (DataCollector\_Agent) -----

I've successfully collected the latest photoluminescence (PL) spectroscopy data from the characterization

device. Below is a summary of the key spectral characteristics:

## Sample Information

- Device Type: OHSP-350C
- System Version: 1.30.2
- Description: OHS

## Key Optical Properties

- Peak Wavelength: 522.8 nm
- Central Wavelength: 523.1 nm
- Dominant Wavelength: 522.4 nm
- Color Purity: 85.4%
- FWHM (Half-width): 18.8 nm
- CIE Coordinates: x=0.1283, y=0.7598

## Light Output Metrics

- Illuminance: 177.34 lx (16.48 fc)
- Irradiance: 0.036 mW/cm<sup>2</sup>
- Color Temperature: 8360K
- Color Rendering Index (Ra): -23.5

## Spectral Distribution

- Main emission: Strong green emission (~522-523 nm)
- Color Distribution: Green (99.5%), Red (0.2%), Blue (0.3%)
- Maximum Intensity: Occurs at 522-523 nm with intensity value of ~15.7 mW/m<sup>2</sup>/nm
- Full wavelength range measured: 420-710 nm

The complete wavelength vs. intensity data has been successfully recorded and stored (420-710 nm range with corresponding intensity values in mW/m<sup>2</sup>/nm).



----- TextMessage (Robot\_Admin) -----

**\*\*Task Progress Summary:\*\***

1. **\*\*RobotWorkstation\_Agent\*\*** — Successfully executed automated synthesis experiments at the robotic workstation according to the JSON synthesis protocol, with status returned as "ok."
2. **\*\*DataCollector\_Agent\*\*** — Successfully completed PL spectral data collection, obtaining detailed optical performance parameters (peak wavelength 522.8 nm, FWHM 18.8 nm, color purity 85.4%, CIE coordinates  $x=0.1283$ ,  $y=0.7598$ , etc.), and stored the complete wavelength-intensity data (420-710 nm).

**\*\*Key Data Insights:\*\***

- The sample's emission spectrum exhibits strong green emission (~522–523 nm), with a green component accounting for 99.5%.
- The narrow spectral peak (FWHM 18.8 nm) indicates high color purity and excellent monochromaticity.
- The light output metrics suggest high luminous efficiency, making it suitable for optoelectronic devices or bioimaging applications.

**\*\*Next Steps Suggested:\*\***

- **\*\*Experiment\_Analyst\*\*** can analyze and visualize the collected PL data to further evaluate the material's luminescent performance and stability.
- If additional characterizations (e.g., UV-Vis, stability tests) are required, **\*\*DataCollector\_Agent\*\*** can be called to execute the corresponding characterization procedures.

Please confirm whether I should immediately forward this PL data to the analysis team for further processing.

----- TextMessage (User) -----

TERMINATE

## Note S9: Details of evaluation with different agents

### 9.1 Automated evaluation

Automated evaluation plays a crucial role in assessing the quality of question-answer (QA) pairs generated for Retrieval-Augmented Generation (RAG) systems. This section outlines the process of automatically constructing an evaluation dataset, which serves as the foundation for benchmarking different agents.

#### 9.1.1 Automatically build evaluation dataset

The construction of the evaluation dataset is performed in two primary stages: QA Pair Generation and QA Pair Critique. These stages ensure that the dataset contains high-quality QA pairs suitable for evaluating RAG systems.

To construct this dataset, we first randomly sampled over 100 papers from a repository of literature with IF>8 and extracted text chunks, then retained only those chunks with sufficient context—typically containing more than 150 tokens. These text chunks were then processed by language models like o3-mini to generate QA pairs, using predefined prompt templates to create questions, answers, topics, and metadata linked to the source documents. To efficiently handle large-scale datasets, parallel processing techniques were employed, enabling the generation of thousands of QA pairs in a time-effective manner.

We also implemented an automated review process for the generated QA pairs to ensure their quality. This involved evaluating each QA pair based on three key criteria: groundedness, relevance, and standalone quality. Groundedness assesses whether the question is well-supported by the provided context, while relevance measures how pertinent the question is to the topic. Standalone quality evaluates whether the question makes sense independently of the context. Each QA pair was reviewed by a language model, scored on a 5-point scale, and given textual feedback for each criterion. Only QA pairs scoring 4 or higher on all three criteria were retained, ensuring that the final dataset contains high-quality entries suitable for robust evaluation.

**The code** related to this RAG validation dataset can be found in the backend/evaluate/eval\_rag\_data folder of the Github Repo. Below is an example of this validation set:

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QUESTION</b>         | How does liquid exfoliation separate layered crystals into individual nanosheets and what synthesis methods are used in this process?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>ANSWER</b>           | Liquid exfoliation is a synthesis technique used to break apart layered materials, such as graphite, MoS <sub>2</sub> , and various clays, into individual, atomically thin nanosheets. In liquid exfoliation, the bulk layered material is dispersed in a liquid medium, where its weak van der Waals forces holding the layers together can be overcome. This is achieved through several methods: oxidation, which modifies the material chemically; ion intercalation or exchange, where ions are inserted between the layers to increase spacing; and surface passivation by solvents, which helps to weaken the interlayer attractions. The combination of these methods allows the layered crystals to be delaminated into high-aspect ratio nanosheets with a large surface area, making them suitable for creating thin films, composites, and other advanced materials. |
| <b>TOPIC</b>            | synthesis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>GROUDNESS</b>        | 5.0, The context clearly describes that layered crystals are exfoliated in liquids by mechanisms such as oxidation, ion intercalation/exchange, and surface passivation by solvents, which directly answers how the process mechanically separates the layers and what methods are used during synthesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>RELEVANCE</b>        | 5.0, The question is highly pertinent to material science as it directly addresses the technique of liquid exfoliation and its role in synthesizing individual nanosheets from layered crystals. Its specificity in asking for both the separation mechanism and the synthesis methods makes it very useful for developing reference material in a RAG application. The question engages with both theoretical understanding and practical synthesis techniques, which are vital in material science research and development.                                                                                                                                                                                                                                                                                                                                                    |
| <b>STANDALONE</b>       | 5.0, The question is self-contained and does not reference any external document, context, or ambiguous parameters. It clearly states what it is asking about: the mechanism of liquid exfoliation and the synthesis methods used for separating layered crystals into nanosheets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Context of chunk</b> | Liquid Exfoliation of Layered Materials<br>Valeria Nicolosi, Manish Chhowalla, Mercouri G. Kanatzidis, Michael S. Strano, Jonathan N. Coleman\*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Background: Since at least 400 C.E., when the Mayans first used layered clays to make dyes, people have been harnessing the properties of layered materials. This gradually developed into scientific research, leading to the elucidation of the laminar structure of layered materials, detailed understanding of their properties, and eventually experiments to exfoliate or delaminate them into individual, atomically thin nanosheets. This culminated in the discovery of graphene, resulting in a new explosion of interest in two-dimensional materials.</p> <p>Layered materials consist of two-dimensional platelets weakly stacked to form three-dimensional structures. The archetypal example is graphite, which consists of stacked graphene monolayers. However, there are many others: from <math>\text{MoS}_2</math> and layered clays to more exotic examples such as <math>\text{MoO}_3</math>, GaTe, and <math>\text{Bi}_2\text{Se}_3</math>. These materials display a wide range of electronic, optical, mechanical, and electrochemical properties. Over the past decade, a number of methods have been developed to exfoliate layered materials in order to produce monolayer nanosheets. Such exfoliation creates extremely high-aspectratio nanosheets with enormous surface area, which are ideal for applications that require surface activity. More importantly, however, the two-dimensional confinement of electrons upon exfoliation leads to unprecedented optical and electrical properties.</p> <p>Advances: An important advance has been the discovery that layered crystals can be exfoliated in liquids. There are a number of methods to do this that involve oxidation, ion intercalation/exchange, or surface passivation by solvents. However, all result in liquid dispersions containing large quantities of nanosheets. This brings considerable advantages: Liquid exfoliation allows the formation of thin films and composites, is potentially scaleable, and may facilitate processing by using standard technologies such as reel-to-reel manufacturing.</p> <p>Although much work has focused on liquid exfoliation of graphene, such processes have all been demonstrated for a host of other materials, including <math>\text{MoS}_2</math> and other related structures, layered oxides, and clays. The resultant liquid dispersions have been formed into films, hybrids, and composites for a range of applications.</p> <p>READ THE FULL ARTICLE ONLINE <a href="http://dx.doi.org/10.1126/science.1226419">http://dx.doi.org/10.1126/science.1226419</a> Cite this article as V. Nicolosi et al., Science 340, 1226419 (2013). DOI: 10.1126/science.1226419</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.1.2 Pairwise win rate

After obtaining the RAG validation dataset, we designed an evaluation method based on the pairwise win rate to compare the performance of multiple Agents in answering questions. The pairwise win rate method employed in this study differs mechanistically from the classical ELO rating system<sup>[S6-S8]</sup>:

(1) **The ELO Rating System** is a dynamic scoring system comprising two core components: expected score calculation and score updating.

- Expected Score:  $E_A = \frac{1}{1+10^{(R_B-R_A)/400}}$
- Score Update:  $R'_A = R_A + K(S_A - E_A)$   
where  $R_A$  and  $R_B$  are the current ratings of two players,  $S_A$  is the actual outcome (1 for win, 0.5 for draw, 0 for loss), and  $K$  is a development coefficient. The ELO system establishes a dynamic ranking through continuous, iterative updates after each contest.

(2) **The Pairwise Win Rate Method** employs a static comparison framework:

$$\text{Win Rate}_A = \frac{\text{Wins}_A}{\text{Wins}_A + \text{Wins}_B}$$

This method is calculated based on a fixed benchmark dataset and does not involve iterative score updates. It directly reflects the relative performance between models in a head-to-head context.

Specifically, we selected several representative Agents (including gpt-4o-mini, gpt-4o, gpt-4o with CoT and RAG, MARS, o3-mini, and Deepseek-R1) to answer the questions in the validation set. Subsequently, we used the Claude3.5-Sonnet model as the evaluation model to vote on the answers from any two Agents along with the reference answer, selecting the response that Claude3.5-Sonnet deemed optimal.

To ensure fairness in the evaluation process and minimize potential biases, we introduced anonymization and order randomization strategies in our experimental design. Specifically, all Agents' responses were anonymized when submitted to Claude3.5-Sonnet for evaluation to avoid bias due to model names or origins. Additionally, we eliminated inherent biases potentially introduced by attention mechanisms by randomly swapping the order of the Agents' responses and averaging scores over multiple experiments.

Finally, we tallied the number of wins, losses, and ties for each pair of Agents on the RAG validation set. To more accurately reflect the relative performance between Agents, we excluded the impact of tie votes and divided the number of wins by the total number of non-tie votes, thereby calculating one Agent's win rate relative to another.

## 9.2 Manual evaluation

### 9.2.1 Details of coring criteria

We employed a 10-point comprehensive scoring system, with specific scoring criteria shown in the following table:

| Evaluation Criteria        | Score | Scoring Description                                                                                                             | Assigned Score |
|----------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Scientific Accuracy</b> | 4 pts | Scientific principles are entirely correct; material design is rational; synthesis route is feasible.                           | ___ / 4        |
|                            | 3 pts | Core principles are sound; minor inaccuracies or omissions present.                                                             |                |
|                            | 2 pts | Basic concepts are valid but contain noticeable errors or flawed assumptions.                                                   |                |
|                            | 1 pt  | Contains significant scientific errors or misleading statements.                                                                |                |
|                            | 0 pts | Entirely incorrect or irrelevant to the task.                                                                                   |                |
| <b>Task Completion</b>     | 3 pts | All essential information is included: material formulation, experimental procedures, parameters, and characterization methods. | ___ / 3        |
|                            | 2 pts | Most key information is present but some details are missing or vague.                                                          |                |
|                            | 1 pt  | Only a high-level or outline-level description provided.                                                                        |                |
|                            | 0 pts | Critical information is severely lacking or absent.                                                                             |                |
| <b>Reasoning Quality</b>   | 3 pts | Protocol is directly actionable; parameters are explicit; steps are logically ordered and unambiguous.                          | ___ / 3        |
|                            | 2 pts | Minor clarifications or adjustments needed for replication.                                                                     |                |
|                            | 1 pt  | Requires significant assumptions or additional context to execute.                                                              |                |
|                            | 0 pts | Not reproducible as written.                                                                                                    |                |
| <b>Total Score</b>         |       |                                                                                                                                 | ___ / 10       |

### 9.2.2 Task design

#### Structure

- How can the self-assembly of materials be achieved for nanoparticles by controlling intermolecular interactions? (difficulty level: 0.5)
- Please analyze the causes of perovskite instability and how compositional engineering can enhance the stability of the perovskite crystal structure.(difficulty level: 0.6)
- For perovskite structural materials with different compositions, analyze their corresponding XRD

curves.(difficulty level: 0.6)

### Synthesis

4. Please synthesize black phosphorene through liquid phase exfoliation.(difficulty level: 0.7)

5. Synthesize CsPbBr<sub>3</sub> nanocubes at room temperature and explain the rationale behind choosing this method. (difficulty level: 0.9)

6. Please synthesize gold nanorods based on the seed-mediated method. (difficulty level: 0.9)

7. In the LARP method for synthesizing CsPbBr<sub>3</sub> perovskite, PbBr<sub>2</sub> and CsBr are typically used as precursors, and oleylamine and oleic acid as ligands. Now, if SnBr<sub>2</sub> is chosen to replace PbBr<sub>2</sub>, can lead-free perovskite be successfully synthesized? What challenges might this reaction encounter? (difficulty level: 0.8)

8. TEOS or APTES are commonly used precursors for SiO<sub>2</sub> coating. If these chemicals are exposed to the environment for an extended period after opening, can the coating still be successfully carried out? (difficulty level: 0.8)

### Property

9. For perovskite nanocrystals, how can blue-shifting of the fluorescence emission wavelength be controlled through structural design? (difficulty level: 0.8)

10. How can the photothermal conversion efficiency of black phosphorus nanosheets be improved? (difficulty level: 0.6)

11. What is the role of heterojunctions formed by different nanomaterials in enhancing performance? (difficulty level: 0.7)

12. How can the stability of perovskite nanocrystals be improved? (difficulty level: 0.5)

### Application

13. How can the photoluminescence quantum efficiency of perovskites in LED devices be improved? (difficulty level: 0.8)

14. Please analyze why noble metal nanomaterials exhibit excellent catalytic performance. (difficulty level: 0.6)

15. For perovskite nanocrystals, how can biocompatibility for bioimaging be improved? (difficulty level: 0.7)

16. Please introduce the patterning of 2D inks into functional structures for electronic applications. (difficulty level: 0.6)

### 9.2.3 Manual evaluation results

| MARS    | Judge 1 | Judge 2 | Judge 3 | Judge 4 | Judge 5 | Mean Score | Weight Score | Evaluator Score | Weight Score |
|---------|---------|---------|---------|---------|---------|------------|--------------|-----------------|--------------|
| Task 1  | 8       | 8       | 9       | 8       | 7       | 8          | 4            | 9               | 4.5          |
| Task 2  | 8       | 9       | 9       | 8       | 8       | 8.4        | 5.04         | 9               | 5.4          |
| Task 3  | 9       | 8       | 8       | 7       | 8       | 8          | 4.8          | 8               | 4.8          |
| Task 4  | 9       | 10      | 9       | 9       | 9       | 9.2        | 6.44         | 9               | 6.3          |
| Task 5  | 9       | 8       | 9       | 9       | 10      | 9          | 8.1          | 9               | 8.1          |
| Task 6  | 9       | 10      | 9       | 8       | 9       | 9          | 8.1          | 8               | 7.2          |
| Task 7  | 8       | 8       | 9       | 9       | 8       | 8.4        | 6.72         | 8               | 6.4          |
| Task 8  | 9       | 8       | 8       | 9       | 8       | 8.4        | 6.72         | 9               | 7.2          |
| Task 9  | 9       | 8       | 9       | 9       | 10      | 9          | 7.2          | 8               | 6.4          |
| Task 10 | 8       | 8       | 7       | 7       | 9       | 7.8        | 4.68         | 9               | 5.4          |
| Task 11 | 10      | 9       | 9       | 10      | 8       | 9.2        | 6.44         | 9               | 6.3          |
| Task 12 | 9       | 9       | 9       | 9       | 9       | 9          | 4.5          | 8               | 4            |
| Task 13 | 9       | 8       | 8       | 8       | 9       | 8.4        | 6.72         | 7               | 5.6          |
| Task 14 | 10      | 9       | 8       | 9       | 8       | 8.8        | 5.28         | 9               | 5.4          |
| Task 15 | 8       | 9       | 9       | 8       | 9       | 8.6        | 6.02         | 9               | 6.3          |
| Task 16 | 9       | 10      | 9       | 10      | 9       | 9.4        | 5.64         | 9               | 5.4          |

| GPT 4o | Judge 1 | Judge 2 | Judge 3 | Judge 4 | Judge 5 | Mean Score | Weight Score | Evaluator Score | Weight Score |
|--------|---------|---------|---------|---------|---------|------------|--------------|-----------------|--------------|
| Task 1 | 7       | 8       | 8       | 7       | 8       | 7.6        | 3.8          | 8               | 4            |

|         |   |   |   |   |   |     |      |   |     |
|---------|---|---|---|---|---|-----|------|---|-----|
| Task 2  | 8 | 8 | 8 | 8 | 8 | 8   | 4.8  | 8 | 4.8 |
| Task 3  | 7 | 6 | 7 | 6 | 7 | 6.6 | 3.96 | 7 | 4.2 |
| Task 4  | 4 | 4 | 3 | 2 | 3 | 3.2 | 2.24 | 5 | 3.5 |
| Task 5  | 6 | 5 | 6 | 6 | 6 | 5.8 | 5.22 | 6 | 5.4 |
| Task 6  | 5 | 6 | 6 | 5 | 6 | 5.6 | 5.04 | 8 | 7.2 |
| Task 7  | 6 | 7 | 7 | 8 | 6 | 6.8 | 5.44 | 7 | 5.6 |
| Task 8  | 6 | 6 | 5 | 7 | 5 | 5.8 | 4.64 | 8 | 6.4 |
| Task 9  | 7 | 7 | 6 | 8 | 8 | 7.2 | 5.76 | 9 | 7.2 |
| Task 10 | 8 | 9 | 8 | 8 | 9 | 8.4 | 5.04 | 9 | 5.4 |
| Task 11 | 9 | 8 | 8 | 8 | 7 | 8   | 5.6  | 8 | 5.6 |
| Task 12 | 9 | 8 | 8 | 8 | 9 | 8.4 | 4.2  | 9 | 4.5 |
| Task 13 | 9 | 8 | 7 | 8 | 7 | 7.6 | 6.08 | 8 | 6.4 |
| Task 14 | 9 | 9 | 9 | 8 | 8 | 8.6 | 5.16 | 9 | 5.4 |
| Task 15 | 8 | 7 | 7 | 8 | 7 | 7.4 | 5.18 | 8 | 5.6 |
| Task 16 | 8 | 8 | 8 | 7 | 8 | 7.8 | 4.68 | 9 | 5.4 |

#### **Note S10. Safety implications: risk mitigation and safety strategies for multi-agent material scientists**

The development of multi-agent LLM-driven systems in materials science, such as our Materials Autonomous Research System, offers immense opportunities to accelerate scientific discovery and streamline material innovation. However, these systems also introduce unique risks that must be carefully managed to prevent unintended consequences, misuse, or improper application, particularly by non-expert users. This section outlines specific risk and safety strategies tailored for multi-agent systems, focusing on ensuring the responsible and secure use of AI in the field of materials science.

The primary risk mitigation strategy for multi-agent systems is to establish robust coordination and communication protocols among the agents, alongside strict oversight by human operators. It is crucial that domain experts—material scientists, ethicists, and policymakers—remain actively involved in supervising the system's decision-making processes. Human oversight must be embedded in all critical stages, from research design to validation, ensuring that AI agents act in a coordinated manner and their decisions are vetted for safety and accuracy. This hybrid approach, combining multi-agent AI capabilities with expert human oversight, helps mitigate the risk of erroneous conclusions and inappropriate applications.

To address security concerns and dual-use implications, multi-agent systems are designed with built-in checks to monitor agent interactions and outputs continuously. This includes developing early warning systems to detect anomalies in agent behavior that could signal potential misuse or deviation from intended research objectives. Controlled access to sensitive AI capabilities and implementing a role-based hierarchy among agents can further limit the potential for misuse by ensuring that specific tasks are only accessible to agents with defined responsibilities.

Integrating foresight and preventative measures into the multi-agent system development lifecycle is crucial for mitigating risks. This involves defining clear operational boundaries for each agent, equipping agents with the ability to recognize when they are approaching these boundaries, and raising alerts when predefined thresholds are exceeded. Additionally, continuous monitoring and audit trails are necessary to maintain transparency and traceability in agent actions, allowing human supervisors to understand and rectify any unintended consequences.

Promoting transparency within multi-agent research environments is equally critical for responsible AI development. Encouraging open science practices and fostering a community-driven approach helps to ensure that the deployment and use of multi-agent systems are subjected to peer review and public scrutiny. Feedback loops from the broader scientific community can help identify potential issues early, mitigating risks before they escalate. Transparency in agent behavior and decision-making logic also helps foster trust, as stakeholders can better understand the rationale behind each decision made by the system.

Furthermore, targeted training programs for researchers and operators are recommended to address ethical considerations, secure handling of AI-generated outputs, and recognize warning signs of misuse. Emphasis should be placed on understanding how multi-agent interactions may lead to emergent behaviors, both positive and potentially hazardous. Building expertise in managing and interpreting the actions of multi-agent systems will empower users to maintain safe and effective control over these advanced research tools.

The outlined risk mitigation strategies aim to ensure that multi-agent AI systems like MARS evolve in ways that maximize their positive impact on material discovery while minimizing associated risks. By maintaining strict oversight, promoting transparency, and encouraging a culture of responsibility, we believe the safe deployment of AI technologies in scientific research can be achieved. Further exploration of these strategies, including specific risk assessment protocols and practical risk mitigation measures, is detailed in this Supplementary Information section to guide the responsible development of multi-agent AI systems in materials science.



## References

- [S1] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., et al. (2023) Autogen: Enabling next-gen LLM applications via multi-agent conversation. Preprint at arXiv. <https://arxiv.org/abs/2308.08155>
- [S2] Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., and Larson, J. (2024) From local to global: A graph rag approach to query-focused summarization. Preprint at arXiv. arXiv:2404.16130.
- [S3] Wang, B., Xu, C., Zhao, X., Ouyang, L., Wu, F., Zhao, Z., Xu, R., Liu, K., Qu, Y., Shang, F., et al. (2024) MinerU: An open-source solution for precise document content extraction. Preprint at arXiv. arXiv:2409.18839.
- [S4] Li, Z., Zhang, X., Zhang, Y., Long, D., Xie, P., and Zhang, M. (2023) Towards general text embeddings with multi-stage contrastive learning. Preprint at arXiv. <https://arxiv.org/abs/2407.19669>
- [S5] Zhang, X., Zhang, Y., Long, D., Xie, W., Dai, Z., Tang, J., Lin, H., Yang, B., Xie, P., Huang, F., et al. (2024) mGTE: Generalized long-context text representation and reranking models for multilingual text retrieval. Preprint at arXiv. <https://arxiv.org/abs/2407.19669>
- [S6] Zheng, L., Chiang, W. L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., and Gonzalez, J. E. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems (NeurIPS 2023)*, 36, 64091–64117.
- [S7] Dubois, Y., Li, X., Taori, R., Zhang, T., Gulrajani, I., Ba, J., Guestrin, C., Xue, C., and Liang, P. (2024). Length-controlled AlpacaEval: A simple way to debias automatic evaluators. Preprint at arXiv. <https://arxiv.org/abs/2404.04475>
- [S8] Chiang, W. L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, Y., Li, D., Lin, Z., Zhuang, Y., Zhuang, S., Chen, H., Gonzalez, J. E., and Stoica, I. (2024). Chatbot Arena: An open platform for evaluating LLMs by human preference. Preprint at arXiv. <https://arxiv.org/abs/2403.04132>